

The Idle Margin: Why Everyday Conservation Goes Undone

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Abstract

Vehicle idling—leaving the engine running while stopped—is a repeated conservation decision in which private and social incentives can point in the same direction: for sufficiently long stops, switching off saves fuel while reducing emissions. Why, then, do drivers persist in idling? In partnership with a leading telematics firm in Pakistan, I collect 10-second GPS data on ignition status, speed, and location from 6,621 private vehicles. The average vehicle idles for 27 minutes per observed day. Baseline beliefs reveal widespread misperceptions about whether warming up an engine extends engine life and how long a stop must last before shutting off saves fuel or reduces environmental harm. I use a randomised additive design that varies private information, social information, payments, and an in-car reminder technology. Private information reduces daily idling from stops longer than 30 seconds by 1.8 minutes, or 9% of the control mean, with effects persisting after treatment ends. Estimated benefits are 3.7 times programme costs over the eight-week intervention, with about half accruing directly to drivers through fuel savings. Adding social information or payments produces no detectable additional reduction. By contrast, activating a device that provides repeated in-car reminders *increases* idling by 1.0 minute, with the increase concentrated in long idling spells—a pattern consistent with procrastination. The results identify idling as a malleable and consequential behavioural margin, while showing that interventions designed to reduce it can backfire.

Keywords: vehicle idling, air pollution, transport emissions, information, limited attention, reminders, field experiment.

JEL codes: D91, C93, Q53, Q54, Q58, R41.

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1 Introduction

Particulate air pollution is one of the largest threats to human health. Its burden is especially severe in South Asia, home to roughly one-quarter of the global population (UN, 2024) and some of the world’s most polluted air. Pakistan is among the region’s most affected countries: the average resident could live an additional 3.3 years if particulate pollution met the World Health Organization guideline for $PM_{2.5}$ (Greenstone et al., 2025b). Road transport is a major contributor, accounting for 43% of aggregate local air-pollutant emissions in Punjab—the country’s most populous province—and 83% in Lahore—its most polluted city (FAO, 2018; Ilyas and Nissar, 2023).

Vehicle idling—leaving the engine running while stopped—is a margin of vehicle emissions that may depart from the standard economic view in which privately optimal choices are socially inefficient because individuals do not internalise the costs they impose on others. For stops longer than 10 seconds, shutting off and restarting the engine generally saves fuel and reduces emissions relative to continued idling (Gaines and Rask, 2012). Yet in the study data, idling in spells longer than this threshold accounts for nearly one-quarter of total engine-on time across major Pakistani cities, pointing to potentially large private and social gains from reducing idling. Some of this behaviour may reflect genuine private benefits, such as maintaining cabin comfort, but baseline beliefs also reveal substantial misperceptions about the private costs and benefits of idling. Idling may therefore persist not only because drivers fail to internalise its social costs through local air pollutants and greenhouse-gas emissions, but also because they misperceive its private costs (Carrico et al., 2009).

Vehicle idling therefore provides a useful setting to study a broader economic question: when conservation offers both private and social benefits but requires repeated action, what prevents people from acting, and which scalable interventions can change behaviour?

I study this question using an additive treatment design that varies private information, social information, payments, and an in-car reminder technology to distinguish the roles of misperceptions, social motivation, private preferences, and inattention. I partner with a leading telematics firm in Pakistan to conduct a large-scale randomised field experiment with 6,621 private vehicles across the Islamabad Capital Territory and the provinces of Punjab and Sindh—jurisdictions that together account for over three-quarters of Pakistan’s population (PBS, 2023). GPS trackers record ignition status, speed, location, and other vehicle telemetry every 10 seconds, allowing me to measure idling passively rather than through self-reports. I classify a vehicle as idling when its ignition is on and its recorded speed is zero and aggregate the resulting idling spells to the vehicle-day level. I randomly assign vehicles to five treatment arms and a control arm.

During the eight-week intervention, drivers assigned to *Private Info* receive weekly messages correcting two common misperceptions: that modern engines benefit from prolonged warm-up and that shutting off during short stops wastes fuel or increases engine wear. The messages also report each driver’s recent idling and translate potential reductions into personalised fuel savings. *Private & Social Info* adds information about the local health and global climate costs of idling. *Private & Social Info + Pay* additionally rewards drivers for reducing weekly idling below their baseline

benchmark, using a per-minute payment calibrated to the estimated social cost of idling. The reminder arm combines the same information and financial incentive with *IdleGuard*, an in-vehicle device that beeps after 30 seconds of continuous idling and repeats every 30 seconds while idling continues. A matched no-reminder arm receives identical information and payments but no active reminder, isolating the incremental effect of *IdleGuard*. The control arm receives weekly road-safety messages through the same communication channels.

First, baseline misperceptions are widespread. Among drivers, 76% believe that warming up the engine increases engine life, while only 39% believe that warm-up idling harms the environment. Beliefs about when shutting off begins to save fuel or reduce environmental harm are similarly well above the 10-second benchmark. Even using a more conservative 30-second cutoff, 94% of drivers report a higher fuel-saving threshold, with a mean of 6.6 minutes, and 85% report a higher environmental-harm threshold, with a mean of 5.0 minutes.

Second, *Private Info* reduces the primary outcome—daily idling from stops longer than 30 seconds, the threshold used for personalised feedback—by 1.8 minutes per observed vehicle-day relative to the control arm during the eight-week intervention, or 9% of the control post-period mean. It also reduces the daily number of such spells by 0.5, or 5% of the control mean. Private information therefore reduces idling on both the intensive and extensive margins. All remaining treatment arms produce negative point estimates relative to the control arm, but their magnitudes vary. Daily idling falls by 1.3 minutes under *Private & Social Info*, 1.9 minutes under *Private & Social Info + Pay*, and 1.7 minutes under the matched *Private & Social Info + Pay + No-Reminder* arm. The reduction under *Private & Social Info + Pay + Reminder* is smaller, at 0.7 minutes, and statistically indistinguishable from zero.

Third, the incremental effects show little evidence that social information or payments reduce idling beyond the preceding arm. Adding social information increases daily idling from stops longer than 30 seconds by 0.5 minutes relative to *Private Info*, but the estimate is statistically indistinguishable from zero and its confidence interval rules out large additional reductions. Adding payments decreases idling by 0.6 minutes relative to *Private & Social Info*. The estimate is statistically indistinguishable from zero, but its confidence interval does not rule out a meaningful additional reduction. By contrast, activating *IdleGuard* increases daily idling from stops longer than 30 seconds by 1.0 minute relative to the matched no-reminder arm, or 5% of the control post-period mean. It also increases the daily number of such spells by 0.4, or 4% of the control mean. Because this comparison holds information and payments fixed, it isolates the incremental effect of the real-time reminder.

Fourth, both the reduction from *Private Info* and the increase caused by activating *IdleGuard* are concentrated in sustained idling. Approximately 79% of the *Private Info* effect on daily idling from spells longer than 30 seconds comes from spells lasting more than two minutes. The corresponding share of the incremental *IdleGuard* effect is approximately 81%. The reminder therefore does not merely shift behaviour around its first beep at 30 seconds; its adverse effect arises primarily in long idling spells.

Fifth, the mechanism evidence points to private misperceptions as an important constraint. *Private Info* moves both targeted endline beliefs in the intended direction and produces larger idling reductions among drivers whose baseline beliefs leave more scope for correction, although the differences across belief groups are imprecisely estimated. The weak average effect of social information also masks informative heterogeneity. Adding social information tends to strengthen the private-information effect among less-mistaken drivers but weaken it among those with stronger fuel-saving or environmental misperceptions. These patterns are consistent with broader social information diluting a targeted private-cost correction or conflicting with strongly held prior beliefs. Endline credibility responses provide additional, though suggestive, support for the latter interpretation: social information is perceived as more credible among drivers with less contrary environmental priors, but not among those with more contrary priors. Payments likewise produce little average additional reduction, suggesting limited residual price-responsive idling at the implemented incentive level.

Sixth, the evidence suggests that *IdleGuard* induces a two-stage behavioural response. Drivers appear to substitute away from preventive self-monitoring when they expect to rely on the device, allowing more idling spells to reach its 30-second reminder threshold. Activating *IdleGuard* increases the daily number of such spells relative to the matched no-reminder arm, with approximately 87% of the increase arising in *traffic-like* idling. After the reminder sounds, drivers also fail to correct promptly: only about 19% of the resulting increase in idling duration comes from the first 30 seconds of threshold-crossing spells, while 56% comes from idling beyond two minutes. Endline reports provide complementary evidence for both stages: some drivers report becoming more comfortable leaving the engine running because the beep would remind them, while many report not usually correcting promptly after it sounded. This delayed corrective response is consistent with procrastination when switching off entails an immediate action cost but the benefits accrue subsequently (O’Donoghue and Rabin, 1999).

This paper builds on several literatures and makes three main contributions. First, it contributes to the economics of air pollution by studying an important but comparatively neglected source: emissions generated by everyday vehicle use. Transport accounts for a substantial share of urban air pollution, yet field evidence on the behavioural choices that determine vehicle emissions remains limited (Jacobsen et al., 2023). Much of the environmental and development literature instead studies pollution from industrial production, household energy use, agricultural burning, or the regulation and monitoring of large emitters (Duflo et al., 2013, 2018; Greenstone et al., 2025a; Hanna et al., 2016; Berkouwer and Dean, 2022; Jack et al., 2025). Research on vehicles has largely examined policies targeting externalities—including emissions standards, driving restrictions, and congestion pricing (Jacobsen et al., 2023; Davis, 2008; Kreindler, 2024; Cook et al., 2025)—with a smaller literature studying drivers’ own fuel-conserving choices through speed and acceleration (Langer and McRae, 2014; Knittel and Tanaka, 2021). I study idling, a distinct and highly repeated margin of vehicle use for which private and social incentives point in the same direction: switching off saves fuel while reducing local and global emissions. To the best of my knowledge, this is the

first large-scale field experiment to identify why drivers persist in idling and to test interventions designed to reduce it.

Second, the paper contributes to the behavioural public economics of corrective energy and environmental policy, which asks how policy should respond when privately chosen energy use reflects both misoptimisation and unpriced social costs (Allcott et al., 2014; Allcott and Taubinsky, 2015; Allcott and Sunstein, 2015; Heutel, 2015; Tsvetanov and Segerson, 2013; Allcott et al., 2015; Allcott and Mullainathan, 2010). One influential empirical approach uses information to distinguish misperceptions from residual preferences and thereby assess the scope for corrective policy (Allcott and Taubinsky, 2015). Existing information experiments have largely studied durable purchases—including lighting, vehicles, water heaters, and air conditioners—and generally find limited effects on actual purchases (Allcott and Taubinsky, 2015; Rodemeier and Löschel, 2025; Allcott and Knittel, 2019; Allcott and Sweeney, 2017; Newell and Siikamäki, 2014; Davis and Metcalf, 2016). I bring this logic to repeated, realised use of a durable that drivers already own. Personalised information about the private fuel savings from reducing each driver’s own idling produces persistent changes in behaviour, showing that misperceptions can matter even after the durable-purchase decision has been made.

The additive design then asks what explains the idling that remains after private-cost misperceptions are targeted. Adding social information tests whether making external costs salient produces further conservation; adding payments tests whether residual idling responds to stronger private incentives; and activating a real-time reminder tests whether inattention at the moment of choice remains important. Social information and payments add little on average, while the reminder increases idling relative to an otherwise identical no-reminder arm. The experiment therefore uses interventions mapped to distinct explanations for behaviour to diagnose why privately and socially costly actions persist and to inform corrective policy.

Third, the paper contributes to the economics of attention and reminders by showing that externally supplied attention can substitute for, rather than simply augment, people’s own monitoring. Existing work studies how salience, reminders, planning prompts, and other attention aids affect behaviour, while related work in psychology shows that people may strategically offload prospective memory onto external cues (Chetty et al., 2009; Karlan et al., 2016; Carrera et al., 2018; Bronchetti et al., 2023; Damgaard and Gravert, 2018; Koch et al., 2024; Gilbert, 2015). I provide causal evidence from a behaviour-triggered reminder whose timing allows preventive and corrective responses to be distinguished. Activating *IdleGuard* makes idling spells more likely to reach the 30-second reminder threshold, consistent with drivers relying on the device in place of their own preventive monitoring. Yet the reminder does not induce prompt correction once it sounds: most of the resulting increase in idling occurs well after the first beep and is concentrated in sustained spells. The evidence therefore reveals a two-stage response in which external monitoring weakens preventive self-monitoring and delayed action limits subsequent correction, consistent with procrastination when responding entails an immediate action cost (O’Donoghue and Rabin, 1999).

The rest of the paper is organised as follows. Section 2 describes the setting and experimental

design; Section 3 presents the data and measurement; Section 4 develops the empirical framework; Section 5 reports the experimental results; Section 6 examines mechanisms; Section 7 discusses welfare and policy implications; and Section 8 concludes.

2 Setting and Experimental Design

2.1 Vehicle Idling in Pakistan

Pakistan has some of the most polluted air in the world. The average resident could live an additional 3.3 years if particulate pollution were reduced to the World Health Organization guideline (Greenstone et al., 2025b). Recent emissions-inventory studies identify road transport as the largest source of aggregate air-pollutant emissions, accounting for an estimated 43% of the total in Punjab and 83% in Lahore (FAO, 2018; Ilyas and Nissar, 2023). The Urban Unit of the Government of Punjab therefore recommends that “abatement measures must be focused primarily on the transport sector” and that “public awareness of behavioral modification is very crucial in this aspect” (Ilyas and Nissar, 2023). This paper studies one such behavioural margin: vehicle idling—the practice of leaving the engine running while the vehicle is stationary.

Vehicle idling can be broadly divided into three exhaustive categories: (1) *warming*, at cold start; (2) *waiting*, such as during passenger pickup or at a drive-through; and (3) *traffic*, such as at stoplights or in congestion (Carrico et al., 2009). Policy guidance notes that modern engines do not require prolonged warming and that, for stops lasting more than 10 seconds, restarting the engine generally saves fuel, reduces emissions, and limits engine wear (U.S. EPA, 2025; U.S. DOE, 2026; Gaines and Rask, 2012). In the study’s baseline data, the median vehicle idles for approximately 20 minutes per observed day, and nearly 25% of engine-on time across major Pakistani cities is spent in avoidable idling. These magnitudes highlight the potential private and social gains from anti-idling interventions.

2.2 Partner and Study Sample

I partnered with a leading vehicle-tracking company in Pakistan to design and test anti-idling interventions using GPS tracker data. The trackers can be configured to record vehicle activity at an unusually fine 10-second resolution, allowing idling episodes and their duration to be measured passively and precisely. GPS trackers are widespread among cars in Pakistan because banks and other financial institutions commonly require them for financed vehicles as a means of protecting the underlying asset. Vehicle owners may also install them independently for theft protection, vehicle recovery, and general monitoring (TPL Trakker, 2024). Bank financing has accounted for a substantial share of new-car sales in Pakistan—estimated at approximately 40–45% during the recent expansion in automobile lending—so tracker-equipped vehicles span a broad segment of the ordinary passenger-car market rather than a narrow specialist population (Khan, 2021).

I applied four filters to the partner’s universe of 72,076 active subscriptions to define the study’s eligible sample. The first restricted the sample to *passenger four-wheelers*: I excluded motorcycles, rickshaws, tractors, trucks, and other commercial vehicle types, retaining light passenger vehicles such as sedans, hatchbacks, and SUVs. This kept the study focused on ordinary passenger cars, the vehicle class to which standard anti-idling guidance—including the 10-second shutoff rule—most directly applies. The second restricted the sample to *owner-driven vehicles*: I excluded cars linked to companies because drivers who do not own the vehicle may have weaker incentives to reduce fuel and maintenance costs. The third imposed a *tracking-history requirement*: I retained vehicles with sufficient subscription time remaining to stay observable through the study period, while also retaining vehicles with at least 18 months of prior tracking history, whose continued relationship with the partner suggested a greater likelihood of subscription renewal. The fourth imposed a *geographic restriction*: I used the tracker installation city as a proxy for where the vehicle was typically driven and retained vehicles in the Islamabad Capital Territory and the provinces of Punjab and Sindh. This yielded a sufficiently large recruitment pool while maintaining a focus on the country’s most populous and heavily polluted regions. The four filters yielded 42,101 eligible vehicles. Excluding the 1,400 vehicles that had participated in an earlier pilot left a recruitment frame of 40,701 vehicles, which I randomly ordered for the survey team.

I established a survey unit at the firm’s in-house call centre to recruit participants by phone. Starting from the randomly ordered frame, the unit proceeded sequentially through the list towards a powered target of 6,000 drivers who had consented and completed the baseline survey. This approach avoided drawing a fixed sample upfront that could fall below the target because of non-contact, non-consent, or incomplete interviews.

After obtaining consent to participate in a research study on driving behaviour conducted jointly by the firm and researchers at Tufts University, enumerators informed participants that they could be recontacted to receive further study information, report weekly fuel use, and complete an endline survey. The baseline survey collected demographic and contact information, including the WhatsApp number and email address used for treatment delivery, as well as measures of driving and idling beliefs and social and environmental preferences. All participants received a small participation payment, while the weekly fuel survey and an incentivised social-preference measure carried additional payments; these payments were made at the end of the study.

Because treatment was delivered to an individual driver while GPS outcomes were measured for the vehicle, enumerators were trained to identify and enrol the person who drove the vehicle most often. The baseline survey therefore recorded the main driver’s contact information, aligning treatment delivery as closely as possible with the driver responsible for most of the vehicle’s observed driving behaviour.

Most vehicles in the 40,701-vehicle recruitment frame initially reported GPS data at intervals longer than 10 seconds. In parallel with recruitment, I worked with the partner’s diagnostics team to remotely reconfigure the trackers of newly recruited vehicles to report at 10-second intervals using over-the-air technology. Because reconfiguration could be delayed by temporary connectivity

constraints—for example, when a vehicle was in an area with weak network coverage and could not immediately receive the update—recruitment continued beyond the powered target rather than waiting for every enrolled tracker to be reconfigured. Recruitment ultimately yielded 10,491 baseline records. After applying the remaining respondent and GPS eligibility restrictions, the randomisation sample comprised 6,621 vehicles with complete baseline data and 10-second GPS reporting.

2.3 Experimental Treatments

The design includes five treatment arms alongside a control arm. Four treatment arms form an additive sequence in which each successive arm adds one component. Comparing the first three arms in this sequence identifies the incremental effects of social information and payments. The fifth treatment arm is a no-reminder counterpart to the fourth. Before assignment to these two arms, drivers completed the same incentivised demand elicitation for a new tracker that would remind them to switch off the engine after the vehicle had remained stopped with the engine running for some time. The no-reminder arm therefore provides a matched comparison that isolates the incremental effect of activating the reminder while holding fixed any effect of this prior elicitation. Together, these comparisons allow me to distinguish among misperceptions, social motivation, preferences, and attention as candidate determinants of idling. I also estimate treatment effects for each treatment arm relative to the control arm.

Across the six arms, participants received weekly messages through the partner’s client-facing mobile app, WhatsApp Business account, and official email platform. Messages were sent on Mondays for eight weeks beginning in early April 2026. The content of these messages varied across arms. Table 1 summarises the treatment components included in each arm. I describe each arm in detail below.

Table 1: Experimental arms and treatment components

Arm	Treatment component			
	Private Info	Social Info	Pay	Reminder
<i>Private Info</i>	✓			
<i>Private & Social Info</i>	✓	✓		
<i>Private & Social Info + Pay</i>	✓	✓	✓	
<i>Private & Social Info + Pay + Reminder</i>	✓	✓	✓	✓
<i>Private & Social Info + Pay + No-Reminder</i>	✓	✓	✓	
<i>Control</i>				

Notes: Check marks indicate the components included in each arm. Comparing the first three arms in the additive sequence identifies the incremental effects of social information and payments. Comparing the *Reminder* and *No-Reminder* arms identifies the incremental effect of the in-vehicle reminder. The control arm received general road-safety messages.

Private Info. This treatment arm received only anti-idling information designed to address the three main forms of idling: *warming*, *waiting*, and *traffic*. The message informed drivers that modern

engines do not require prolonged warm-up, so they can begin driving sooner to save fuel and reduce engine wear, and that turning off the engine during stops longer than 10 seconds generally saves fuel and limits engine wear (U.S. EPA, 2025; U.S. DOE, 2026; Gaines and Rask, 2012). It also reported each driver’s typical weekly idling at stops longer than 30 seconds and translated a one-minute reduction into personalised estimates of per-minute, weekly, and monthly fuel savings. The same message was repeated each week.

Private & Social Info. Drivers in this treatment arm received the same information as those in *Private Info*, augmented with information about the social costs of idling. The message explained that reducing *warming, waiting, and traffic* idling lowers local air pollution and CO₂ emissions. As with fuel savings, it translated the climate damages avoided by a one-minute reduction in idling into a personalised rupee value and reported the corresponding weekly and monthly totals alongside the private fuel savings (Carleton and Greenstone, 2021; Rennert et al., 2022). The message also described the health harms from PM_{2.5} (Greenstone et al., 2025b) and included photographs of a smog-covered road in Lahore and a man stranded with his dog during a flood. Captions linked tailpipe PM_{2.5} emissions to shorter lives and tailpipe CO₂ emissions to flooding, making the environmental consequences of idling more salient.

Private & Social Info + Pay. Drivers in this treatment arm received the same content as those in *Private & Social Info*, supplemented with a financial incentive. I used the treatment arm’s average weekly idling at stops longer than 30 seconds and the associated estimated climate damages (Carleton and Greenstone, 2021; Rennert et al., 2022) to calibrate a fixed Pigouvian-style per-minute payment rate (Pigou, 1920; Baumol and Oates, 1971). Drivers were told that they would receive this amount for each minute by which they reduced their weekly idling below their individual baseline benchmark. Each weekly message reported the fixed per-minute payment rate and the maximum payment available. Payments earned in each week were processed at the start of the following week and reported in that week’s message. Because payments were made in arrears, drivers received a final payment notification on the Monday after the eight-week treatment period ended. At the fixed payment rate of Rs. 4 per minute, the average maximum weekly payment across the three treatment arms with a payment component was Rs. 538. This was equivalent to 5.6% of estimated average weekly earnings for salaried participants in those arms, using official age-specific wages reweighted to match their observed age distribution (Pakistan Bureau of Statistics, 2025).

Private & Social Info + Pay + Reminder. Drivers in this treatment arm received the same information and financial incentive as those in *Private & Social Info + Pay* and were offered *IdleGuard*, a GPS tracker developed for this study to deliver a real-time anti-idling reminder. When the vehicle remained stationary with the engine running for 30 seconds, *IdleGuard* emitted a beep and repeated the alert every 30 seconds while idling continued. Drivers were informed that the device would remind them to switch off the engine during stops longer than 30 seconds. The

reminder was active throughout the eight-week treatment period.

Private & Social Info + Pay + No-Reminder. Drivers in this treatment arm received the same information and financial incentive as those in *Private & Social Info + Pay + Reminder*, but did not receive an *IdleGuard* reminder. Comparing the two arms therefore isolates the incremental effect of the real-time reminder.

Control. The control arm received general road-safety messages throughout the treatment period, such as reminders to wear a seatbelt or use indicators before changing lanes.

2.4 Randomisation

I randomised the 6,621-vehicle randomisation sample in two stages to construct the six experimental arms.

The first stage took place immediately after baseline and assigned vehicles to five groups: (1) *Private Info*, with 1,030 vehicles; (2) *Private & Social Info*, with 1,030 vehicles; (3) *Private & Social Info + Pay*, with 1,030 vehicles; (4) *Private & Social Info + Pay + Tech*, with 2,500 vehicles; and (5) *Control*, with 1,031 vehicles. I stratified the first-stage assignment by vehicle age (0–4 versus 5+ years), which is closely related to vehicle emissions (Jacobsen et al., 2023), and certainty about main-driver status (certain versus uncertain). Main-driver status was uncertain for 621 baseline respondents.

The second stage was conducted through treatment-briefing phone calls, which had been described during the baseline survey as possible follow-up calls providing additional information about the study. These calls began shortly after the first-stage randomisation and were conducted separately for each of the five first-stage groups. During calls to the 2,500 drivers assigned to the *Private & Social Info + Pay + Tech* group, participants completed an incentivised demand elicitation; after a randomly selected subset had their choices implemented, the remaining drivers entered a lottery assigning them to the *Private & Social Info + Pay + Reminder* or *Private & Social Info + Pay + No-Reminder* arm.

The procedure had two steps. First, drivers made an incentivised choice between a new tracker device provided at no cost and Rs. 10,000, using an elicitation approach in the spirit of the Becker–DeGroot–Marschak mechanism (Becker et al., 1964). The new tracker was described as identical to the respondent’s existing tracker, except that it would emit a gentle beep after the vehicle had remained stopped with the engine running for some time, reminding the driver to switch off the engine. Drivers were also asked how much they believed the device could reduce their weekly fuel expenditure and whether they would use it if provided free of charge.

Care was taken to minimise any pre-treatment behavioural response to the description of the device. In particular, enumerators did not disclose the precise interval after which the device would beep and did not use the terms “idling” or *IdleGuard*. To support incentive-compatible elicitation,

enumerators received rigorous training, and call recordings were back-checked daily, following implementation practices used in related field work (Berkouwer and Dean, 2022). Respondents were instructed clearly, and twice, to state their true preference. They were informed that they had a small probability of receiving their chosen option and that stating a particular preference could not affect whether their choice was implemented, since implementation was determined by a random draw. Before the choice was locked in, each driver was told that it could not subsequently be changed and was asked whether they wished to retain that choice, encouraging careful consideration. The stated choice was implemented for a randomly selected 1% of drivers, corresponding to 25 participants, who were then removed from the experiment (Allcott et al., 2020).

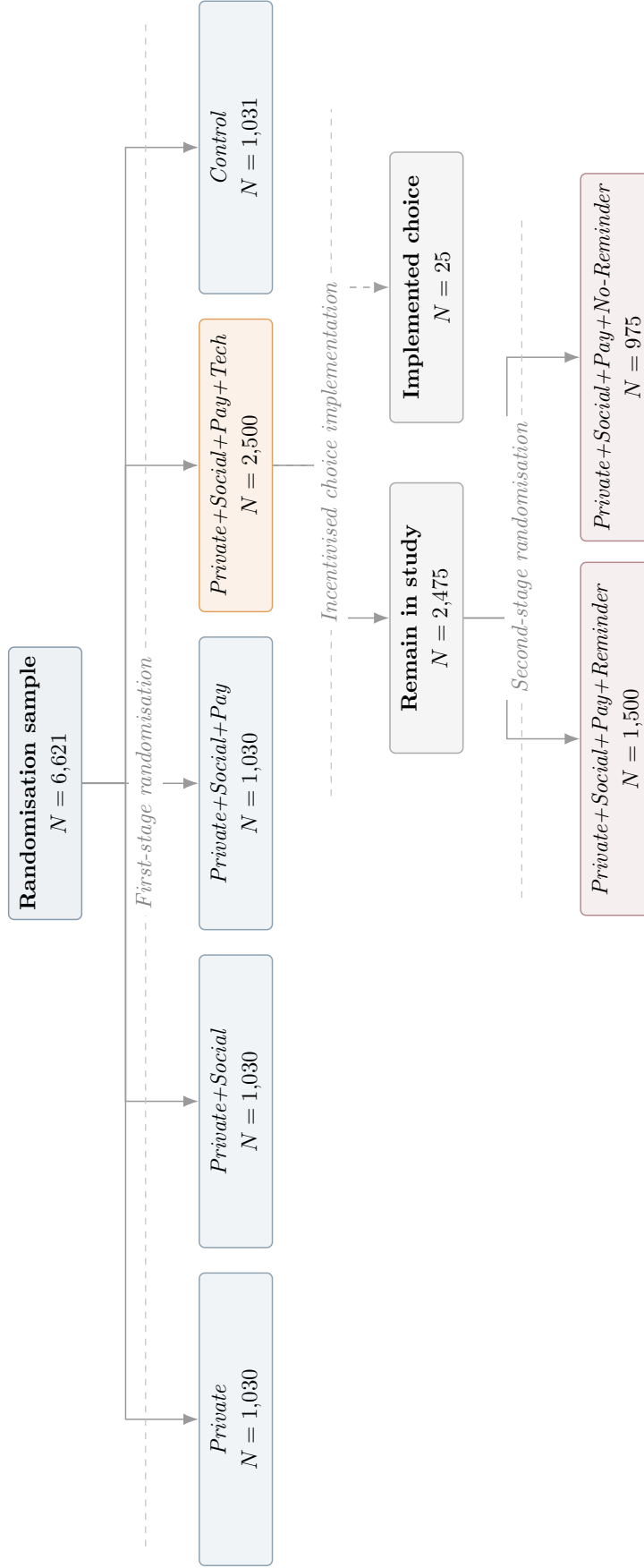
Second, the remaining 2,475 drivers were told that a computerised lottery conducted during the call would determine whether they were assigned to receive the device. The lottery assigned 1,500 drivers to the *Private & Social Info + Pay + Reminder* arm and 975 drivers to the *Private & Social Info + Pay + No-Reminder* arm. Drivers assigned to the reminder arm were informed that a technician would contact them to arrange installation. Drivers assigned to the no-reminder arm were told that they would continue participating in the study as planned.

2.5 Implementation

Treatment briefing. Following the first-stage randomisation in early November 2025, the survey unit began conducting separate treatment-briefing calls for each of the five first-stage groups: *Private Info*, *Private & Social Info*, *Private & Social Info + Pay*, *Private & Social Info + Pay + Tech*, and *Control*. The calls served three purposes: (1) to provide the follow-up study information that participants had been told at baseline they might receive and to prepare them for the subsequent treatment-delivery phase; (2) to collect bank-account information from drivers assigned to first-stage groups with a payment component, allowing any anti-idling payments earned during treatment to be processed; and (3) within the *Private & Social Info + Pay + Tech* group, to conduct the demand elicitation and second-stage randomisation described above and to collect the contact and location information required for *IdleGuard* installation. To allow sufficient time for installation, drivers in the *Private & Social Info + Pay + Tech* group were prioritised for briefing, alongside a randomly selected 20% of drivers from each of the remaining first-stage groups. This sequencing allowed installations to begin while briefing was still under way. Treatment-briefing calls continued through mid-January 2026.

Drivers in the *Private Info* and *Private & Social Info* groups were told that, later in the study, they would begin receiving weekly messages about driving behaviour, including tips intended to improve their driving habits. They were informed that the messages would be sent every Monday for eight weeks through the partner’s mobile app, WhatsApp, and email, and were asked to look out for them. Drivers in the *Control* group were told that they would receive general road-safety tips. To avoid any premature treatment response, idling was not mentioned at any stage of these calls. Indeed, the survey team itself had not been informed that the study focused on idling.

Figure 1: Two-stage randomisation and construction of the final experimental arms



Notes: The first stage assigned the 6,621-vehicle randomisation sample to five groups, stratified by vehicle age and certainty about main-driver status. Drivers in the *Private + Social + Pay + Tech* group then chose between a new tracker device and Rs. 10,000. The incentivised choice was implemented for 25 randomly selected drivers, who were removed from the experiment; the remaining 2,475 were randomised between the *Private + Social + Pay + Reminder* and *Private + Social + Pay + No-Reminder* arms.

Drivers assigned to receive a payment component—those in *Private & Social Info + Pay* and, following the second-stage randomisation, those in *Private & Social Info + Pay + Reminder* and *Private & Social Info + Pay + No-Reminder*—were also informed that they would have an opportunity to earn money based on their driving behaviour. Idling was again not mentioned during these calls; the subsequent treatment messages instead explained how payments could be earned by reducing idling. Drivers were asked to provide bank-account or mobile-wallet details for receiving any payments earned. Those assigned to the *Private & Social Info + Pay + Reminder* arm were additionally asked to provide an address and contact details for the installation visit, as well as the city in which the vehicle was primarily driven. I used this information to coordinate *IdleGuard* installations in batches with the partner’s nationwide field-force team as briefing calls progressed.

***IdleGuard* rollout.** Of the 1,500 drivers assigned to the *Private & Social Info + Pay + Reminder* arm, 1,129 (75%) accepted the *IdleGuard* offer. By August 2025, I had worked with the partner’s diagnostics and laboratory teams to import and locally procure the required components and to develop, test in the laboratory, and field-test sufficient *IdleGuard* devices for the rollout. As briefing calls progressed, devices were dispatched in staggered batches from the partner’s Karachi head office to regional field offices, with each shipment matched to the assigned drivers’ reported installation cities and the technicians responsible for serving those locations. The rollout was coordinated through 30 regional dispatch locations and involved 80 field technicians serving drivers across 164 cities and towns. Field technicians received a per-installation incentive. In total, 678 devices were installed by the end of January 2026, corresponding to 60% of drivers who had accepted the offer.

Each installation took approximately 45–60 minutes. The technician first tested that the device was functioning correctly, including that it emitted the intended beep. A diagnostics lead at the head office then remotely disabled the beep using over-the-air technology while remaining on a call with the technician on site, allowing the lead to hear and confirm that the device had fallen silent. Drivers were therefore left with a fully functioning tracker whose reminder feature had been disabled and were informed that the beep would be activated only at a later launch date.

Treatment delivery. Although treatment briefing and *IdleGuard* installations had been completed by the end of January 2026, I delayed treatment delivery until after Ramadan because driving patterns could differ during the month. The eight-week intervention therefore began in early April 2026, simultaneously across all six experimental arms, and continued through the end of May. Of the 6,596 vehicles in the final six-arm experimental sample, 35 drivers had requested during treatment briefing not to receive further messages, leaving a treatment-delivery sample of 6,561 vehicles. To study the persistence of treatment effects and distinguish learning from continued message salience, I randomly assigned half of the *Private Info* arm to receive treatment for two additional weeks after the scheduled eight-week intervention, while treatment ended as scheduled for the remaining half.

Weekly treatment delivery involved three main activities. First, treatment messages were sent every Monday at 9:00 a.m. Pakistan Standard Time through the partner’s client-facing mobile app,

WhatsApp Business account, and official email platform. Second, beginning in the second treatment week, anti-idling payments earned during the preceding week were calculated and processed each Monday. Third, at treatment launch, the reminder feature of installed *IdleGuard* devices was activated remotely using over-the-air technology.

Each Sunday, I used GPS data from the preceding Monday–Saturday period to construct the driver-level statistics required to populate the message templates shown in Online Appendix Section A.2. I worked with the partner’s information technology team to develop an automated system for transferring these data, populating the treatment templates, and delivering the resulting messages. WhatsApp and email messages were sent directly to enrolled main drivers using contact details collected at baseline and reconfirmed during briefing. Because the partner could not reliably identify the main driver’s app user ID when multiple user IDs were linked to the same vehicle, app messages were sent to all user IDs associated with the randomised vehicle. In each of the eight common-treatment weeks, the system sent approximately 22,500 messages across the three channels and recorded message-level delivery and engagement information for each platform.

Sending app messages to multiple user IDs associated with the same vehicle was consistent with vehicle-level treatment assignment and outcome measurement. Contamination could arise, however, if a single user ID were linked to study vehicles assigned to different experimental arms. Such overlap was rare: in the baseline data, only 0.6% of app user IDs were linked to more than one study vehicle.

Using the account details collected during treatment briefing, I prepared weekly payment files alongside the treatment-message files and sent them to the partner’s finance team for processing each Monday. I reconciled the resulting payment-status reports and resubmitted failed transactions for processing. Across the eight treatment weeks, drivers in the three treatment arms with a payment component earned an average of Rs. 78 per week, or 0.8% of the same estimated weekly earnings benchmark. The partner successfully processed 8,106 weekly payment transactions over the intervention, totalling approximately Rs. 2.05 million (USD 7,400).

To activate the *IdleGuard* reminder remotely, I shared the tracker details for all 678 successful installations with the partner’s diagnostics team on the first day of treatment. The reminders had been activated on almost all devices by the following day; a small number came online later because temporary network problems delayed receipt of the remote activation command. Drivers could contact the partner’s call centre by phone, email, or social media. During the eight-week treatment period, only 39 drivers requested that the reminder be deactivated because they found the beep annoying, corresponding to 6% of successful installations.

Fuel survey and endline. During the first seven treatment weeks, I administered an incentivised weekly fuel-expenditure survey through a WhatsApp chatbot using the WATI API. At 12:00 p.m. Pakistan Standard Time, the chatbot asked drivers to report their fuel expenditure for the previous day. Drivers who had not responded received a reminder at 8:00 p.m. the same day. Before launch, I randomly assigned the 6,596 drivers in the final six-arm experimental sample across Tuesday

through Sunday, stratified by experimental arm. I re-randomised survey days each week so that reported expenditure would reflect a range of driving days rather than a fixed day of the week.

Following the intervention, I worked with the survey unit to conduct a phone endline survey beginning in mid-June 2026 and continuing through August 2026. The survey repeated selected baseline measures of beliefs and self-reported behaviour and collected new measures of message recall and engagement, perceived treatment credibility and usefulness, idling contexts, intentions to reduce future idling, and treatment-specific experiences with payments and *IdleGuard*. It also included incentivised choices regarding whether drivers wished to continue receiving the weekly messages.

2.6 Balance and Pre-Treatment Comparability

The five first-stage assignment groups are balanced across baseline survey and administrative characteristics. Because the second-stage randomisation created the final six experimental arms, I also assess their comparability in GPS observability and idling immediately before treatment launch. The estimated differences are generally small, and joint tests do not reject balance or pre-treatment comparability at conventional significance levels. See Online Appendix Tables [C.1](#) and [C.2](#).

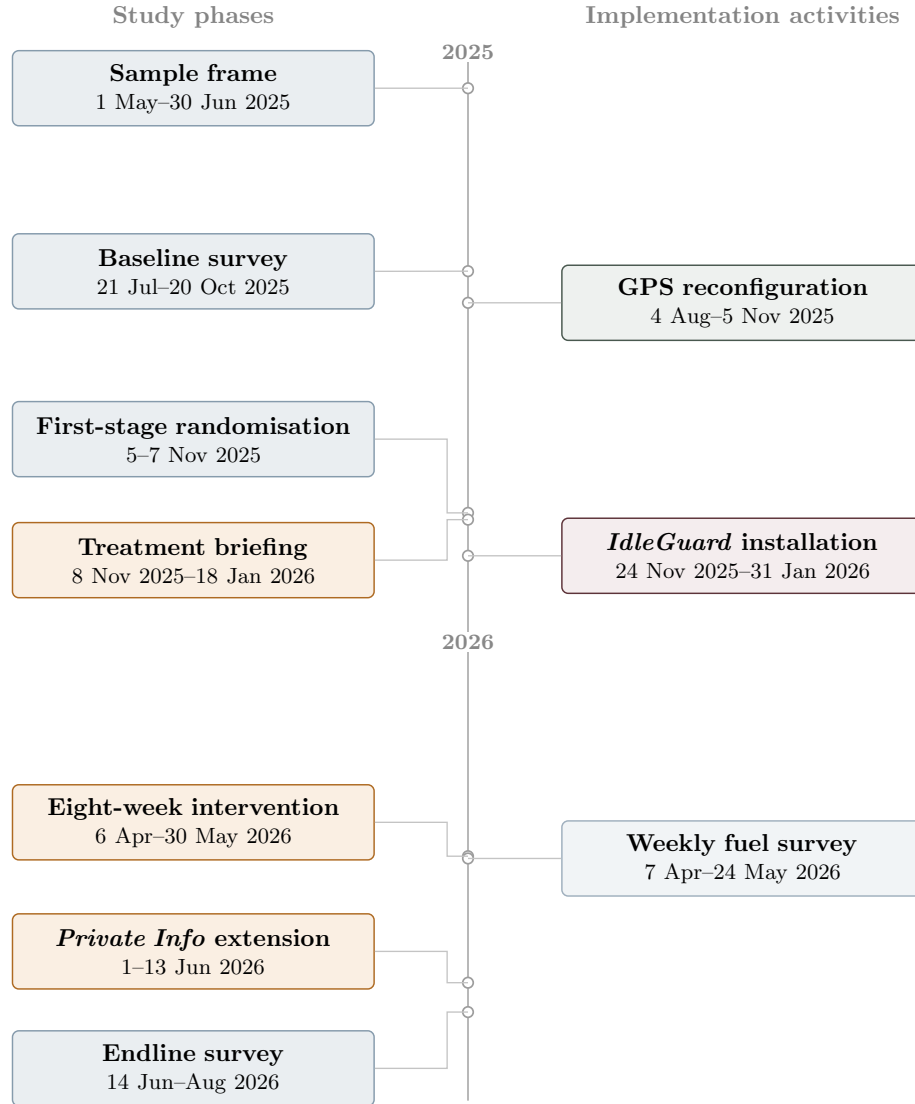
3 Data and Measurement

3.1 GPS Tracker Data

Each of the 6,621 vehicles in the randomisation sample was equipped with a GPS tracker installed beneath the dashboard and powered by the vehicle’s battery, allowing it to transmit data to the partner’s servers even when the engine was off. The tracker’s GPS receiver used signals from orbiting satellites to determine the vehicle’s location ([GPS.gov](#), [n.d.](#)), while an integrated cellular modem and subscriber identity module (SIM) card transmitted the resulting records to the partner’s servers over the mobile network. Each row, or ping, in the raw GPS tracker data contains a timestamp, latitude, longitude, speed, ignition status, battery voltage, and other tracker variables.

The partner’s trackers ordinarily reported data at a coarser frequency while vehicles were idling because its operational requirements placed greater emphasis on high-frequency reporting during movement. As baseline data collection progressed, I worked with the partner’s diagnostics team to reconfigure the trackers installed in recruited vehicles to report at 10-second intervals while idling. A tracker was considered successfully reconfigured if, during the post-reconfiguration diagnostic period, at least 95% of its idling pings were no more than 15 seconds apart. The resulting data provide approximately 10-second-resolution records for observed vehicles from Monday through Saturday. Sundays were excluded from the weekly data extracts to allow sufficient time to process the preceding week’s data and prepare treatment-delivery files for the following Monday. The GPS data cover the period from 17 November 2025 through 22 August 2026 and span 81 tracker-installation cities

Figure 2: Experimental timeline and implementation activities



Notes: The central spine uses a linear calendar scale based on the start of each phase or activity; boxes are displaced only to avoid overlap. GPS trackers were reconfigured to report at 10-second intervals before randomisation, and *IdleGuard* installation was completed before treatment launch. Half of the *Private Info* arm was randomly assigned to receive messages for two additional weeks. The incentivised fuel survey was fielded through a WhatsApp chatbot during the first seven treatment weeks.

across Pakistan.

3.2 Measuring Idling

Ping-level idling classification. I classify a ping as idling when the vehicle’s ignition status is on and its recorded speed is zero. Online Appendix [B.1](#) validates ignition status as a strong proxy for whether the engine is running and documents the GPS-displacement correction used to remove false positives from slow movement.

Idling-spell construction. Within each vehicle-day and trip, I group consecutive idling pings into spells and aggregate their duration, excluding intervals spanning telemetry blackouts. Online Appendix B.2 provides complete construction details.

Vehicle-day outcome construction. I aggregate idling spells within each vehicle-day to construct daily idling duration and spell counts. The primary outcome is daily idling duration from spells longer than 30 seconds. The treatment informed drivers that shutting off the engine during stops longer than 10 seconds generally saves fuel and reduces emissions, while personalised feedback and payments were based on idling spells longer than 30 seconds (U.S. EPA, 2025; U.S. DOE, 2026; Gaines and Rask, 2012). I also construct outcomes using the 10-second threshold and total idling without a minimum spell-duration threshold. I screen implausible vehicle-days and manually audit extreme observations; results are unchanged under conservative winsorisation. Online Appendix B.3 provides complete construction and validation details.

3.3 Analysis Sample and Outcomes

Of the 6,596 vehicles in the final six-arm randomised sample, 6,451 appear at least once in the cleaned GPS tracker data over the full panel period, from 17 November 2025 to 22 August 2026. These vehicles constitute the GPS analysis panel. Online Appendix Figure A.1 summarises the construction of the experimental sample and GPS analysis panel. The unit of analysis is the vehicle-day, organised on a balanced vehicle-by-date grid. Because trackers report whether vehicles are parked, moving, or idling, behavioural outcomes are coded as zero when the corresponding activity does not occur. They remain missing only during temporary periods of tracker non-reporting caused by connectivity problems.

The primary outcome is daily idling duration from spells longer than 30 seconds. Secondary outcomes include total daily idling duration, daily idling duration from spells longer than 10 seconds, daily counts of spells longer than 10 and 30 seconds, daily moving minutes, and trip counts. I also examine effects across alternative spell-duration thresholds and bins and decompose idling into *warming-like*, *traffic-like*, and *waiting-like* contexts using the telemetry data. The first idling spell after ignition-on is classified as *warming-like* if the vehicle has not yet moved. A non-warming spell is classified as *traffic-like* if the vehicle moved within 120 seconds before the spell began and moves again within 300 seconds after its start. All remaining spells are classified as *waiting-like*.

3.4 Survey and Fuel-Expenditure Data

Baseline survey. The baseline survey collected demographic information on the vehicle’s main driver, beliefs about idling and fuel-saving and environmental-harm thresholds, confidence in those beliefs, and stated and revealed measures of pro-social preferences, the latter elicited through an incentivised choice. I use these measures to assess statistical balance across assignment groups, characterise baseline misperceptions, and examine heterogeneity in treatment effects.

Weekly fuel-expenditure survey. The weekly fuel-expenditure survey asked the vehicle’s main driver to report fuel expenditure on the previous day, with respondents instructed to report zero if none was incurred. I use this measure to estimate treatment effects on daily fuel expenditure. The survey-based measure provides a self-reported complement to engineering estimates of fuel use and emissions constructed from GPS inputs.

Endline survey. The endline survey re-elicited idling beliefs, fuel-saving and environmental-harm thresholds, and confidence in those beliefs to study belief updating. It also repeated the baseline stated measure of pro-social preferences to examine whether stated preferences changed over the study. The survey collected measures of message engagement, comprehension, perceived credibility, and incentivised revealed demand for continued treatment messages to study mechanisms. Finally, treatment-specific modules collected information to help interpret the incremental effects of social information and *IdleGuard*.

3.5 Data Quality and Attrition

Post-launch GPS data quality and observability. Behavioural outcomes remain missing during periods of tracker non-reporting, raising the possibility that estimated treatment effects partly reflect differential selection into the observed GPS sample. Following the GPS data-quality attrition check in Kreindler (2024), Online Appendix Table C.6 tests for treatment-arm differences in tracker reporting and GPS observability after treatment launch. The estimated differences are small and statistically insignificant, and the joint tests provide no evidence of differential post-launch data availability across study arms.

Survey attrition. Online Appendix Table C.7 tests for differential attrition in the weekly fuel-expenditure survey. Drivers in the three treatment arms with payments were more likely than those in the control arm to provide valid fuel-expenditure outcomes. I therefore do not report treatment effects on fuel expenditure for these arms relative to the control arm or the incremental effect of payments. Online Appendix Table C.8 finds no evidence of differential attrition for *Private Info* relative to the control arm or for the incremental effect of *IdleGuard*.

4 Empirical Framework

4.1 Treatment Logic and Predictions

Why do drivers privately choose to idle? Does idling reflect a mistake, a genuine preference, or inattention in the moment? And does concern for social harms also matter? The study uses a sequential additive treatment design to diagnose which of these candidate determinants are behaviourally important.

At a sufficiently long stop, the driver chooses between continuing to idle and switching off the engine to restart later. The driver chooses to idle if the utility from idling is at least as high as the utility from restarting:

$$U(\text{Idle}) \geq U(\text{Restart}).$$

The utility difference between the two actions can be written as

$$U(\text{Idle}) - U(\text{Restart}) \equiv B^p - C^p + D - \alpha C^s. \quad (1)$$

B^p denotes the genuine private benefit of idling relative to restarting. I normalise the private benefit of restarting to zero. The private benefits of idling include comfort and convenience, such as avoiding the hassle of switching off and restarting. Similarly, C^p denotes the private fuel and maintenance costs of idling relative to those of restarting. D captures behavioural distortions, including mistakes and inattention. Mistakes arise from failures of private-cost information, including beliefs that warming up benefits engine life and overestimation of the stop duration at which the fuel and maintenance costs of continued idling exceed those of restarting. Inattention means that, at the moment of choice, the driver fails to notice the relevant stop, track its duration, consider switching off, or act on that option. C^s denotes the social costs of idling relative to restarting, including damages from local air pollutants and greenhouse-gas emissions. Finally, α measures the weight the driver places on these social costs.

Private-cost information on warming requirements and on the fuel- and maintenance-saving thresholds for waiting and traffic idling, delivered through *Private Info*, targets the mistake component of D . Its effect is identified relative to the control arm and is expected to reduce idling. *Private & Social Info* adds information about the social costs of idling while holding private-cost information fixed. Its effect is therefore identified relative to *Private Info* and tests whether making C^s salient further reduces idling. Absent countervailing responses to the additional information, the predicted effect is more negative when drivers place greater weight, α , on these social costs. Similarly, comparing *Private & Social Info + Pay* with *Private & Social Info* holds the information fixed and tests whether residual idling is responsive to financial incentives, as would be expected if it reflects genuine private value from idling. The predicted effect is negative. Finally, comparing *Private & Social Info + Pay + Reminder* with *Private & Social Info + Pay + No-Reminder* holds information and financial incentives fixed and tests whether residual idling reflects inattention at the moment of choice. The predicted effect is negative unless the reminder technology crowds out drivers' own monitoring.

4.2 Main Specification

The main estimation sample comprises 6,451 vehicles observed from 17 November 2025 through 30 May 2026, the final day of the eight-week treatment period, with the vehicle-day as the unit

of observation. I use the following two-way fixed-effects difference-in-differences (TWFE-DiD) specification to estimate treatment effects for each treatment arm relative to the control arm. The same specification also underlies the incremental treatment effects defined in Section 4.3:

$$Y_{it} = \alpha_i + \delta_t + \sum_{g \neq P} \beta_g (\mathbb{1}\{G_i = g\} \times \text{Post}_t) + \varepsilon_{it}. \quad (2)$$

The vehicle and day are indexed by i and t , respectively. Y_{it} denotes the behavioural outcome for vehicle i on day t . α_i denotes vehicle fixed effects, which absorb time-invariant differences across vehicles, such as persistent differences in baseline idling or regular travel patterns. δ_t denotes day fixed effects, which absorb shocks common to all vehicles on a given day, such as weather or congestion. Because treatment assignment is random, the fixed effects are not required for causal identification; they improve precision by absorbing persistent vehicle-level and common day-level variation in idling. $\mathbb{1}\{G_i = g\}$ is an indicator for vehicle i 's assignment to arm g , and Post_t equals one from 6 April 2026, when treatment delivery began for all arms, and zero otherwise. The control arm is the omitted category. Each β_g therefore compares the pre-to-post change in outcomes for vehicles assigned to treatment arm g with the corresponding change for vehicles assigned to the control arm. ε_{it} is the error term. The paper's primary diagnostic comparisons are the effect of *Private Info* relative to the control arm and the incremental treatment effects introduced in Section 4.3. Behavioural outcomes remain missing on unobserved vehicle-days rather than being coded as zero. Standard errors are clustered at the vehicle level throughout to account for serial correlation within vehicles over time.

4.3 Incremental Treatment Effects

The incremental effects of social information, payments, and *IdleGuard* are linear combinations of the treatment-arm coefficients from Equation 2. They compare *Private & Social Info* with *Private Info*, *Private & Social Info + Pay* with *Private & Social Info*, and *Private & Social Info + Pay + Reminder* with *Private & Social Info + Pay + No-Reminder*, respectively. The incremental effect of social information tests whether making social costs salient matters beyond private-cost information. The incremental effect of payments tests whether residual idling is responsive to financial incentives, as would be expected if it reflects genuine private value from idling. The incremental effect of *IdleGuard* tests whether residual idling reflects inattention at the moment of choice.

4.4 Dynamics and Persistence

I use the following treatment-by-week specification to examine how the treatment effects estimated in Equation 2 evolve over the eight treatment weeks. The estimation sample and time window are unchanged:

$$Y_{it} = \alpha_i + \delta_t + \sum_{g \neq P} \sum_{w=13}^{20} \beta_{gw} (\mathbb{1}\{G_i = g\} \times \mathbb{1}\{W_t = w\}) + \varepsilon_{it}, \quad (3)$$

$\mathbb{1}\{W_t = w\}$ is an indicator equal to one when day t falls in treatment week w . The control arm is the omitted category, and the twelve pre-treatment study weeks form the pooled reference period. Each β_{gw} therefore compares the change in outcomes for vehicles assigned to treatment arm g in treatment week w , relative to the pooled pre-treatment period, with the corresponding change for vehicles assigned to the control arm. All other terms are defined as in Equation 2.

At the end of the eight-week intervention, I randomly assigned the *Private Info* arm to *Early-Off* and *Late-On* subgroups. The *Early-Off* subgroup stopped receiving messages after 30 May 2026, while the *Late-On* subgroup continued receiving them for two additional weeks. To study the persistence of treatment effects across all treatment arms, I modify Equation 2 by replacing the single treatment-by-post interaction with treatment-arm interactions for the active-treatment period, the first two weeks after the common intervention ended, and the remaining ten post-treatment weeks. I also estimate a week-specific version over all twelve weeks after the common intervention ended to trace the evolution of post-treatment effects more finely. For the sharper comparison within the *Private Info* arm, I directly compare the randomly assigned *Early-Off* and *Late-On* subgroups during the two-week continuation window. I complement this comparison with a stacked switch-off specification reported in Online Appendix Table D.13.

5 Experimental Results

5.1 Baseline Idling and Beliefs

Daily idling levels are high before treatment, and baseline survey responses reveal widespread misperceptions about both the consequences of idling and when shutting off the engine becomes beneficial. As shown in Table 2, GPS coverage is high during the two weeks immediately preceding treatment launch: vehicles are observed on 87% of possible days. Over observed vehicle-days, total idling is also substantial, averaging 27 minutes per day. Idling from spells longer than 30 seconds accounts for more than 20 of these minutes. Although nearly all vehicles in the sample have modern engines that do not require prolonged warm-up, 76% of drivers believe that warming up the engine increases engine life (U.S. EPA, 2025; U.S. DOE, 2026). Only 39% believe that warming up the engine harms the environment. Policy guidance further indicates that restarting after stops longer than 10 seconds generally saves fuel and reduces emissions (U.S. EPA, 2025; U.S. DOE, 2026; Gaines and Rask, 2012), yet only 6% of drivers report a fuel-saving threshold of 30 seconds or less, and only 15% report an environmental-harm threshold of 30 seconds or less.

The potential returns to reducing sustained idling are substantial. Idling in spells longer than 30 seconds accounts for nearly 18% of total engine-on time across the 12 pre-treatment weeks among vehicles in the analysis sample.

Table 2: Summary statistics for the experimental sample

	Median	Mean	SD	p10	p90	N
<i>Panel A. Driver and vehicle profile</i>						
Age (years)	40.00	41.02	11.40	28.00	57.00	6,596
Bachelor’s degree or higher (=1)	1.00	0.66	0.47	0.00	1.00	6,589
Engine size $\leq 1000\text{cc}$ (=1)	0.00	0.49	0.50	0.00	1.00	6,483
Vehicle age ≥ 5 years (=1)	0.00	0.40	0.49	0.00	1.00	6,596
<i>Panel B. Idling beliefs and preferences</i>						
Stop duration very important for shutoff decision (=1)	0.00	0.21	0.40	0.00	1.00	6,596
Believes warm-up increases engine life (=1)	1.00	0.76	0.43	0.00	1.00	6,596
Believes warm-up harms environment (=1)	0.00	0.39	0.49	0.00	1.00	6,596
Fuel-saving shutoff threshold ≤ 30 seconds (=1)	0.00	0.06	0.24	0.00	0.00	6,466
Environmental-harm shutoff threshold ≤ 30 seconds (=1)	0.00	0.15	0.36	0.00	1.00	6,452
WWF-Pakistan donation out of Rs. 100	100.00	89.46	28.38	50.00	100.00	6,479
<i>Panel C. GPS coverage and behaviour</i>						
Share of days observed	1.00	0.87	0.32	0.00	1.00	6,596
Total idling minutes	19.99	27.11	34.74	5.54	54.83	5,846
Idling minutes from stops $> 30\text{s}$	13.98	20.44	30.06	3.34	42.84	5,846
Idling spells from stops $> 30\text{s}$	8.08	10.26	11.47	2.17	20.42	5,846
Trips	7.33	11.54	35.61	1.92	19.17	5,846

Notes: The unit of observation is the vehicle. The sample comprises the 6,596 vehicles in the final six-arm randomised sample, excluding the 25 removed before the second-stage *Reminder* versus *No-Reminder* randomisation. Panels A and B report baseline characteristics and beliefs; Panel C reports vehicle-level averages of GPS coverage and behaviour over the two clean weeks beginning on 23 and 30 March 2026. Share of days observed is computed over all possible vehicle-days, with unobserved days coded as zero; GPS behavioural outcomes are averaged over observed vehicle-days. Indicators equal zero or one.

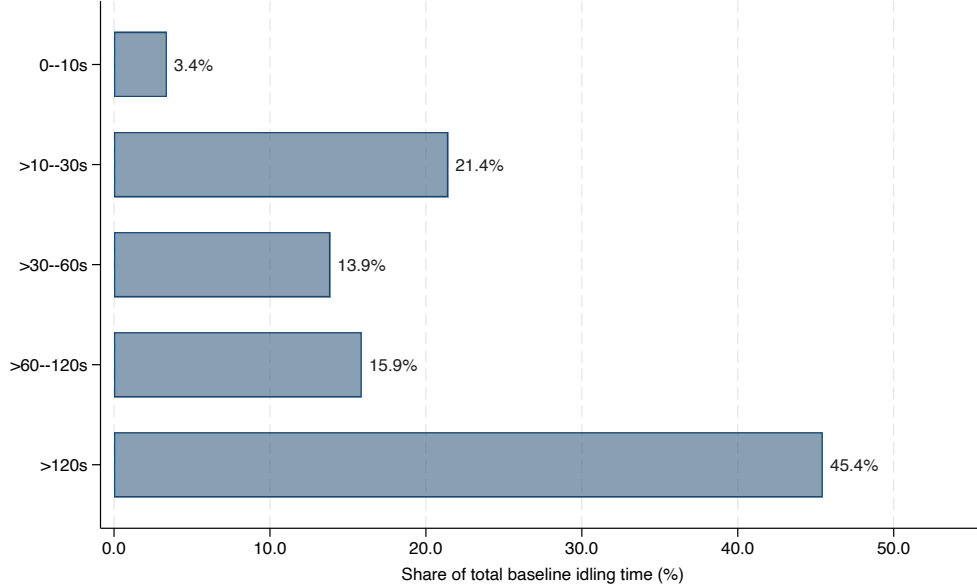
Baseline idling is concentrated in sustained spells rather than brief stops. Figure 3 assigns each idling spell to a mutually exclusive bin based on its total duration and reports the time contributed by each bin as a share of total idling time. Spells lasting more than two minutes alone account for 45% of all idling time, while spells longer than 30 seconds collectively account for roughly three-quarters. This concentration in sustained stops also supports the use of 30 seconds as the threshold for the paper’s primary idling outcome.

5.2 Treatment Effects by Arm

Table 3 reports treatment effects for each treatment arm relative to the control arm. All five estimates for daily idling from stops longer than 30 seconds are negative. The largest reductions—for *Private Info*, *Private & Social Info + Pay*, and *Private & Social Info + Pay + No-Reminder*—range from 8% to 9% of the control post-period mean.

The *Private Info* arm reduces daily idling from stops longer than 30 seconds by 1.8 minutes, or 9% of the control mean, with the estimate statistically significant at the 1% level. The estimate is similar under a vehicle-level ANCOVA specification (Online Appendix Table D.1). It also reduces the daily number of such spells by 0.5, or 5% of the control mean, with the estimate statistically

Figure 3: Composition of baseline idling time by spell duration



Notes: The figure reports the share of total baseline idling time contributed by mutually exclusive spell-duration bins. The sample comprises the 6,596 vehicles in the final six-arm randomised sample, excluding the 25 whose incentivised choices were implemented before the second-stage *Reminder* versus *No-Reminder* randomisation. Baseline idling is measured over observed GPS vehicle-days during the two clean weeks beginning on 23 and 30 March 2026.

significant at the 5% level. The effects for *Private & Social Info* and *Private & Social Info + Pay* are broadly similar. The main exception is the smaller duration effect for *Private & Social Info*, which reduces daily idling from stops longer than 30 seconds by 1.3 minutes, or 6% of the control mean.

The estimated effect for *Private & Social Info + Pay + Reminder* is also negative, but it is the only treatment-arm estimate that is statistically indistinguishable from zero, with a 95% confidence interval of $[-1.87, 0.41]$. By contrast, *Private & Social Info + Pay + No-Reminder* produces larger reductions in both idling duration and spell counts. The next subsection examines this comparison directly, while Section 6 investigates why activating the reminder appears to weaken the effect of the otherwise identical no-reminder arm. Overall, private information meaningfully reduces idling on both the intensive and extensive margins, and these reductions are broadly preserved when social information or payments are added, but not when the real-time reminder is activated.

5.3 Incremental Treatment Effects

Figure 4 reports the incremental effects of social information, payments, and *IdleGuard*. Adding social information increases daily idling from stops longer than 30 seconds by 0.5 minutes relative to *Private Info*, but the estimate is statistically indistinguishable from zero. Its 95% confidence interval of $[-0.70, 1.69]$ rules out large additional reductions in idling. Adding payments decreases idling by 0.6 minutes relative to *Private & Social Info*, but this estimate is also statistically indistinguishable

Table 3: Treatment effects across behavioural margins

	Idling duration (minutes)			Idle spells (count)		Vehicle use	
	Total	> 10s	> 30s	> 10s	> 30s	Moving minutes	Trips
Private Info $\times$ Post	-1.993*** (0.737)	-1.952*** (0.719)	-1.809*** (0.654)	-1.067 (0.705)	-0.532** (0.229)	-1.321 (1.739)	-0.325 (0.559)
Private + Social Info $\times$ Post	-1.523** (0.722)	-1.467** (0.702)	-1.316** (0.629)	-1.089 (0.737)	-0.479** (0.242)	-1.707 (1.967)	-0.551 (0.448)
Private + Social Info + Pay $\times$ Post	-2.087*** (0.756)	-2.029*** (0.737)	-1.873*** (0.673)	-1.058 (0.716)	-0.415* (0.223)	-2.470 (1.793)	-0.779 (0.563)
Private + Social Info + Pay + Reminder $\times$ Post	-0.826 (0.665)	-0.773 (0.643)	-0.730 (0.582)	-0.024 (0.637)	-0.066 (0.206)	-0.470 (1.629)	-0.341 (1.123)
Private + Social Info + Pay + No-Reminder $\times$ Post	-1.876*** (0.718)	-1.854*** (0.699)	-1.725*** (0.631)	-0.915 (0.660)	-0.453** (0.222)	-2.545 (1.709)	-0.153 (0.457)
Control post-period mean	27.721	26.770	20.972	32.590	10.238	80.756	10.567
Observations	701441	701441	701441	701441	701441	701441	701441
Vehicles	6428	6428	6428	6428	6428	6428	6428
Vehicle FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Calendar-date FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Joint p -value	0.023	0.020	0.017	0.280	0.051	0.568	0.661

Notes: The unit of observation is the vehicle-day. Each column reports treatment-arm-by-post coefficients from a regression absorbing vehicle and calendar-date fixed effects; the control arm is omitted. Post equals one from 6 April through 30 May 2026; the pooled pre-treatment period comprises 12 study weeks observed between 17 November 2025 and 4 April 2026. Columns 1–3 report idling duration, Columns 4–5 idling-spell counts, and Columns 6–7 vehicle-use margins; thresholds refer to total spell duration. Outcomes are computed over observed GPS vehicle-days. The sample excludes the 25 vehicles removed before the second-stage *Reminder* versus *No-Reminder* randomisation. Standard errors are clustered by vehicle. The joint p -value tests whether all displayed coefficients equal zero. Significance levels are denoted by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

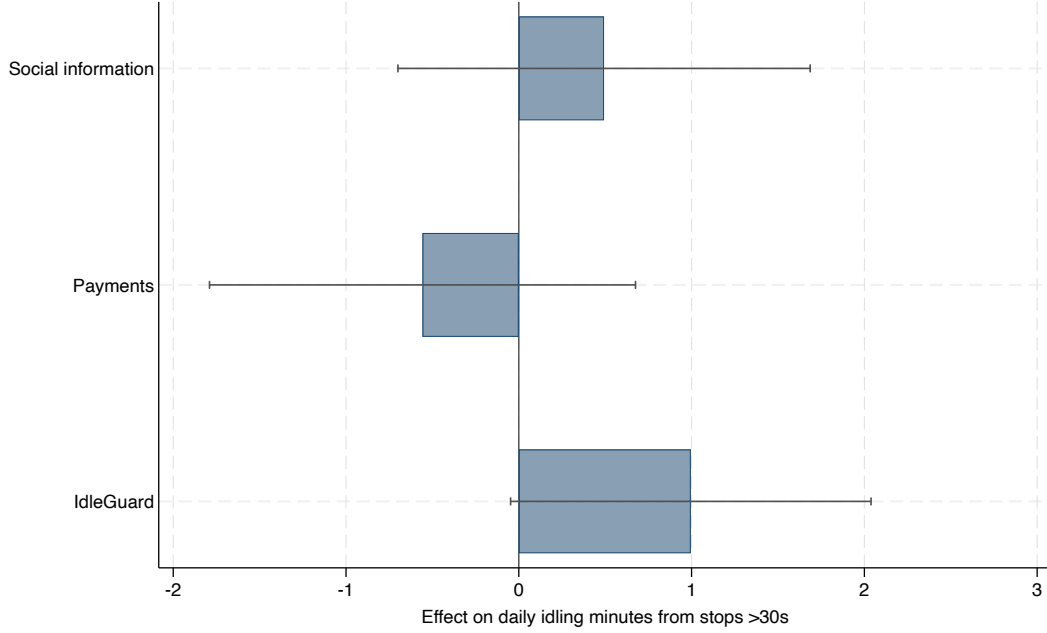
from zero. Its 95% confidence interval of $[-1.79, 0.68]$ does not rule out a meaningful additional reduction.

By contrast, activating *IdleGuard* increases daily idling from stops longer than 30 seconds by 1.0 minute relative to the matched no-reminder arm, equal to 5% of the control post-period mean, with the estimate statistically significant at the 10% level. It also increases the daily number of such spells by 0.4, equal to 4% of the control mean, with the estimate again statistically significant at the 10% level. Thus, social information and payments produce no detectable additional reductions, while the real-time reminder weakens the otherwise identical no-reminder arm. Section 6 investigates why.

5.4 Treatment Dynamics

Average treatment effects reported in Section 5.2 may mask meaningful variation across treatment weeks. I therefore examine whether effects emerge immediately or build gradually, fade with repeated exposure, or remain stable over the intervention (Figure 5). The effects of *Private Info* appear early, remain broadly stable, and are statistically significant in most treatment weeks. The estimates for *Private & Social Info + Pay* are generally more negative than those for *Private Info* in the

Figure 4: Incremental treatment effects on idling minutes from stops longer than 30 seconds



Notes: The figure reports the incremental effects of social information, payments, and *IdleGuard* on daily idling minutes from stops longer than 30 seconds. Positive coefficients indicate increased idling relative to the preceding arm or matched no-reminder arm. Horizontal bars report point estimates, and capped lines report 95% confidence intervals. Online Appendix Table D.2 reports the corresponding estimates, comparison definitions, specification, and inference.

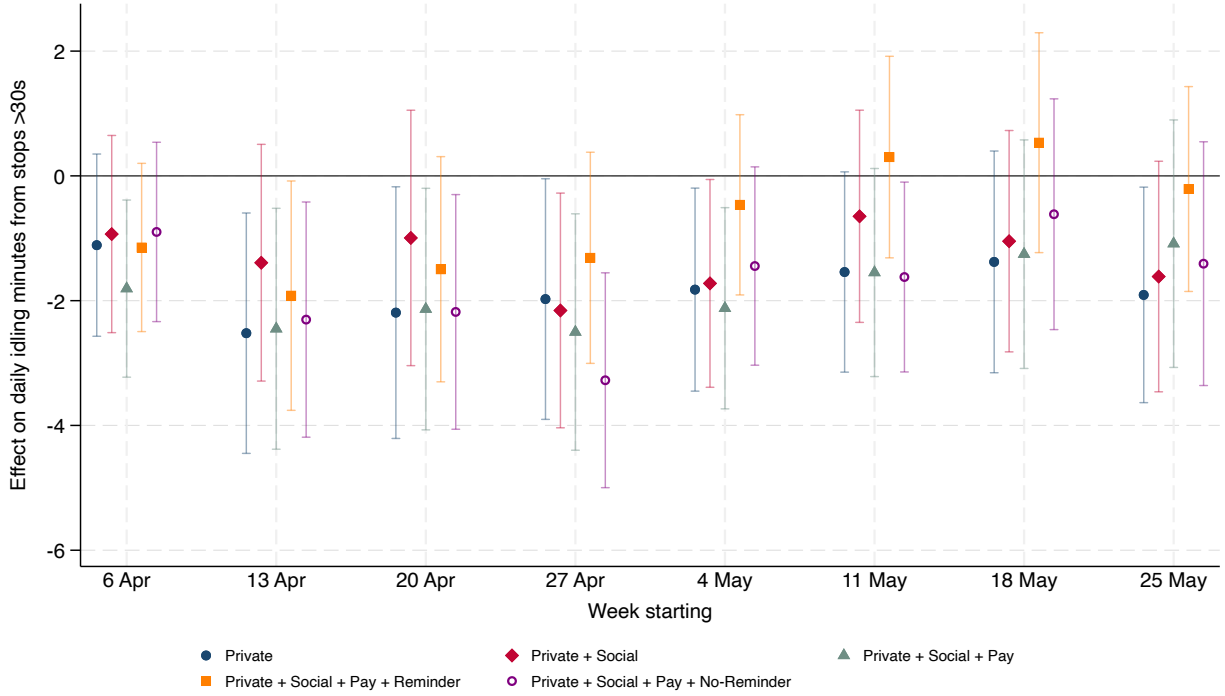
initial weeks. The estimated effects for *Private & Social Info + Pay + Reminder* gradually move towards zero and become positive in the latter weeks, whereas those for the matched *Private & Social Info + Pay + No-Reminder* arm remain negative throughout. Overall, the average treatment effects are not driven by isolated weeks, the weaker performance of the reminder arm becomes more pronounced over the intervention, and there is no general pattern of progressive treatment-effect decay. Online Appendix Figure D.6 extends this analysis to the full pre-treatment study-week path and shows no evidence of differential pre-treatment dynamics.

5.5 Persistence after Treatment Ends

Table 4 reports treatment effects for each treatment arm relative to the control arm across three periods: the eight treatment weeks, the first two post-treatment weeks, and the remaining ten post-treatment weeks. With the exception of *Private & Social Info + Pay + Reminder*, the estimates remain negative over the remaining ten post-treatment weeks, although their magnitude and precision vary across arms.

In the first two post-treatment weeks, *Private Info* reduces daily idling from stops longer than 30 seconds by 2.0 minutes, or 9% of the control mean in the same period, with the estimate statistically significant at the 1% level. This reduction is similar to the estimated effect during the treatment period. The post-treatment *Private Info* estimates pool vehicles whose messages ended after eight

Figure 5: Dynamic treatment effects on idling minutes from stops longer than 30 seconds



Notes: The figure reports week-specific treatment effects on daily idling minutes from stops longer than 30 seconds relative to the control arm. Points report treatment-arm-by-week coefficients, and capped lines report 95% confidence intervals. The 12 pre-treatment study weeks observed between 17 November 2025 and 4 April 2026 form the pooled reference period. Estimates are horizontally offset within each treatment week for readability. Regressions absorb vehicle and calendar-date fixed effects; outcomes are computed over observed GPS vehicle-days, and standard errors are clustered by vehicle. The sample excludes the 25 vehicles whose incentivised choices were implemented before the second-stage *Reminder* versus *No-Reminder* randomisation.

weeks with those randomly assigned to receive messages for two additional weeks. The direct continuation and switch-off tests reported in Online Appendix Tables D.12 and D.13 are imprecisely estimated and provide no detectable evidence that continuing the messages for two additional weeks further reduced idling or that idling rebounded when messages stopped. Over the remaining ten post-treatment weeks, the pooled *Private Info* estimate remains negative at 1.1 minutes per day, but is no longer statistically significant. The effect of *Private & Social Info + Pay + No-Reminder* also remains negative at 1.8 minutes per day, or 8% of the contemporaneous control mean, and is statistically significant at the 5% level. For the *Early-Off* subgroup whose messages ended after the scheduled eight-week campaign, the corresponding pooled twelve-week post-switch-off estimate remains negative and is reported in Online Appendix Table D.14.

Online Appendix Table D.11 reports the persistence of the incremental treatment effects. Prior assignment to *IdleGuard* is associated with 1.5 minutes more idling relative to the matched no-reminder arm in the first two post-treatment weeks, equal to 7% of the control mean in the same period, with the estimate statistically significant at the 10% level. This difference grows to 2.1 minutes over the remaining ten post-treatment weeks, equal to 10% of the control mean, and is

Table 4: Persistence of treatment effects on idling minutes from stops longer than 30 seconds

	Idling minutes from stops >30s		
	Treatment weeks	First 2 post weeks	Remaining post weeks
Private Info $\times$ Post	-1.870*** (0.660)	-2.025*** (0.782)	-1.142 (0.778)
Private + Social Info $\times$ Post	-1.329** (0.631)	-0.755 (0.914)	-0.785 (0.691)
Private + Social Info + Pay $\times$ Post	-1.910*** (0.672)	-1.335 (0.873)	-1.020 (0.769)
Private + Social Info + Pay + Reminder $\times$ Post	-0.750 (0.581)	0.122 (0.771)	0.337 (0.631)
Private + Social Info + Pay + No-Reminder $\times$ Post	-1.698*** (0.631)	-1.420 (0.885)	-1.768** (0.709)
Control period mean	20.972	22.032	22.070
Observations	1,076,021	1,076,021	1,076,021
Vehicles	6,429	6,429	6,429
Vehicle FE	Yes	Yes	Yes
Calendar-date FE	Yes	Yes	Yes
Joint p -value	0.015	0.043	0.027

Notes: The unit of observation is the vehicle-day, and the outcome is daily idling minutes from stops longer than 30 seconds. Each column reports treatment-arm-by-period coefficients from a regression absorbing vehicle and calendar-date fixed effects; the control arm is omitted. The pooled pre-treatment period comprises 12 study weeks observed between 17 November 2025 and 4 April 2026. The treatment period runs from 6 April through 30 May 2026, followed by two post-treatment weeks from 1 through 13 June and ten from 15 June through 22 August 2026. The *Private Info* coefficients pool *Early-Off* and *Late-On* vehicles in both post-treatment periods. Outcomes are computed over observed GPS vehicle-days, and standard errors are clustered by vehicle. Significance levels are denoted by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

statistically significant at the 1% level.

Taken together, the treatment effects show substantial but heterogeneous persistence across the twelve weeks after the common intervention ended. The estimates for *Private Info* and *Private & Social Info + Pay + No-Reminder* remain negative relative to the control arm over the remaining ten post-treatment weeks. By contrast, the estimate for *Private & Social Info + Pay + Reminder* is close to zero over this period. The resulting positive Reminder–No-Reminder difference persists after *IdleGuard* is no longer active, consistent with more durable behaviour change in the no-reminder arm than in the otherwise identical reminder arm.

5.6 Where in the Idling Distribution Do Effects Arise?

Baseline idling is concentrated in sustained spells. As shown in Figure 3, spells lasting more than two minutes account for 45% of total baseline idling time, while spells longer than 30 seconds collectively account for roughly three-quarters. I therefore examine where along the spell-duration distribution the treatment effects arise. This pattern is informative for both the behavioural interpretation of the results and the policy relevance of the interventions.

Online Appendix Table D.5 shows that treatment effects relative to the control arm are concentrated in spells lasting more than two minutes. For *Private Info*, approximately 79% of the total effect on idling from spells longer than 30 seconds comes from the > 120 -second bin: 1.43 of the 1.81 minutes reduced per vehicle-day. The other treatment arms broadly follow the same pattern. The corresponding shares are approximately 71% for *Private & Social Info*, 88% for *Private & Social Info + Pay*, 85% for *Private & Social Info + Pay + Reminder*, and 83% for *Private & Social Info + Pay + No-Reminder*. Consistent with this concentration, the joint test rejects the null of no treatment effects in the > 120 -second bin ($p = 0.026$), while the corresponding tests for the shorter-duration bins do not reject.

For the incremental treatment effects, Online Appendix Table D.6 shows that approximately 81% of the increase in idling caused by activating *IdleGuard*, relative to the matched no-reminder arm, arises in the > 120 -second bin. In sum, the treatment effects are concentrated in longer idling spells, and the increase in idling caused by activating *IdleGuard* does not appear to reflect merely mechanical bunching around its first beep at 30 seconds. Instead, the increase arises primarily in sustained idling. I examine the mechanisms underlying this pattern in the next section.

6 Mechanisms

6.1 Evidence of Belief Correction

Did information work through beliefs? Table 2 documents widespread misperceptions about both the consequences of idling and when shutting off the engine becomes beneficial. Table 5 directly tests whether *Private Info* corrected the two private-cost beliefs targeted by the treatment. Relative to the control arm, *Private Info* reduces by 6.3 percentage points the share of respondents who believe that warming up the engine increases engine life, from a control mean of 81.1%. It also increases by 5.2 percentage points the share reporting that shutting off the engine during a stop of 30 seconds or less saves fuel, from a control mean of 9.2%. Both estimates are statistically significant at the 1% level and adjust for the corresponding baseline belief.

These effects show that the information treatment moved posterior beliefs in the intended direction. Consistent with this interpretation, Online Appendix Table E.2 shows larger behavioural reductions among drivers whose baseline beliefs left more scope for correction, although the differences across belief groups are imprecisely estimated. Taken together with the reduction in idling reported in Section 5.2, the evidence indicates that *Private Info* changed both the targeted beliefs and behaviour. It does not, however, establish that individual-level belief updating fully mediates the behavioural response.

Table 5: Effects of *Private Info* on endline private-cost beliefs

	Endline private-cost beliefs	
	Believes warm-up increases engine life	Fuel-saving shutoff threshold ≤ 30 seconds
<i>Private Info</i>	-0.063*** (0.020)	0.052*** (0.016)
Control mean	0.811	0.092
Observations	1628	1558
Baseline outcome control	Yes	Yes

Notes: The unit of observation is the vehicle. The sample comprises vehicles assigned to *Private Info* or *Control* with a usable endline outcome and the corresponding baseline belief. Column 1 equals one if the respondent believes that warming up the engine increases engine life. Column 2 equals one if the respondent reports a fuel-saving shutoff threshold of 30 seconds or less. Each column compares *Private Info* with *Control* and adjusts for the corresponding baseline belief. A negative estimate in Column 1 and a positive estimate in Column 2 indicate belief correction. Endline availability does not differ significantly between the two groups for either outcome. Robust standard errors are shown in parentheses. Significance levels are denoted by $*p < 0.10$, $**p < 0.05$, and $***p < 0.01$.

6.2 Why Do Social Information and Payments Add Little?

Although the incremental effect of social information is statistically indistinguishable from zero on average, this masks informative heterogeneity by baseline fuel-saving and environmental misperceptions. Table 6 reports effects on daily idling minutes from stops longer than 30 seconds by whether the respondent’s baseline fuel-saving shutoff threshold was below or at least 300 seconds. It shows three comparisons: (1) the effect of *Private Info* relative to the control arm, (2) the effect of *Private & Social Info* relative to the control arm, and (3) the incremental effect of social information. Among less-mistaken drivers, who reported a threshold below 300 seconds, *Private Info* reduces idling relative to the control arm, while *Private & Social Info* produces a larger reduction that is statistically significant at the 5% level. Adding social information therefore does not offset the private-information effect and may strengthen it, although the incremental effect is imprecisely estimated. More-mistaken drivers respond about twice as strongly to *Private Info* as less-mistaken drivers, consistent with belief updating. Among these drivers, however, adding social information increases idling relative to *Private Info*, reversing much of the reduction generated by private information. The incremental effect of social information differs significantly across the two belief groups at the 5% level. This pattern is consistent with a dilution mechanism: adding the broader social-information layer weakens the targeted private-cost correction precisely where that correction is most valuable.

Online Appendix Table E.5 reveals a similar pattern for beliefs about the shutoff threshold for environmental harm. The two environmental-belief groups respond similarly to private information alone. Once environmental information is added, however, drivers with more contrary baseline beliefs respond substantially less, a pattern suggestive of belief conflict. I probe this interpretation using respondents’ confidence in their threshold beliefs. If belief conflict drives this pattern, social information should weaken the private-information effect most among drivers whose mistaken

environmental beliefs are held with greater confidence. Consistent with this prediction, the increase in idling caused by adding social information is larger among drivers with strong environmental misperceptions—those who reported a threshold of at least 300 seconds and were confident or very confident in that response.

Endline evidence sharpens these interpretations. Adding social information does not measurably improve either of two environmental beliefs relevant to its content: whether engine warm-up harms the environment or whether shutting off within 30 seconds reduces environmental harm (Online Appendix Table E.7). Direct endline measures of message overload, action relevance, and demand for continued messages provide no independent support for dilution (Online Appendix Table E.8). By contrast, among respondents who report reading the treatment messages, adding social information reduces low-credibility ratings among drivers with less contrary environmental priors, but not among those with more contrary priors; the difference in effects is marginally statistically significant ($p = 0.059$; Online Appendix Table E.9). This pattern provides suggestive evidence that more contrary priors limit the credibility of social information, consistent with the belief-conflict interpretation.

The incremental effect of payments is also statistically indistinguishable from zero, although its point estimate is negative. To reduce concerns that this result reflects limited power, Online Appendix Table E.10 pools the two non-*IdleGuard* payment arms. The pooled estimate remains modest and statistically insignificant. Conceptually, payments were designed to test whether idling that remained after information provision reflected genuine private benefits from continued idling. Taken together, the results suggest that private information addresses the principal binding constraint, leaving limited scope for payments to reduce idling further at the implemented incentive level.

Table 7 supports this interpretation. It defines environmental scepticism using beliefs about the environmental harm from engine warm-up and the corresponding shutoff threshold, and examines heterogeneity in the effect of *Private Info* and the incremental effects of social information and payments. Among less-sceptical drivers, *Private Info* reduces idling, while adding social information modestly strengthens this effect. Consistent with limited residual price-responsive idling at the implemented incentive level, adding payments yields little further reduction. Among more-sceptical drivers, however, adding social information increases idling relative to *Private Info*, consistent with the belief-conflict mechanism described above. This creates scope for payments to substitute for ineffective social motivation: adding payments largely restores the reduction achieved by private information alone. The limited average gain from payments therefore appears to reflect less a general ineffectiveness of financial incentives than the fact that there is usually little residual idling for them to address.

Table 6: Social-information effects by baseline fuel-saving threshold beliefs

	Idling minutes from stops >30s
<i>Panel A. Fuel-saving threshold below 300 seconds</i>	
<i>Private Info</i>	-1.204 (0.803)
<i>Private & Social Info</i>	-2.061** (0.849)
Social information	-0.856 (0.949)
<i>Panel B. Fuel-saving threshold at least 300 seconds</i>	
<i>Private Info</i>	-2.310*** (0.791)
<i>Private & Social Info</i>	-0.649 (0.689)
Social information	1.661** (0.798)
<i>Panel C. At least 300 seconds minus below 300 seconds</i>	
<i>Private Info</i>	-1.106 (0.898)
<i>Private & Social Info</i>	1.411* (0.855)
Social information	2.517** (1.240)
Control post-period mean, threshold below 300s	20.858
Control post-period mean, threshold at least 300s	21.097
Observations	325308
Vehicles	2976
Vehicle FE	Yes
Calendar-date FE	Yes

Notes: The unit of observation is the vehicle-day, and the outcome is daily idling minutes from stops longer than 30 seconds. Panels A and B report effects for drivers whose baseline fuel-saving shutoff threshold was below and at least 300 seconds, respectively; Panel C reports the corresponding at-least-300-seconds-minus-below-300-seconds differences. Within each panel, rows report the effects of *Private Info* and *Private & Social Info* relative to the control arm and the incremental effect of social information. Regressions absorb vehicle and calendar-date fixed effects. Post equals one from 6 April through 30 May 2026; the pooled pre-treatment period comprises 12 study weeks observed between 17 November 2025 and 4 April 2026. Outcomes are computed over observed GPS vehicle-days. The sample excludes the 25 vehicles removed before the second-stage *Reminder* versus *No-Reminder* randomisation. Standard errors are clustered by vehicle. Significance levels are denoted by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

6.3 Why Does *IdleGuard* Increase Idling?

IdleGuard was designed to restore attention at the moment of choice, but it appears to have triggered a competing pathway: reduced self-monitoring. Activating *IdleGuard* increases daily idling from stops longer than 30 seconds by roughly one minute relative to the matched no-reminder arm, equal to about 5% of the control post-period mean. The estimate is statistically significant at the 10% level (Online Appendix Table D.2). Heterogeneity and telemetry analyses provide internally consistent evidence that substitution away from self-monitoring may explain this effect.

Who drives the increase? Table 8 shows that it is concentrated among drivers with high baseline overall idling and high baseline *warming-like* idling. Table 9 also shows larger adverse point estimates among drivers with greater pre-treatment demand for the device: those who chose the device over

Table 7: Information and payment effects by environmental scepticism

	Idling minutes from stops >30s
<i>Panel A. Not environmentally sceptical</i>	
<i>Private Info</i>	-1.462** (0.740)
Social information	-0.366 (0.753)
Payments	0.432 (0.722)
<i>Panel B. Environmentally sceptical</i>	
<i>Private Info</i>	-2.591*** (0.985)
Social information	2.563** (1.151)
Payments	-3.053** (1.500)
<i>Panel C. Sceptical minus not sceptical</i>	
<i>Private Info</i>	-1.128 (1.013)
Social information	2.929** (1.375)
Payments	-3.486** (1.665)
Control post-period mean, not environmentally sceptical	20.388
Control post-period mean, environmentally sceptical	22.160
Observations	663203
Vehicles	6076
Vehicle FE	Yes
Calendar-date FE	Yes

Notes: The unit of observation is the vehicle-day, and the outcome is daily idling minutes from stops longer than 30 seconds. Environmentally sceptical drivers did not report that warming up the engine harms the environment and reported an environmental-harm shutoff threshold of at least 300 seconds; unsure warm-up responses are excluded. Panels A and B report effects for non-sceptical and sceptical drivers, and Panel C reports sceptical-minus-non-sceptical differences. Rows report the effect of *Private Info* relative to the control arm and the incremental effects of social information and payments. Regressions absorb vehicle and calendar-date fixed effects. Post equals one from 6 April through 30 May 2026; the pooled pre-treatment period comprises 12 study weeks observed between 17 November 2025 and 4 April 2026. Outcomes are computed over observed GPS vehicle-days. The sample excludes the 25 vehicles removed before the second-stage *Reminder* versus *No-Reminder* randomisation. Standard errors are clustered by vehicle. Significance levels are denoted by $*p < 0.10$, $**p < 0.05$, and $***p < 0.01$.

cash, said they would use it if offered for free, or expected it to generate positive fuel savings. The corresponding differences relative to lower-demand drivers are statistically indistinguishable from zero.

Taken together, the increase appears strongest among drivers who had more idling to monitor and who expressed greater demand for, or confidence in, the reminder technology. This pattern is difficult to reconcile with the device simply being irrelevant. Instead, it is consistent with drivers reducing their own monitoring when they expected to rely on *IdleGuard*, thereby weakening part of the behavioural response observed in the matched no-reminder arm.

IdleGuard beeped after 30 seconds of idling and repeated the alert every 30 seconds while the spell persisted. If the device substituted for drivers' own monitoring, more spells should persist through

the first 30 seconds and cross the reminder threshold under *IdleGuard* than under the matched no-reminder arm. This is what I find. Online Appendix Table E.14 shows that the no-reminder arm reduces the daily number of spells longer than 30 seconds by 0.5 relative to the control arm, with the estimate statistically significant at the 5% level. Activating *IdleGuard* increases this outcome by 0.4 spells relative to the matched no-reminder arm, with the estimate statistically significant at the 10% level. Approximately 87% of this increase arises in *traffic-like* idling. The pattern is consistent with preventive self-monitoring in the no-reminder arm being weakened by the device.

The threshold-crossing difference persists after the reminder is removed. Over the twelve post-treatment weeks, vehicles previously assigned to the no-reminder arm have 0.6 fewer spells longer than 30 seconds per day than the control arm, while the corresponding estimate for vehicles previously assigned to *IdleGuard* is close to zero (Online Appendix Table E.15). The resulting prior-*IdleGuard*-assignment difference is 0.9 spells per day, is statistically significant at the 1% level, and is concentrated primarily in *traffic-like* idling. Because the reminder was no longer active during this period, the persistence of the gap is consistent with more durable preventive self-monitoring in the no-reminder arm.

Because the primary outcome counts the full duration of every spell longer than 30 seconds, rather than only time beyond 30 seconds, the increase caused by *IdleGuard* could arise mechanically from more spells crossing the threshold. For example, if a driver in the no-reminder arm switches off after 20 seconds while an *IdleGuard* driver continues to 35 seconds, the outcome records zero for the former and the full 35 seconds for the latter, even though their behaviour differs by only 15 seconds. Online Appendix Table E.16 shows that this mechanical feature is not the main driver: only about 19% of the increase in idling caused by *IdleGuard* comes from the first 30 seconds of spells crossing the threshold, while about 81% comes from idling beyond 30 seconds. The decomposition also shows that drivers do not generally correct immediately after the first beep by switching off or moving. More than half of the full gap—56%—comes from idling beyond two minutes.

Online Appendix Table E.17 confirms that the increase is concentrated in sustained spells. Spells lasting longer than two minutes account for 81% of the full increase in idling caused by *IdleGuard*. Activating *IdleGuard* increases daily idling from these spells by 0.8 minutes relative to the matched no-reminder arm, equal to about 7% of the corresponding no-reminder post-period mean, with the estimate statistically significant at the 10% level. It also increases their daily number by 0.13 spells, equal to about 6% of the no-reminder post-period mean, with the estimate statistically significant at the 5% level.

Endline responses provide complementary, though descriptive, evidence for both stages. Among drivers assigned to *IdleGuard* who reported whether the reminder changed how comfortable they were leaving the engine running, 42% said that it made them more comfortable doing so because the beep would remind them, consistent with reduced preventive self-monitoring. In a separate question, 60% of drivers who reported how they responded after the beep said that they did not usually correct promptly, consistent with incomplete corrective action. By contrast, random assignment to *IdleGuard*, relative to the matched no-reminder arm, has no detectable effect on endline beliefs

Table 8: Incremental *IdleGuard* effects by baseline idling intensity

	Idling minutes from stops >30s	
	Overall	Warming-like
Low baseline	-0.134 (0.459)	-0.383 (0.513)
High baseline	2.051** (0.957)	2.288** (0.950)
Difference: high minus low	2.185** (1.062)	2.671** (1.080)
No-Reminder post-period mean, low baseline	9.776	12.889
No-Reminder post-period mean, high baseline	30.061	28.080
Observations	254782	254782
Vehicles	2200	2200
Vehicle FE	Yes	Yes
Calendar-date FE	Yes	Yes

Notes: The unit of observation is the vehicle-day, and the outcome is daily idling minutes from stops longer than 30 seconds. The sample comprises vehicles assigned to the *Reminder* and matched *No-Reminder* arms. Each column reports the incremental effect of *IdleGuard* for vehicles with low and high baseline idling. Column 1 defines baseline intensity using the vehicle-level average of total idling over the two clean pre-treatment GPS weeks beginning on 23 and 30 March 2026; Column 2 uses the corresponding average of *warming-like* idling. Low and high denote vehicles at or below and above the median, respectively. Difference rows report high-minus-low effects. Regressions absorb vehicle and calendar-date fixed effects. Post equals one from 6 April through 30 May 2026; the pooled pre-treatment period comprises 12 study weeks observed between 17 November 2025 and 4 April 2026. Outcomes are computed over observed GPS vehicle-days, and standard errors are clustered by vehicle. Significance levels are denoted by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

about the amount of engine warm-up required. Online Appendix Tables E.19 and E.20 report these results.

Taken together, the evidence suggests a two-stage response: activating *IdleGuard* weakens preventive self-monitoring, allowing more spells to cross the 30-second threshold, while corrective action after the reminder is delayed, causing those spells to persist. The remaining question is why *IdleGuard* drivers “snooze” after being reminded. Section 6.4 develops a procrastination model, drawing on O’Donoghue and Rabin (1999), that provides a natural explanation for this delayed response.

6.4 Why Do *IdleGuard* Drivers Snooze?

The telemetry evidence above suggests that drivers “delegate” monitoring to the device: activating *IdleGuard* makes idling spells more likely to reach the first beep relative to the matched no-reminder arm. Yet even after being reminded, drivers do not generally switch off the engine or move promptly enough to prevent these spells from becoming sustained. I explain this post-reminder delay using the simple immediate-cost procrastination model developed by O’Donoghue and Rabin (1999), adapted here to a setting with repeated opportunities to act.

Table 9: Incremental *IdleGuard* effects by pre-treatment demand for the device

	Idling minutes from stops >30s		
	Device choice	Would use if free	Expected fuel savings
Lower demand	0.305 (0.836)	1.158 (1.184)	0.129 (1.528)
Higher demand	1.791** (0.793)	1.133* (0.642)	1.910** (0.793)
Difference: higher minus lower	1.486 (1.151)	-0.025 (1.347)	1.781 (1.721)
No-Reminder post-period mean, lower demand	20.486	19.990	19.984
No-Reminder post-period mean, higher demand	19.470	19.925	20.780
Observations	221962	222801	160666
Vehicles	2021	2028	1461
Vehicle FE	Yes	Yes	Yes
Calendar-date FE	Yes	Yes	Yes

Notes: The unit of observation is the vehicle-day, and the outcome is daily idling minutes from stops longer than 30 seconds. The sample comprises vehicles assigned to the *Reminder* and matched *No-Reminder* arms. Each column reports the incremental effect of *IdleGuard* for drivers with lower and higher pre-treatment demand for the device. In Column 1, lower and higher demand denote choosing cash and the device, respectively, in the incentivised briefing-stage elicitation. In Column 2, they denote reporting that the driver would not or would use the device if offered for free; in Column 3, they denote reporting zero or positive expected fuel savings. Difference rows report higher-minus-lower-demand effects. Regressions absorb vehicle and calendar-date fixed effects. Post equals one from 6 April through 30 May 2026; the pooled pre-treatment period comprises 12 study weeks observed between 17 November 2025 and 4 April 2026. Outcomes are computed over observed GPS vehicle-days. Each column includes drivers with a valid response to the corresponding measure. The sample excludes the 25 vehicles removed before the second-stage *Reminder* versus *No-Reminder* randomisation. Standard errors are clustered by vehicle. Significance levels are denoted by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

Setup. The model begins conditional on an idling spell having reached a reminder beep; it therefore explains post-reminder delay rather than whether the spell reaches the first beep. Time is discrete, and each period corresponds to the 30 seconds between successive *IdleGuard* beeps. Let t denote the instant at which a beep sounds and the driver chooses whether to act, and suppose the driver expects the stop to end at time T . The driver can switch off immediately or continue idling and reconsider at the next beep. Switching off at t entails an immediate one-time action cost $k > 0$, capturing the cognitive and physical effort of responding to the reminder, operating the ignition, and later restarting the vehicle. It also immediately begins generating a constant net benefit flow. I denote by $b > 0$ the net benefit generated over each subsequent 30-second interval, combining fuel and maintenance gains less the flow cost of reduced comfort or convenience. For notational convenience, the flow generated over the interval beginning at t is indexed by its endpoint $t + 1$. The formal exposition focuses on waiting and traffic idling, where the relevant action is switching off; the same timing logic applies to warming idling, where the relevant action is instead to begin moving.

Time-consistent benchmark. Consider first a time-consistent driver with exponential discount factor $\delta \in (0, 1]$. Relative to leaving the engine on for the remainder of the stop, the value at time t of switching off immediately is

$$V_t^{TC}(\text{switch now}) = -k + \sum_{s=t+1}^T \delta^{s-t} b. \quad (4)$$

Switching off therefore has positive conventional net present value whenever

$$\sum_{s=t+1}^T \delta^{s-t} b > k. \quad (5)$$

This condition can hold over a sufficiently long stop even when the net benefit from any single 30-second interval is small relative to the action cost.

Present-biased choice. Now suppose the driver has quasi-hyperbolic (β, δ) preferences, where $\beta \in (0, 1]$ captures present bias. The action cost incurred at t receives full weight, while payoffs accruing after the decision—including the net-benefit flow generated over the ensuing 30-second interval—are discounted by β . At time t , switching off immediately is therefore valued at

$$V_t(\text{switch now}) = -k + \beta \sum_{s=t+1}^T \delta^{s-t} b. \quad (6)$$

This asymmetry may make delaying action by one beep attractive. If the driver plans to switch off at the next beep, the action cost is shifted to $t + 1$ and is therefore discounted by both β and δ , while the net benefit from the intervening 30-second interval is forgone. The value of this plan at time t is

$$V_t(\text{switch next beep}) = -\beta\delta k + \beta \sum_{s=t+2}^T \delta^{s-t} b. \quad (7)$$

The two options yield the same benefits from $t + 2$ onward. Comparing Equations 6 and 7, the driver switches off immediately if and only if

$$\begin{aligned} -k + \beta\delta b &\geq -\beta\delta k, \\ \beta\delta b &\geq (1 - \beta\delta)k. \end{aligned} \quad (8)$$

The left-hand side is the value at time t of the net-benefit flow generated over the first 30-second interval after switching off, which is forgone by waiting. The right-hand side is the current-self gain from postponing the action cost: paying k immediately rather than valuing the same cost at only $\beta\delta k$ next period. The driver therefore switches off immediately only when the former is at least as large as the latter.

Condition 5 can hold while Condition 8 fails: switching off is worthwhile over the remainder of

the stop, even though the current self prefers postponing it until the next beep. Intuitively, the driver delays action not because switching off is undesirable overall, but because moving the action cost out of the present is attractive enough to outweigh the loss of one 30-second net-benefit flow.

Repeated deferral. Repeated delay follows naturally for a naïve present-biased driver. At time t , the driver plans to switch off at the next beep but incorrectly expects the future self to carry out that plan. When $t + 1$ arrives, however, the action cost is again immediate, and the same comparison recurs. A sequence of intended one-period delays can therefore produce a sustained idling spell. A sophisticated driver anticipates this future temptation and may instead switch off earlier to avoid repeated deferral. Because the telemetry data reveal actions rather than beliefs about future behaviour, they cannot distinguish naïve from sophisticated present bias.

Connection to the evidence. The model links the two stages of the *IdleGuard* effect. Activating the device makes idling spells more likely to reach the 30-second reminder threshold, consistent with weaker preventive self-monitoring. Once that threshold is reached, the reminder does not generally prompt immediate correction: most of the resulting increase in idling accrues after 30 seconds, and more than half accrues beyond two minutes. Repeated present-biased deferral therefore offers a parsimonious explanation for how a reminder can restore attention without inducing prompt action. The telemetry evidence is consistent with this mechanism but does not identify present bias or distinguish naïve from sophisticated drivers.

7 Welfare and Policy Implications

7.1 Physical Benefits

I translate the effect of *Private Info* relative to the control arm—a reduction of 1.952 daily idling minutes from stops longer than 10 seconds per observed vehicle-day—into fuel savings and avoided CO₂ and PM_{2.5} emissions. I calculate cumulative impacts over four evaluation horizons: the eight-week treatment period; the 20-week period from treatment launch through the twelve observed post-treatment weeks; and two one-year scenarios with faster and slower decay after treatment ends. Online Appendix Table F.2 reports the resulting fuel savings and avoided emissions per treated vehicle. Although the observed-period savings are modest for an individual vehicle, they become substantial when aggregated across many vehicles. These physical impacts form the basis for the cost–benefit analysis below.

7.2 Monetised Benefits and Cost-Effectiveness

I monetise the private fuel savings and avoided CO₂ and PM_{2.5} emissions and compare the resulting benefits with programme costs. Gross benefits comprise avoided fuel expenditure, global climate damages, and local PM_{2.5}-related health damages. Programme costs comprise app, email, and

WhatsApp delivery over the eight-week messaging campaign. Table 10 reports gross benefits, programme costs, net benefits, and benefit–cost ratios per treated vehicle across the four evaluation horizons. The resulting net benefits are positive under every horizon considered, with benefit–cost ratios ranging from 3.66 over the treatment period to 14.99 under the slower-decay one-year scenario.

Table 10: Cost–benefit results per treated vehicle

Evaluation horizon	Gross benefit	Programme cost	Net benefit	Benefit–cost ratio
8-week treatment period	1,375.55	375.68	999.87	3.66
20-week observed period	3,200.13	375.68	2,824.45	8.52
1 year: faster decay	4,264.47	375.68	3,888.79	11.35
1 year: slower decay	5,632.89	375.68	5,257.21	14.99

Notes: Benefits and costs are reported in PKR per treated vehicle. Gross benefits comprise avoided private fuel expenditure, global climate damages, and local PM_{2.5}-related health damages. Programme cost is the per-vehicle cost of app, email, and WhatsApp delivery over the eight-week campaign. The 20-week horizon includes twelve observed post-treatment weeks. The one-year scenarios assume linear decay to zero either 26 weeks after treatment ends or by campaign week 52. Net benefit equals gross benefit minus programme cost; the benefit–cost ratio equals gross benefit divided by programme cost. The underlying fuel-use, emissions, exposure, valuation, and cost inputs are documented in the Online Appendix.

7.3 Scalability and Policy Design

Digital messaging campaigns are inexpensive to scale. GPS trackers are widespread among cars in Pakistan because banks and other financial institutions commonly require them for financed vehicles as a means of protecting the underlying asset. Bank financing accounted for an estimated 40–45% of new-car sales during the recent expansion in automobile lending, so many passenger vehicles already generate telematics data (Khan, 2021). This existing infrastructure allows personalised anti-idling messages to be constructed from vehicle data that are already collected and delivered at low marginal cost. Less personalised anti-idling information could also be distributed more broadly through WhatsApp or text messages without requiring vehicle-level telemetry.

I illustrate the potential scale of such a campaign by applying the per-vehicle estimates to alternative shares of Pakistan’s registered passenger-vehicle stock. Under the preferred illustrative reach assumption of 25%, corresponding to approximately 1.22 million vehicles, the slower-decay one-year scenario implies fuel savings of about 11.5 million litres and avoided emissions of approximately 28,800 tonnes of CO₂ and 10.7 tonnes of PM_{2.5}. The corresponding estimated national net benefit is approximately PKR 6.4 billion, or USD 23.1 million. These calculations illustrate the potential scale of the programme rather than forecasting realised coverage or aggregate effects.

8 Conclusion

This paper studies why people persist in a repeated conservation behaviour for which reducing activity can generate both private and social gains. In a large-scale field experiment with 6,621 vehicles in Pakistan, I use an additive design to distinguish among misperceptions, social motivation, private preferences, and inattention as explanations for vehicle idling. Correcting private-cost misperceptions reduces idling by 1.8 minutes per vehicle-day, or 9% of the control mean, and the effect persists after treatment ends. Adding social information or payments produces little additional reduction. By contrast, activating a real-time in-vehicle reminder increases idling relative to an otherwise identical no-reminder arm. The effects of both private information and the reminder are concentrated in sustained idling spells.

The results point to two distinct behavioural mechanisms. Private information changes the beliefs it targets and reduces idling, indicating that misperceptions can sustain inefficient behaviour even in the repeated use of a durable that people already own. The reminder generates a different response. Drivers appear to rely on *IdleGuard* in place of their own preventive monitoring, allowing more idling spells to reach the reminder threshold, and then fail to correct promptly once reminded. Most of the resulting increase occurs well after the first beep, consistent with procrastination when acting entails an immediate cost while the benefits accrue subsequently. Externally supplied attention can therefore substitute for self-monitoring rather than simply augment it.

These findings have implications for the design of corrective environmental policy. The private-information intervention is inexpensive to deliver, generates benefits that exceed programme costs, and can be scaled using existing digital and telematics infrastructure. More broadly, the results caution against treating persistent harmful behaviour as evidence that stronger incentives or additional intervention layers are necessarily required. Effective policy instead depends on identifying what sustains the behaviour—whether mistaken beliefs, genuine private benefits, weak social motivation, or failures of attention and follow-through—and targeting that margin directly. In repeated conservation settings, a simple intervention aimed at the relevant distortion may outperform a more intensive one.

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Online Appendix to
The Idle Margin: Why Everyday Conservation Goes Undone

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Contents

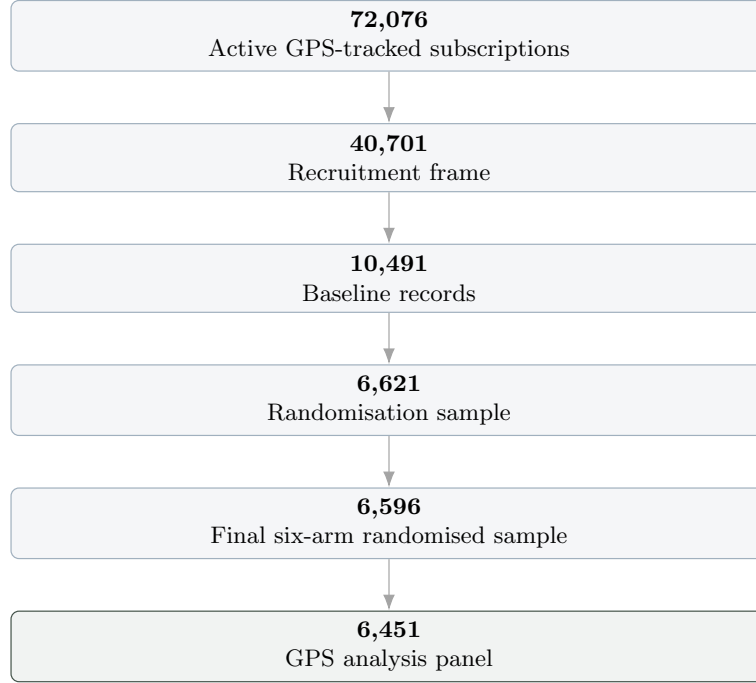
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A Experimental Design and Implementation

A.1 Sample Recruitment and Construction

Figure A.1: Experimental sample flow and GPS analysis panel construction

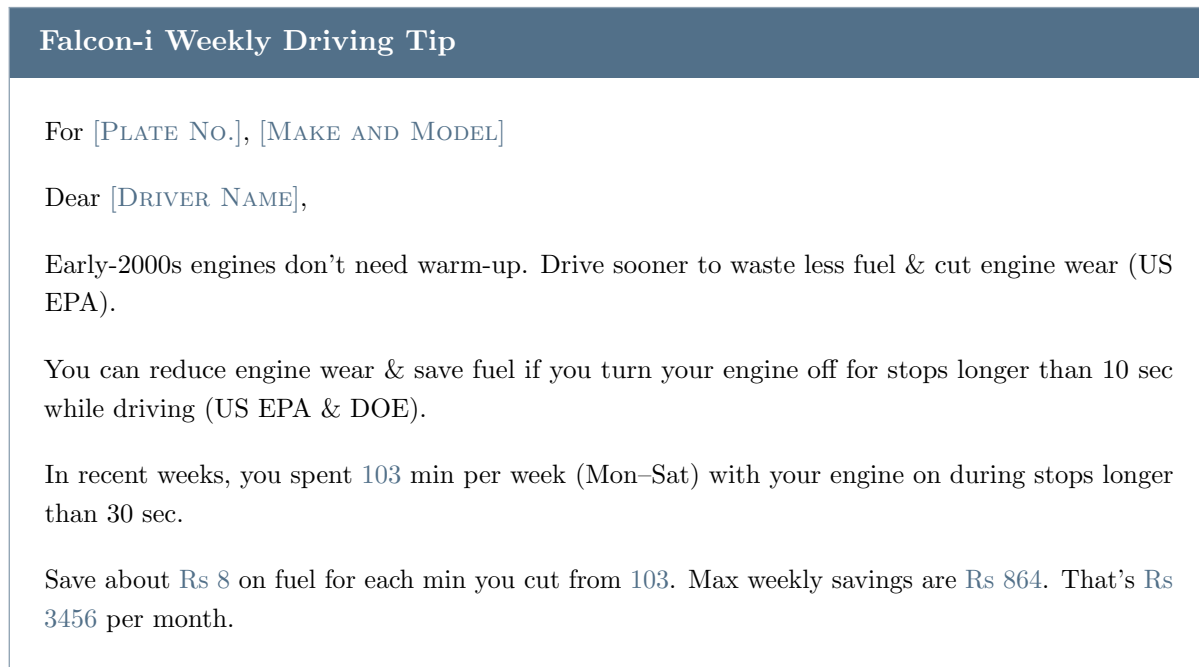


Notes: Counts are vehicles. The recruitment frame applies the passenger-four-wheeler, owner-driver, tracking-history, geographic, and non-pilot restrictions to active subscriptions. Baseline records comprise completed baseline surveys before the final respondent and GPS eligibility restrictions. The 25 drivers whose incentivised choices were implemented were removed before the second-stage *Reminder* versus *No-Reminder* randomisation. The GPS analysis panel comprises vehicles in the final six-arm randomised sample observed at least once in the cleaned GPS data between 17 November 2025 and 22 August 2026.

A.2 Treatment Materials

A.2.1 Private Information

Figure A.2: Illustrative *Private Info* treatment message



Notes: The figure reconstructs the participant-facing *Private Info* message from the final implementation script. Identifying fields are shown as placeholders; numerical fields reproduce the values generated for a driver with the median weekly idling benchmark in the treatment-delivery data. Messages were delivered through the partner's mobile app, WhatsApp Business account, and official email platform.

A.2.2 Social Information

Figure A.3: Illustrative *Private & Social Info* treatment message

Falcon-i Weekly Driving Tip

For [PLATE NO.], [MAKE AND MODEL]

Dear [DRIVER NAME],

Early-2000s engines don't need warm-up. Drive sooner to waste less fuel, pollute less & cut engine wear (US EPA).

You can reduce engine wear, save fuel & release less CO₂ if you turn your engine off for stops longer than 10 sec while driving (US EPA & DOE).

In recent weeks, you spent 101 min per week (Mon–Sat) with your engine on during stops longer than 30 sec.

Save about Rs 8 on fuel for each min you cut from 101, & help lower environmental harm (e.g., floods, heat) worth about Rs 3. Max weekly savings are Rs 1186. That's Rs 4744 per month.

Engine-on stops emit PM_{2.5}, a pollutant that cuts the average Pakistani's life by 3.3 years (Univ of Chicago).



Tailpipe CO₂ adds to flooding



Tailpipe PM_{2.5} shortens lives

Notes: The figure reconstructs the participant-facing *Private & Social Info* message from the final implementation script. Identifying fields are shown as placeholders; numerical fields reproduce the values generated for a driver with the median weekly idling benchmark in the treatment-delivery data. Messages were delivered through the partner's mobile app, WhatsApp Business account, and official email platform.

A.2.3 Payments and *IdleGuard*

Figure A.4: Illustrative *Private & Social Info + Pay + Reminder* treatment message

Falcon-i Weekly Driving Tip

For [PLATE NO.], [MAKE AND MODEL]

Dear [DRIVER NAME],

Early-2000s engines don't need warm-up. Drive sooner to waste less fuel, pollute less & cut engine wear (US EPA).

You can reduce engine wear, save fuel & release less CO₂ if you turn your engine off for stops longer than 10 sec while driving (US EPA & DOE).

In recent weeks, you spent 100 min per week (Mon–Sat) with your engine on during stops longer than 30 sec.

Save about Rs 8 on fuel for each min you cut from 100, & help lower environmental harm (e.g., floods, heat) worth about Rs 4. Max weekly savings are Rs 1179. That's Rs 4716 per month.

For each min you cut from 100 this week (Mon–Sat), we can pay you Rs 4. Max payment for this week: Rs 400. Paid next Mon.

If you forget, *IdleGuard* can remind. It beeps at 30 sec, then every 30 sec, to remind you to turn off the engine if your stop is still over 30 sec.

Engine-on stops emit PM_{2.5}, a pollutant that cuts the average Pakistani's life by 3.3 years (Univ of Chicago).



Tailpipe CO₂ adds to flooding



Tailpipe PM_{2.5} shortens lives

Notes: The figure reconstructs the participant-facing week-one *Private & Social Info + Pay + Reminder* message from the final implementation script. Identifying fields are shown as placeholders; numerical fields reproduce the values generated for a driver with the median weekly idling benchmark in the treatment-delivery data. Messages were delivered through the partner's mobile app, WhatsApp Business account, and official email platform.

B Data Construction

B.1 Engine Status and Movement Classification

Ping-level idling classification. In the raw ping-level data, I initially classify a ping as candidate idling when the vehicle’s ignition status is on and its recorded speed is zero. I then address two potential measurement concerns. First, because ignition status records whether the vehicle’s ignition circuit is powered rather than directly whether the engine is running, I validate it using tracker-recorded battery voltage. Second, because a recorded speed of zero does not guarantee that the vehicle remained stationary between consecutive pings, I use GPS displacement to identify and reclassify false positives arising from slow movement.

Battery-voltage validation. The GPS tracker records a binary ignition-status field indicating whether it detects the vehicle’s ignition circuit as active. I use this field as a proxy for whether the engine is running. A potential concern is that the ignition circuit could remain active while the engine is off, for example when a vehicle is in accessory mode. I validate the interpretation of the ignition-status field using battery voltage recorded independently by the tracker.

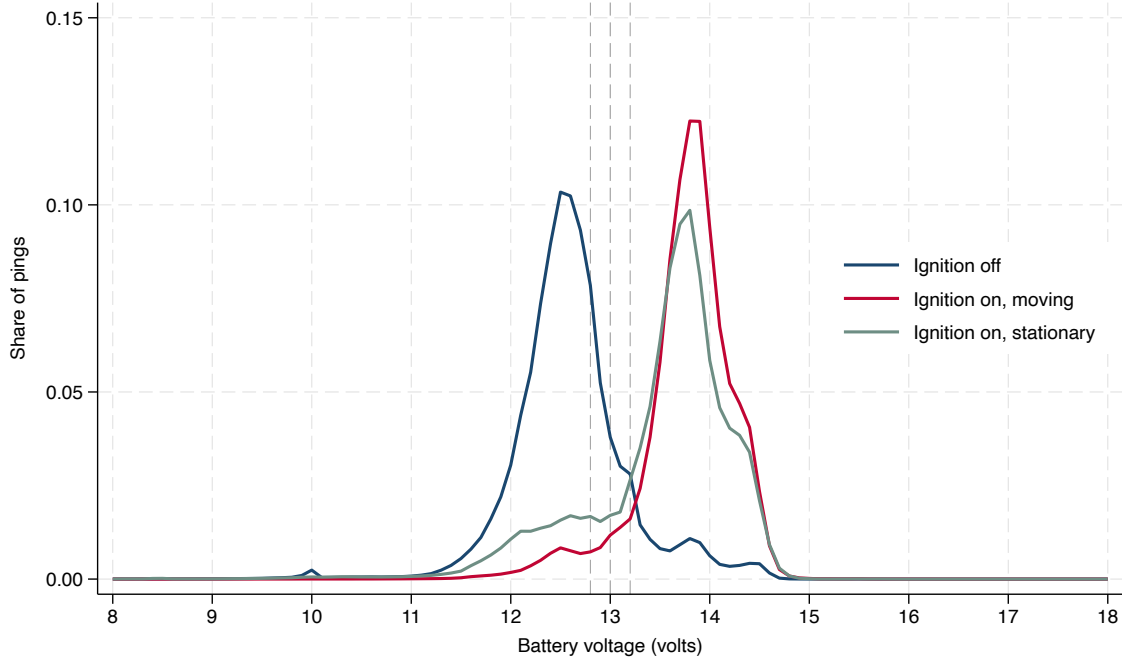
Figure B.1 compares battery-voltage distributions across three mutually exclusive tracker states: ignition off, ignition on while moving, and ignition on while stationary. Moving ignition-on pings provide a benchmark in which the engine is known to be running. The voltage distribution for stationary ignition-on pings closely resembles that of moving ignition-on pings and differs sharply from the ignition-off distribution. In particular, both ignition-on distributions are concentrated in the higher charging-voltage range, whereas ignition-off observations are concentrated at lower voltage.

Table B.1 examines whether fixed battery-voltage cut-offs provide a viable alternative engine-state classification. Battery voltage is observed for 99.5% of pings classified as idling. Depending on the cut-off, voltage screening retains 78.8–85.6% of idling pings and 78.0–84.9% of measured idling minutes. These excluded shares cannot, however, be interpreted as accessory-mode contamination. The same cut-offs also exclude 4.8–10.5% of moving pings, even though the engine must be running while the vehicle is moving. Low voltage therefore does not provide a reliable stand-alone indicator that the engine is off.

Taken together, the battery-voltage evidence validates the ignition-status field as the engine-on measure. Stationary ignition-on observations closely resemble known engine-running moving observations and differ sharply from ignition-off observations. Because universal voltage cut-offs also exclude genuine engine-running observations, I use battery voltage to validate rather than replace the ignition-status measure.

GPS displacement and slow-movement correction. A recorded speed of zero does not necessarily imply that a vehicle remained stationary between consecutive pings. In slow-moving

Figure B.1: Battery voltage by ignition and movement state



Notes: The figure uses ping-level GPS tracker data from the twelve pre-treatment study weeks. Each distribution is normalised separately and reports the share of pings within 0.1-volt bins. Ignition on, moving consists of pings for which ignition status equals one and recorded speed is positive. Ignition on, stationary consists of pings classified as idling under the GPS-adjusted idling definition. The vertical dashed lines mark the alternative battery-voltage cut-offs of 12.8, 13.0, and 13.2 volts examined in Table B.1.

Table B.1: Validation of ignition status using battery voltage

Minimum battery voltage	Idling pings retained	Idling minutes retained	Moving pings excluded
12.8 volts	85.6%	84.9%	4.8%
13.0 volts	82.6%	81.9%	6.3%
13.2 volts	78.8%	78.0%	10.5%
Battery-voltage availability among idling pings		99.5%	

Notes: The table uses ping-level GPS tracker data from the twelve pre-treatment study weeks. Idling pings retained is the share of voltage-observed pings classified as idling whose recorded battery voltage is at or above the indicated cut-off. Idling minutes retained is the corresponding share of measured idling duration. Moving pings excluded is the share of ignition-on pings with positive recorded speed whose battery voltage falls below the indicated cut-off. Because the vehicle is moving in these observations, they provide a benchmark in which the engine is known to be running. Battery-voltage availability is the share of all pings classified as idling with a recorded voltage between 8 and 18 volts.

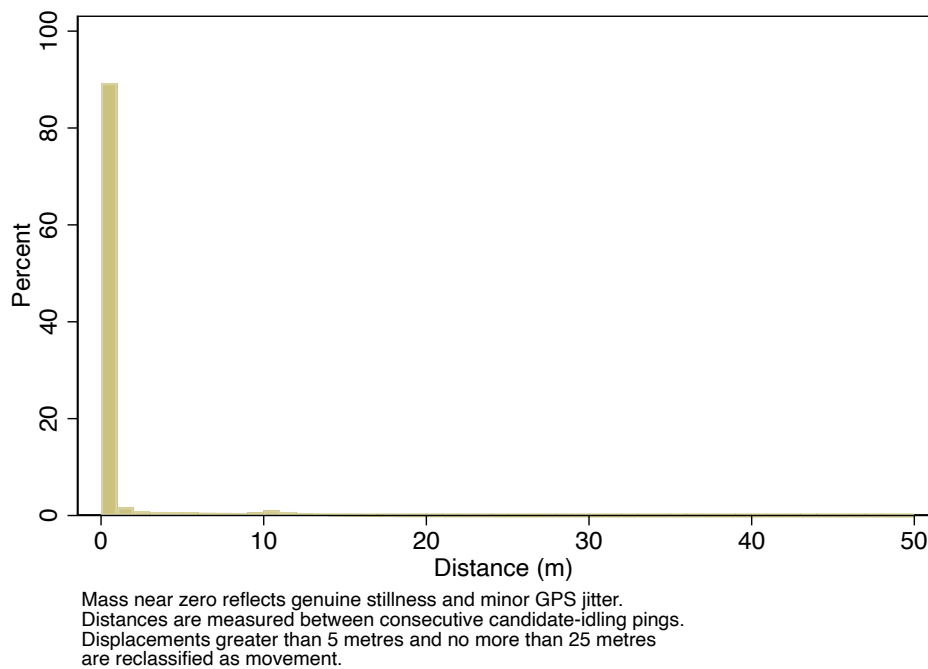
traffic, for example, the tracker may report zero speed even when the vehicle moves between observations. I therefore use GPS displacement to diagnose and correct potential creeping within the speed-based idling classification.

To calibrate this correction, I use the first complete week of high-frequency GPS data, 17–22 November 2025. For each pair of consecutive candidate-idling pings, I calculate the haversine distance d between their recorded coordinates. Candidate-idling pairs are retained for this diagnostic

only when both pings have ignition status on and recorded speed equal to zero, belong to the same vehicle and calendar date, and have valid GPS coordinates. This restriction excludes comparisons that cross vehicle or date boundaries or rely on invalid location records.

Figure B.2 shows the distribution of displacement up to 50 metres. The sharp mass at $d = 0$ indicates that most candidate-idling pairs are genuinely stationary. Small positive displacements are also common and may reflect ordinary GPS jitter rather than actual vehicle movement. I therefore retain observations with displacement of no more than five metres as idling. Displacements satisfying $5 < d \leq 25$ metres are reclassified as movement because they are more consistent with slow rolling or creeping. Values above 25 metres are not automatically reclassified: such large jumps are less plausibly generated by creeping between consecutive candidate-idling pings and may instead reflect GPS or telemetry anomalies.

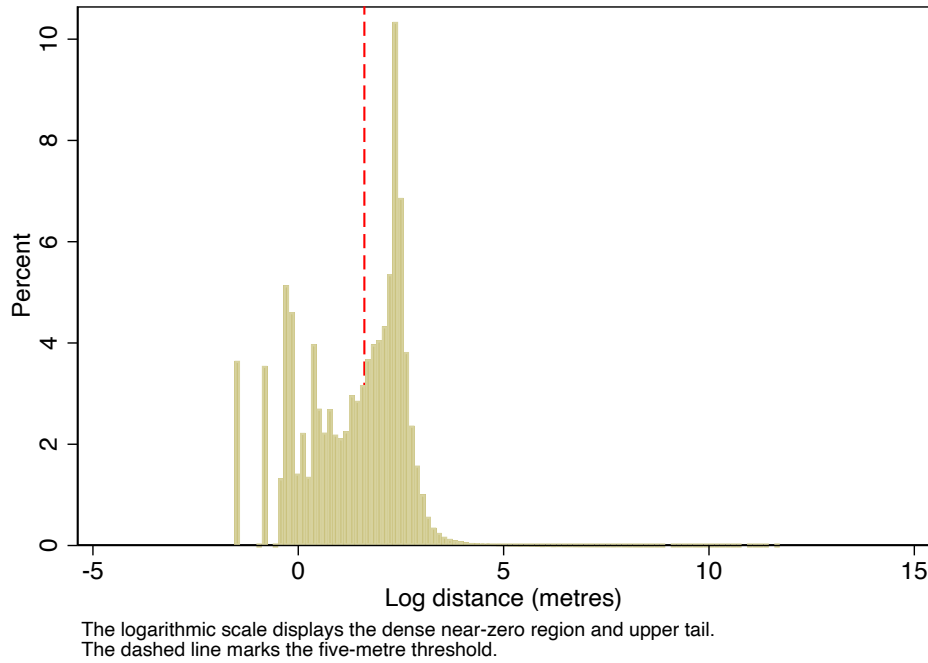
Figure B.2: GPS displacement between consecutive candidate-idling pings



Notes: The figure uses ping-level GPS tracker data from 17–22 November 2025 and reports haversine distance between consecutive candidate-idling pings up to 50 metres. Candidate-idling pings have ignition status on and recorded speed equal to zero. Consecutive pairs are retained only when both pings belong to the same vehicle and calendar date and have valid GPS coordinates. The distribution is measured before applying the GPS-based slow-movement correction. Displacements of no more than five metres are retained as idling, while displacements greater than five and no more than 25 metres are reclassified as movement.

Figure B.3 presents the full displacement distribution on a logarithmic scale. The transformation makes both the dense near-zero region and the much thinner upper tail visible. The figure reinforces the distinction between a narrow GPS-jitter band near zero and larger displacement values that are more consistent with movement. The bounded correction uses this distinction without treating all large GPS jumps as plausible creeping.

Figure B.3: Full distribution of GPS displacement between consecutive candidate-idling pings



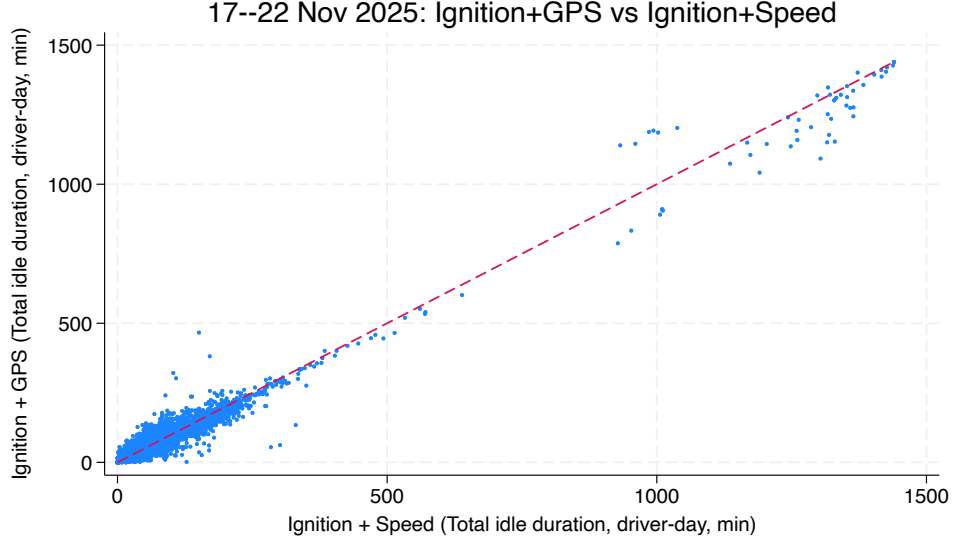
Notes: The figure uses ping-level GPS tracker data from 17–22 November 2025 and reports the log-transformed haversine distance between consecutive candidate-idling pings before applying the GPS-based slow-movement correction. Candidate-idling pings have ignition status on and recorded speed equal to zero. Consecutive pairs are retained only when both pings belong to the same vehicle and calendar date and have valid GPS coordinates. The bounded cleaning rule reclassifies displacements greater than five and no more than 25 metres as movement; larger values are not automatically reclassified.

Comparison with alternative idling definitions. I next assess whether the GPS-based correction materially changes measured idling. Using the same diagnostic week, I construct three alternative vehicle-day measures of total idling duration. The first uses the baseline ignition-and-speed definition, under which ignition-on pings with recorded speed equal to zero are classified as idling. The second uses an ignition-and-GPS benchmark based on realised displacement rather than recorded speed. The third is the preferred cleaned measure, which begins with the ignition-and-speed definition and reclassifies only candidate-idling observations with displacement satisfying $5 < d \leq 25$ metres.

I compare total vehicle-day idling rather than average spell duration because alternative definitions may generate different numbers of detected spells. Vehicle-day totals therefore provide a common outcome that is not mechanically affected by variation in the number of spells.

Figure B.4 compares the ignition-and-GPS benchmark with the baseline ignition-and-speed definition. Observations are concentrated close to the 45-degree line, indicating that the speed-based measure is highly aligned with the displacement-based benchmark. The GPS benchmark nevertheless produces modestly lower idling totals on average, consistent with the ignition-and-speed rule classifying some slow movement as idling.

Figure B.4: Vehicle-day idling under ignition-and-GPS and ignition-and-speed definitions



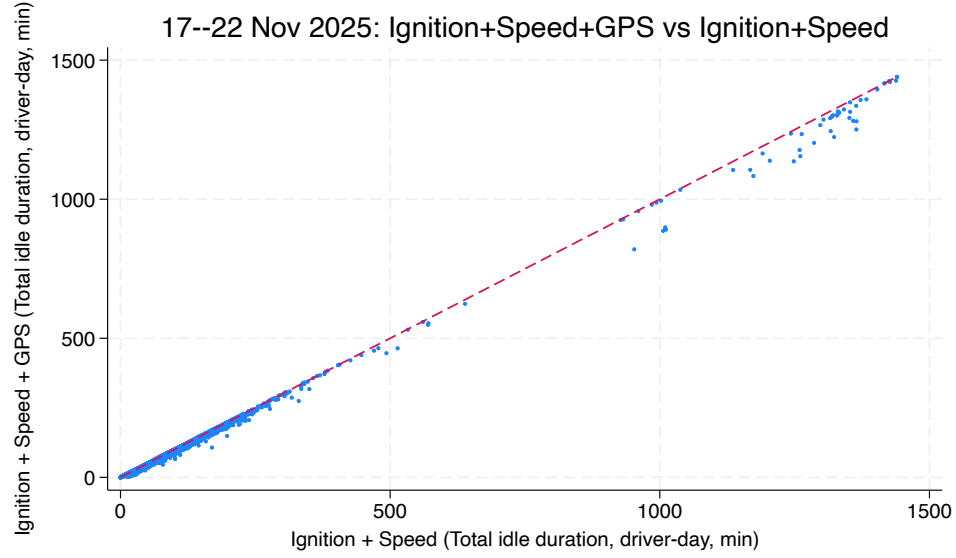
Notes: The figure uses vehicle-day GPS tracker data from 17–22 November 2025. The horizontal axis reports total daily idling duration under the baseline ignition-and-speed definition. The vertical axis reports total daily idling duration under the ignition-and-GPS benchmark. The 45-degree line denotes equality between the two measures.

Figure B.5 compares the preferred cleaned measure with the baseline ignition-and-speed definition. The two measures remain tightly aligned, showing that the GPS correction leaves the baseline measure largely intact while applying a limited downward adjustment to vehicle-days containing likely creeping.

During the diagnostic week, mean vehicle-day total idling is 39.8 minutes under the baseline ignition-and-speed definition, 35.9 minutes under the ignition-and-GPS benchmark, and 38.0 minutes under the preferred cleaned definition. These comparisons support two conclusions. First, the baseline ignition-and-speed measure is broadly valid because it closely tracks the GPS-based benchmark. Second, the bounded correction is not overly aggressive: it removes likely creeping while preserving most of the original ignition-and-speed measure.

The figures in this subsection use one diagnostic week to illustrate and calibrate the measurement correction. The resulting rule is not limited to that week. In the construction of each weekly GPS file, I separately calculate displacement between consecutive candidate-idling pings and reclassify observations with $5 < d \leq 25$ metres as movement. The same correction is therefore applied uniformly throughout the study panel.

Figure B.5: Vehicle-day idling before and after the GPS-based slow-movement correction



Notes: The figure uses vehicle-day GPS tracker data from 17–22 November 2025. The horizontal axis reports total daily idling duration under the baseline ignition-and-speed definition. The vertical axis reports total daily idling duration under the preferred cleaned definition, which reclassifies candidate-idling observations with displacement greater than five and no more than 25 metres as movement. The 45-degree line denotes equality between the two measures.

B.2 Idling Spell Construction

I define a trip as the sequence of pings between an ignition-on event and the subsequent ignition-off event within a vehicle-day. Within each vehicle-day and trip, I group consecutive pings classified as idling into idling spells.

For each idling ping i , observed at time t_i , I assign the elapsed time until the next ping, $t_{i+1} - t_i$, provided that the next ping belongs to the same vehicle-day and trip. The next ping may record either idling or movement because the interval is attributed to the state observed at the current ping. I sum these elapsed-time contributions across the idling pings in each spell to obtain the spell's duration.

I do not assign the interval $t_{i+1} - t_i$ to idling when consecutive pings within the same vehicle-day and trip are separated by more than 30 minutes. Such gaps may reflect interruptions in telemetry rather than continuous idling. This rule does not define the tracker reporting frequency: trackers report at approximately 10-second intervals. It instead prevents a telemetry blackout from being counted as continuous idling.

B.3 Vehicle-Day Panel Construction

I aggregate idling spells within each vehicle-day to construct daily idling duration and spell counts. The primary outcome is daily idling duration from spells longer than 30 seconds. The treatment informed drivers that shutting off the engine during stops longer than 10 seconds generally saves fuel and reduces emissions, while personalised feedback and payments were based on idling spells longer than 30 seconds ([U.S. EPA, 2025](#); [U.S. DOE, 2026](#); [Gaines and Rask, 2012](#)). I also construct outcomes using the 10-second threshold and total idling without a minimum spell-duration threshold.

I exclude vehicle-days with more than 18 recorded hours of idling. This distribution-based screen is intended to capture cases in which a tracker appears to remain stuck in a single ignition state.

I manually audit observations in the far-right tail of the resulting vehicle-day idling distribution. The remaining extreme values appear consistent with genuine idling and are retained in the primary measure. As a conservative robustness check, I also winsorise daily idling at the 99.9th percentile; the results are unchanged.

C Experimental Validity and Data Quality

C.1 Additional Balance and Pre-Treatment Comparability

The first-stage randomisation assigned the 6,621-vehicle randomisation sample to five groups. Table C.1 assesses balance across baseline characteristics plausibly related to idling, drawn from the baseline survey and partner administrative data. For each treatment group, the table reports differences relative to *Control*, together with a p -value from a joint test that all displayed differences are zero. The estimated differences are generally small, and none of the joint tests rejects balance at conventional significance levels.

Table C.1: Pre-randomisation balance across first-stage assignment groups

Variable	Private Info	+ Social Info	+ Pay	+ Tech	N	Control mean
1. Age (years)	0.36 (0.49)	0.49 (0.50)	-0.53 (0.48)	0.31 (0.41)	6621	40.86
2. Bachelor’s degree or higher (=1)	-0.01 (0.02)	-0.01 (0.02)	0.00 (0.02)	-0.00 (0.02)	6614	0.67
3. Engine size $\leq 1000\text{cc}$ (=1)	-0.04 (0.02)	-0.01 (0.02)	-0.02 (0.02)	-0.02 (0.02)	6505	0.51
4. Stop duration very important for shutoff decision (=1)	0.03 (0.02)	0.02 (0.02)	0.02 (0.02)	0.01 (0.01)	6621	0.19
5. Believes warm-up increases engine life (=1)	0.01 (0.02)	0.03 (0.02)	0.02 (0.02)	0.02 (0.02)	6621	0.75
6. Believes warm-up harms environment (=1)	-0.00 (0.02)	0.01 (0.02)	0.04 (0.02)	0.02 (0.02)	6621	0.38
7. Fuel-saving shutoff threshold ≤ 30 seconds (=1)	0.00 (0.01)	-0.01 (0.01)	-0.00 (0.01)	-0.00 (0.01)	6490	0.07
8. Environmental-harm shutoff threshold ≤ 30 seconds (=1)	0.00 (0.02)	0.01 (0.02)	0.02 (0.02)	0.00 (0.01)	6476	0.15
9. WWF-Pakistan donation out of Rs. 100	-2.91 (1.24)	-3.04 (1.24)	-1.15 (1.19)	-1.84 (0.99)	6504	91.27
Joint p-value	0.219	0.193	0.289	0.462		

Notes: The unit of observation is the vehicle. The sample comprises the 6,621 vehicles in the first-stage randomisation. Treatment columns report differences relative to *Control*, with robust standard errors in parentheses. A plus sign indicates that the named component is added to the preceding package. The *Private & Social Info + Pay + Tech* column denotes the original technology group before the subsequent *Reminder* versus *No-Reminder* randomisation. Regressions adjust for the vehicle-age and main-driver-certainty assignment strata. Indicators equal zero or one. The fuel-saving and environmental-harm indicators equal one for reported shutoff thresholds of 30 seconds or less. WWF-Pakistan donation is the amount donated from a Rs. 100 endowment. The final row reports the p -value from a joint test that all displayed differences equal zero.

The final six experimental arms were formed only after the second-stage randomisation conducted during treatment briefing. Table C.2 therefore assesses their comparability in GPS observability and idling duration from stops longer than 30 seconds over the two clean weeks immediately preceding treatment launch. The table reports differences relative to *Control*, together with group-specific joint-test p -values. The estimated differences are generally small, and none of the joint tests rejects comparability at conventional significance levels.

Table C.3 additionally tests for differential daily dynamics across the final six assignment groups during this pre-treatment window. Table C.4 reports the corresponding focused test for the second-

Table C.2: Pre-main-treatment GPS comparability across assignment groups

Variable	Private Info	+ Social Info	+ Pay	+ Reminder	+ No-Reminder	N	Control mean
1. Share of days observed	0.00 (0.01)	0.02 (0.01)	-0.01 (0.01)	-0.01 (0.01)	0.03 (0.01)	6596	0.87
2. Idling minutes from stops >30s	1.58 (1.30)	1.92 (1.36)	1.79 (1.35)	0.69 (1.08)	-0.07 (0.99)	5846	19.45
Joint p-value	0.443	0.186	0.289	0.705	0.153		

Notes: The unit of observation is the vehicle. The sample comprises the 6,596 vehicles in the final six-arm randomised sample. Treatment columns report differences relative to *Control*, with robust standard errors in parentheses. The table uses the two clean immediate pre-main-treatment GPS weeks, starting 23 and 30 March 2026. Share of days observed is computed over all possible vehicle-days, with unobserved days coded as zero; idling minutes from stops longer than 30 seconds are averaged over observed GPS vehicle-days. The *Reminder* and *No-Reminder* columns split the original first-stage technology group; the 25 vehicles whose incentivised choices were implemented are excluded. Regressions adjust for the vehicle-age and main-driver-certainty assignment strata. The final row reports the *p*-value from a joint test that both displayed differences equal zero.

stage *IdleGuard* versus *No-Reminder* comparison. Neither table provides evidence of differential pre-treatment dynamics.

Table C.3: Immediate pre-main-treatment daily dynamics across assignment groups

Outcome	Vehicles	Vehicle-days	Joint <i>p</i> -value
Observed on day (=1)	6596	79152	0.395
Idling minutes from stops >30s	5840	68980	0.564

Notes: The unit of observation is the vehicle-day. The table uses the two clean immediate pre-main-treatment GPS weeks, starting 23 and 30 March 2026, comprising 12 study days before treatment began on 6 April. Observed on day equals one if the vehicle has at least one retained GPS ping and zero otherwise; idling minutes from stops longer than 30 seconds are computed over observed GPS vehicle-days. For each outcome, the reported *p*-value tests whether all assignment-group-by-date interactions equal zero in a regression absorbing vehicle and calendar-date fixed effects. *Control* and the first date are omitted. The final six groups incorporate the second-stage *Reminder* versus *No-Reminder* randomisation; the 25 vehicles whose incentivised choices were implemented are excluded. Standard errors are clustered by vehicle.

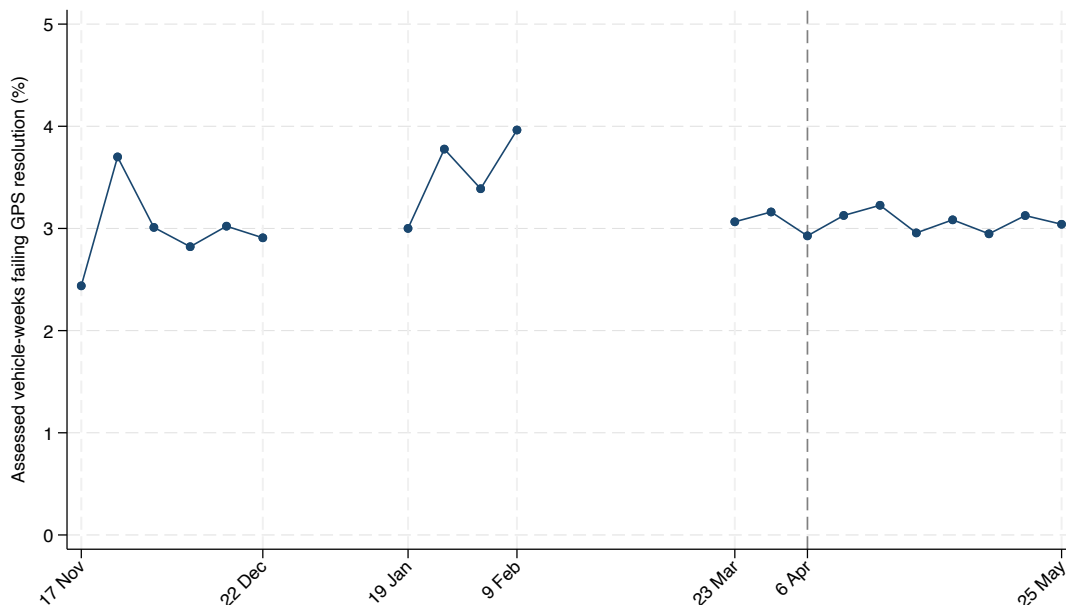
Table C.4: Immediate pre-main-treatment daily dynamics: *IdleGuard* versus *No-Reminder*

Outcome	Vehicles	Vehicle-days	Joint <i>p</i> -value
Observed on day (=1)	2,403	28,836	0.977
Idling minutes from stops >30s	2,198	25,962	0.767

Notes: The unit of observation is the vehicle-day. The sample comprises the *IdleGuard* arm and matched *No-Reminder* arm during the two clean immediate pre-main-treatment GPS weeks, starting 23 and 30 March 2026. Observed on day equals one if the vehicle has at least one retained GPS ping and zero otherwise; idling minutes from stops longer than 30 seconds are computed over observed GPS vehicle-days. For each outcome, the reported *p*-value tests whether all *IdleGuard*-by-date interactions equal zero in a regression absorbing vehicle and calendar-date fixed effects. *No-Reminder* and the first date are omitted. Standard errors are clustered by vehicle.

C.2 GPS Reporting and Resolution Diagnostics

Figure C.1: Weekly share of assessed vehicle-weeks failing the GPS resolution rule



Notes: The figure reports the weekly share of assessed vehicle-weeks failing the GPS resolution rule, defined as having a 95th-percentile idling ping gap greater than 15 seconds. Assessed vehicle-weeks are observed vehicle-weeks with a non-missing 95th-percentile gap. Disconnected segments denote excluded calendar periods. The dashed vertical line marks the start of weekly main treatment delivery on 6 April 2026.

Figure C.1 shows that the weekly failure rate remains broadly stable around treatment launch. Table C.5 formally tests for differential post-launch changes across assignment groups. The estimated treatment-arm-by-post differences are small, and the joint test does not reject equality across treatment groups relative to the control arm.

Behavioural outcomes remain missing during periods of tracker non-reporting rather than being coded as zero. Differential changes in GPS availability after treatment launch could therefore induce differential selection into the observed vehicle-day sample. Following the GPS data-quality attrition check in Kreindler (2024), Table C.6 examines this possibility using two complementary vehicle-week measures.

The first is a stringent indicator equal to one only if the partner’s internal monitoring system never classified the tracker as non-reporting during the week. The second is the share of study days in the week for which the cleaned GPS data contain at least one retained ping. The estimated treatment-arm-by-post differences are small for both measures, and none is individually statistically significant. The omnibus tests likewise fail to reject the joint null of no differential post-launch change in reporting or observability across treatment groups relative to the control arm.

Table C.5: Post-launch treatment-arm differences in GPS resolution failure

	Resolution failure
Private Info $\times$ Post	0.001 (0.005)
Private & Social Info $\times$ Post	0.010 (0.005)
Private & Social Info + Pay $\times$ Post	-0.000 (0.005)
Private & Social Info + Pay + Reminder $\times$ Post	-0.001 (0.005)
Private & Social Info + Pay + No-Reminder $\times$ Post	0.000 (0.005)
Control post-period mean	0.027
Observations	114,170
Vehicles	6,389
Vehicle FE	Yes
Week FE	Yes
Joint p -value	0.354

Notes: The unit of observation is the vehicle-week. The outcome equals one if the vehicle-week's 95th-percentile idling ping gap exceeds 15 seconds. The sample comprises observed vehicle-weeks with a non-missing 95th-percentile gap. The table reports treatment-arm-by-post coefficients from a regression absorbing vehicle and resolution-week fixed effects, with the control arm omitted. Post covers the eight main-treatment resolution weeks, from 6 April through 25 May 2026. Standard errors are clustered by vehicle and shown in parentheses. The joint p -value tests that all displayed coefficients equal zero.

Table C.6: Post-launch treatment-arm differences in tracker reporting and GPS observability

	Partner tracker reporting	Cleaned GPS observability
Private Info $\times$ Post	-0.007 (0.008)	-0.008 (0.007)
Private & Social Info $\times$ Post	-0.003 (0.007)	-0.006 (0.007)
Private & Social Info + Pay $\times$ Post	-0.001 (0.008)	-0.010 (0.007)
Private & Social Info + Pay + Reminder $\times$ Post	-0.008 (0.007)	0.003 (0.006)
Private & Social Info + Pay + No-Reminder $\times$ Post	-0.010 (0.008)	0.001 (0.007)
Control post-period mean	0.929	0.843
Observations	105536	65960
Vehicles	6596	6596
Vehicle FE	Yes	Yes
Week FE	Yes	Yes
Joint p -value	0.694	0.325

Notes: The unit of observation is the vehicle-week. Column 1 equals one if the partner's monitoring system never classified the tracker as non-reporting during the week; Column 2 is the share of possible study days with at least one retained GPS ping. Each column reports treatment-arm-by-post coefficients from a regression absorbing vehicle and week fixed effects, with the control arm omitted. Post begins on 6 April 2026. Column 1 uses partner tracker-status data from 5 February through 24 May; Column 2 uses the two clean immediate pre-main-treatment weeks and the available main-treatment weeks through 25 May. Both samples exclude the 25 vehicles whose incentivised choices were implemented. The control post-period mean is reported separately by outcome. Standard errors are clustered by vehicle and shown in parentheses. The joint p -value tests that all displayed coefficients in the corresponding column equal zero.

C.3 Fuel-Survey Availability and Attrition

I assess differential attrition in the weekly chatbot-based fuel-expenditure survey using two indicators. The first equals one if the respondent answers “Yes” to the weekly consent question and proceeds to the fuel-expenditure question, and zero following an explicit refusal or no response. The second equals one if the subsequent response yields a valid cleaned fuel-expenditure outcome, including a valid report of zero expenditure.

Table C.7 shows that drivers in the three payment-containing arms were more likely than those in the control arm to provide valid fuel-expenditure outcomes during the seven-week survey period. This pattern may reflect confusion among drivers receiving anti-idling payments about whether responding to the fuel survey was connected to those payments, despite the survey being separately incentivised. Alternatively, receiving anti-idling payments may have made the separate fuel-survey incentives more credible.

I therefore do not report fuel-expenditure effects for these arms relative to the control arm or the incremental effect of payments. Table C.8 finds no evidence of differential availability for *Private Info* relative to the control arm or for the incremental effect of *IdleGuard*.

Table C.7: Treatment-arm differences in weekly fuel-survey availability

	Any response	Valid outcome
Private Info	-0.007 (0.011)	-0.010 (0.011)
Private & Social Info	-0.013 (0.011)	-0.013 (0.011)
Private & Social Info + Pay	0.045*** (0.012)	0.039*** (0.012)
Private & Social Info + Pay + Reminder	0.043*** (0.011)	0.037*** (0.011)
Private & Social Info + Pay + No-Reminder	0.044*** (0.012)	0.037*** (0.012)
Control mean	0.133	0.122
Vehicle-weeks	46144	46144
Vehicles	6592	6592
Survey-week FE	Yes	Yes
Assigned-day FE	Yes	Yes
Randomisation-stratum FE	Yes	Yes
Joint p -value	0.000	0.000

Notes: The unit of observation is the vehicle-week. Any response equals one if the respondent answers “Yes” to the weekly consent question and proceeds to the fuel-expenditure question, and zero otherwise. Valid outcome equals one if the subsequent response yields a usable cleaned daily fuel-expenditure measure, including a valid report of zero. Each column reports treatment-arm coefficients from a linear probability model with the control arm omitted. Regressions include survey-week, assigned survey-day, and original randomisation-stratum fixed effects. Survey day was randomised across Tuesday through Sunday within treatment group and survey week. The control mean is calculated in the corresponding regression sample. Standard errors are clustered by vehicle and shown in parentheses. The joint p -value tests that all five displayed coefficients equal zero. Significance levels are denoted by $*p < 0.10$, $**p < 0.05$, and $***p < 0.01$.

Table C.8: Fuel-survey availability for experimental fuel-effect estimands

	Any response	Valid outcome
Private Info	-0.007 (0.011)	-0.010 (0.011)
Social information	-0.006 (0.011)	-0.003 (0.010)
Payments	0.058*** (0.012)	0.052*** (0.011)
IdleGuard	-0.001 (0.012)	-0.001 (0.012)
Vehicle-weeks	46144	46144
Vehicles	6592	6592
Survey-week FE	Yes	Yes
Assigned-day FE	Yes	Yes
Randomisation-stratum FE	Yes	Yes
Joint p -value	0.000	0.000

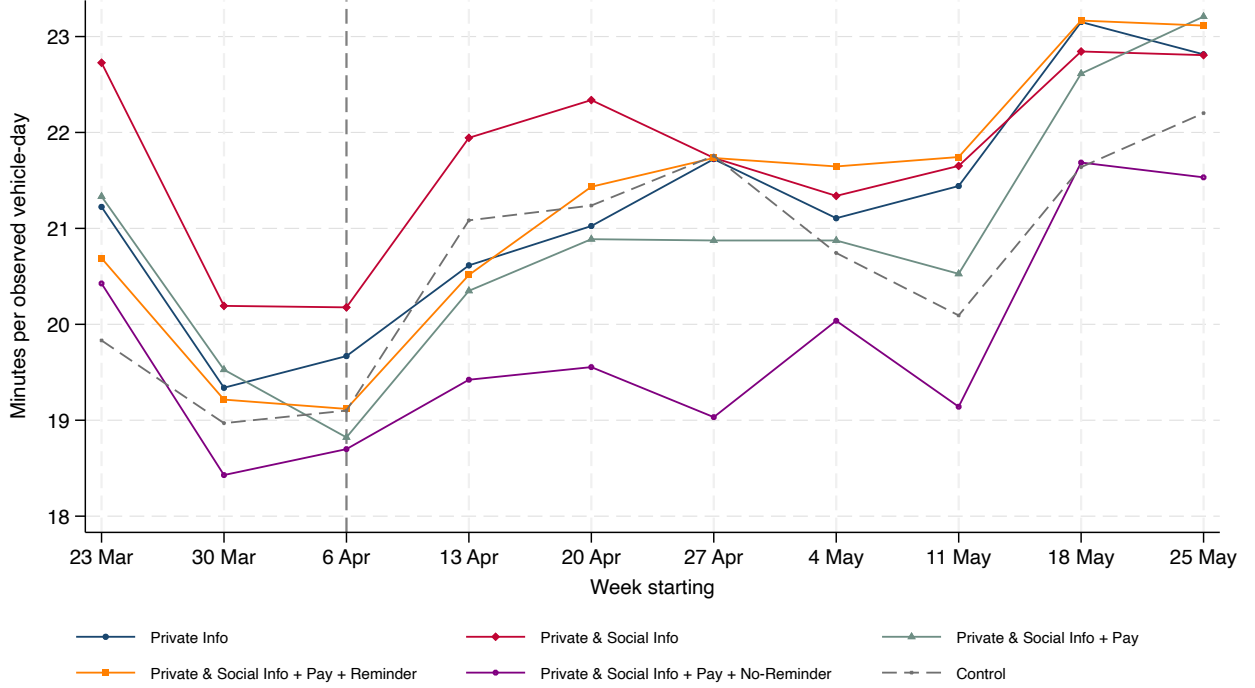
Notes: The unit of observation is the vehicle-week. Outcomes are defined as in Appendix Table C.7. *Private Info* compares that assignment group with the control arm; social information compares *Private & Social Info* with *Private Info*; payments compare *Private & Social Info + Pay* with *Private & Social Info*; and *IdleGuard* compares *Private & Social Info + Pay + Reminder* with the matched *Private & Social Info + Pay + No-Reminder* arm. Estimates come from the corresponding six-arm linear probability models in Appendix Table C.7. Regressions include survey-week, assigned survey-day, and original randomisation-stratum fixed effects. Standard errors are clustered by vehicle. The joint p -value tests that all four displayed effects equal zero. Significance levels are denoted by $*p < 0.10$, $**p < 0.05$, and $***p < 0.01$.

D Additional Experimental Results

D.1 Raw Weekly Trends

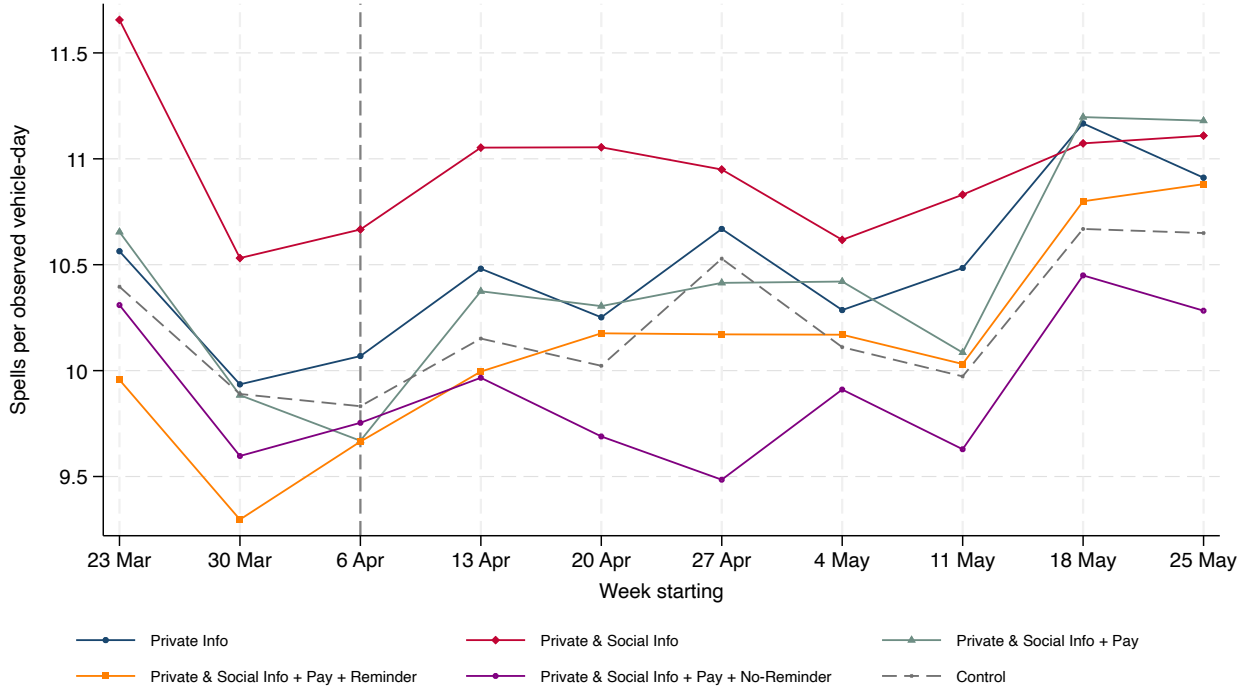
Figures D.1 and D.2 report unadjusted weekly means across assignment groups for daily idling minutes and spell counts from stops longer than 30 seconds.

Figure D.1: Raw weekly trends in idling minutes from stops longer than 30 seconds



Notes: The figure reports unadjusted assignment-group-by-week means of daily idling minutes from stops longer than 30 seconds, computed over observed GPS vehicle-days. It covers the two clean immediate pre-main-treatment weeks and eight main-treatment weeks, from 23 March through 25 May 2026. The dashed vertical line marks the start of weekly main treatment delivery on 6 April 2026. The sample excludes the 25 vehicles whose incentivised choices were implemented before the second-stage *Reminder* versus *No-Reminder* randomisation.

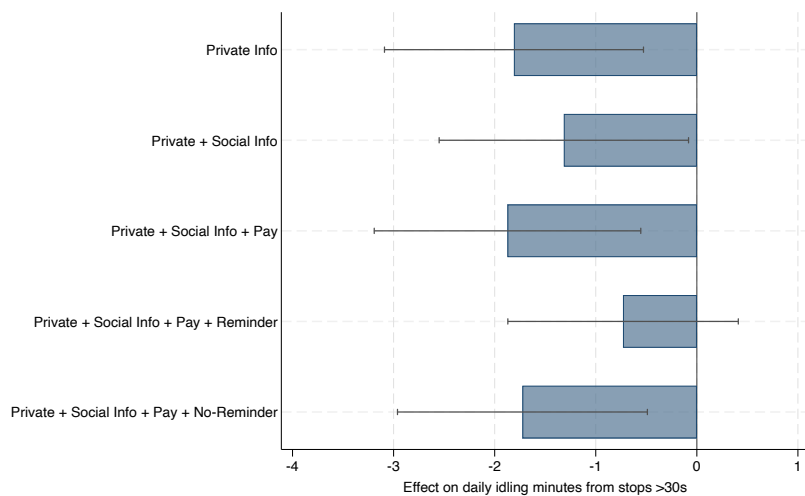
Figure D.2: Raw weekly trends in idling spells from stops longer than 30 seconds



Notes: The figure reports unadjusted assignment-group-by-week means of the daily number of idling spells from stops longer than 30 seconds, computed over observed GPS vehicle-days. It covers the two clean immediate pre-main-treatment weeks and eight main-treatment weeks, from 23 March through 25 May 2026. The dashed vertical line marks the start of weekly main treatment delivery on 6 April 2026. The sample excludes the 25 vehicles whose incentivised choices were implemented before the second-stage *Reminder* versus *No-Reminder* randomisation.

D.2 Additional Main Treatment Effects

Figure D.3: Treatment effects on idling minutes from stops longer than 30 seconds



Notes: The figure reports treatment-arm-by-post coefficients for daily idling minutes from stops longer than 30 seconds, estimated relative to the control arm. Bars show point estimates and capped lines show 95% confidence intervals. Regressions absorb vehicle and calendar-date fixed effects. Post covers the eight main-treatment weeks, from 6 April through 30 May 2026; the pooled pre-treatment period comprises 12 study weeks observed between 17 November 2025 and 4 April 2026. Outcomes are computed over observed GPS vehicle-days. The sample excludes the 25 vehicles removed before the second-stage *Reminder* versus *No-Reminder* randomisation. Standard errors are clustered by vehicle.

Table D.1: Treatment effects across behavioural margins using vehicle-level ANCOVA

	Idling duration (minutes)			Idle spells (count)		Vehicle use	
	Total	> 10s	> 30s	> 10s	> 30s	Moving minutes	Trips
Private Info	-2.090**	-2.043**	-1.899***	-0.487	-0.459*	-0.551	-1.256
	(0.827)	(0.808)	(0.737)	(0.940)	(0.237)	(1.674)	(1.174)
Private + Social Info	-0.978	-0.887	-0.687	-0.868	-0.344	-0.003	-1.661
	(0.727)	(0.708)	(0.647)	(0.792)	(0.259)	(1.878)	(1.108)
Private + Social Info + Pay	-1.663**	-1.571**	-1.378**	-0.814	-0.269	-1.337	-1.853
	(0.747)	(0.728)	(0.666)	(0.767)	(0.236)	(1.687)	(1.148)
Private + Social Info + Pay + Reminder	-0.511	-0.415	-0.397	0.702	0.108	-0.330	-1.085
	(0.679)	(0.660)	(0.612)	(1.039)	(0.261)	(1.556)	(1.279)
Private + Social Info + Pay + No-Reminder	-1.910***	-1.857***	-1.694***	-0.901	-0.441**	-2.497	-1.315
	(0.705)	(0.686)	(0.620)	(0.678)	(0.223)	(1.627)	(1.119)
Control treatment-period mean	27.743	26.767	20.986	32.488	10.198	80.207	11.791
Vehicles	5833	5833	5833	5833	5833	5833	5833
Baseline outcome control	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Randomisation-stratum FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Vehicle FE	No	No	No	No	No	No	No
Calendar-date FE	No	No	No	No	No	No	No
Joint p -value	0.023	0.020	0.021	0.564	0.192	0.600	0.381

Notes: The unit of observation is the vehicle. Each column reports treatment-arm coefficients from a regression of the vehicle's average outcome during the eight treatment weeks on treatment-arm indicators, its average outcome during the twelve pre-treatment study weeks, and randomisation-stratum fixed effects; the control arm is omitted. Vehicle-level means are computed over observed GPS vehicle-days. The estimation sample in each column includes vehicles with a non-missing outcome in both the pre-treatment and treatment periods. Columns 1–3 report idling duration, Columns 4–5 idling-spell counts, and Columns 6–7 vehicle-use margins; thresholds refer to total spell duration. The sample excludes the 25 vehicles removed before the second-stage *Reminder* versus *No-Reminder* randomisation. Regressions do not include vehicle or calendar-date fixed effects. Robust standard errors are shown in parentheses. The joint p -value tests whether all displayed treatment-arm coefficients equal zero. Significance levels are denoted by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

Table D.2: Incremental treatment effects across behavioural margins

	Idling duration (minutes)			Idle spells (count)		Vehicle use	
	Total	> 10s	> 30s	> 10s	> 30s	Moving minutes	Trips
Social information	0.471 (0.691)	0.485 (0.679)	0.493 (0.609)	-0.022 (0.758)	0.053 (0.249)	-0.387 (1.890)	-0.226 (0.472)
Payments	-0.564 (0.712)	-0.562 (0.698)	-0.557 (0.629)	0.030 (0.768)	0.064 (0.244)	-0.762 (1.940)	-0.228 (0.476)
IdleGuard	1.050* (0.609)	1.081* (0.595)	0.996* (0.532)	0.891 (0.613)	0.387* (0.207)	2.075 (1.502)	-0.188 (1.086)
Control post-period mean	27.721	26.770	20.972	32.590	10.238	80.756	10.567
Observations	701441	701441	701441	701441	701441	701441	701441
Vehicles	6428	6428	6428	6428	6428	6428	6428
Vehicle FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Calendar-date FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Joint p -value	0.293	0.251	0.212	0.549	0.291	0.499	0.888

Notes: The unit of observation is the vehicle-day. Social information compares *Private & Social Info* with *Private Info*; payments compare *Private & Social Info + Pay* with *Private & Social Info*; and *IdleGuard* compares *Private & Social Info + Pay + Reminder* with the matched *Private & Social Info + Pay + No-Reminder* arm. Regressions absorb vehicle and calendar-date fixed effects. Post covers 6 April through 30 May 2026; the pooled pre-treatment period comprises 12 study weeks observed between 17 November 2025 and 4 April 2026. Outcomes are computed over observed GPS vehicle-days. The sample excludes the 25 vehicles removed before the second-stage randomisation. Standard errors are clustered by vehicle. The joint p -value tests whether all three displayed effects equal zero. Significance levels are denoted by $*p < 0.10$, $**p < 0.05$, and $***p < 0.01$.

D.3 Alternative Outcome Definitions and Duration Decompositions

D.3.1 Excess-over-Threshold Outcomes

Appendix Tables D.3–D.4 report alternative duration outcomes that count only time beyond the relevant threshold. The primary outcomes count the full duration of qualifying spells; the excess-over-threshold outcomes instead exclude the first 10 or 30 seconds of each qualifying spell.

Table D.3: Treatment effects using excess-over-threshold idling duration

	All idling	Excess-beyond-threshold duration		Idle spells (count)		Vehicle use	
	Total minutes	> 10s	> 30s	> 10s	> 30s	Moving minutes	Trips
Private Info $\times$ Post	-1.993*** (0.737)	-1.775*** (0.647)	-1.543*** (0.576)	-1.067 (0.705)	-0.532** (0.229)	-1.321 (1.739)	-0.325 (0.559)
Private & Social Info $\times$ Post	-1.523** (0.722)	-1.286** (0.622)	-1.076** (0.542)	-1.089 (0.737)	-0.479** (0.242)	-1.707 (1.967)	-0.551 (0.448)
Private & Social Info + Pay $\times$ Post	-2.087*** (0.756)	-1.853*** (0.667)	-1.666*** (0.601)	-1.058 (0.716)	-0.415* (0.223)	-2.470 (1.793)	-0.779 (0.563)
Private & Social Info + Pay + Reminder $\times$ Post	-0.826 (0.665)	-0.769 (0.577)	-0.697 (0.512)	-0.024 (0.637)	-0.066 (0.206)	-0.470 (1.629)	-0.341 (1.123)
Private & Social Info + Pay + No-Reminder $\times$ Post	-1.876*** (0.718)	-1.701*** (0.625)	-1.499*** (0.551)	-0.915 (0.660)	-0.453** (0.222)	-2.545 (1.709)	-0.153 (0.457)
Control post-period mean	27.721	21.339	15.853	32.590	10.238	80.756	10.567
Observations	701441	701441	701441	701441	701441	701441	701441
Vehicles	6428	6428	6428	6428	6428	6428	6428
Vehicle FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Calendar-date FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Joint p -value	0.023	0.021	0.024	0.280	0.051	0.568	0.661

Notes: The unit of observation is the vehicle-day. The table repeats the main treatment-arm specification, replacing the duration outcomes in Columns 2–3 with excess-over-threshold measures that exclude the first 10 or 30 seconds of each qualifying spell. Column 1 reports total idling minutes, Columns 4–5 idling-spell counts, and Columns 6–7 vehicle-use margins. Each column reports treatment-arm-by-post coefficients from a regression absorbing vehicle and calendar-date fixed effects; the control arm is omitted. Post covers 6 April through 30 May 2026; the pooled pre-treatment period comprises 12 study weeks observed between 17 November 2025 and 4 April 2026. Outcomes are computed over observed GPS vehicle-days. The sample excludes the 25 vehicles whose incentivised choices were implemented before the second-stage *Reminder* versus *No-Reminder* randomisation. Standard errors are clustered by vehicle. The joint p -value tests whether all displayed coefficients equal zero. Significance levels are denoted by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

Table D.4: Incremental treatment effects using excess-over-threshold idling duration

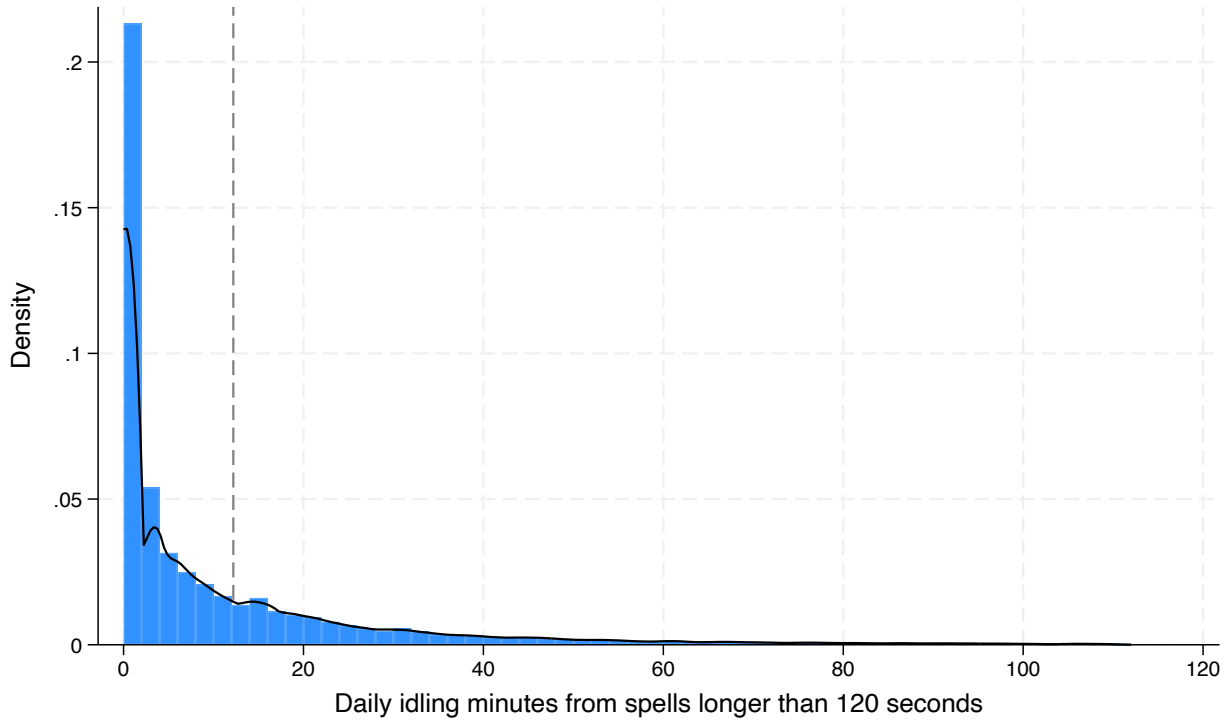
	All idling	Excess-beyond-threshold duration		Idle spells (count)		Vehicle use	
	Total minutes	> 10s	> 30s	> 10s	> 30s	Moving minutes	Trips
Social information	0.471 (0.691)	0.489 (0.600)	0.467 (0.518)	-0.022 (0.758)	0.053 (0.249)	-0.387 (1.890)	-0.226 (0.472)
Payments	-0.564 (0.712)	-0.567 (0.621)	-0.589 (0.546)	0.030 (0.768)	0.064 (0.244)	-0.762 (1.940)	-0.228 (0.476)
IdleGuard	1.050* (0.609)	0.932* (0.527)	0.802* (0.457)	0.891 (0.613)	0.387* (0.207)	2.075 (1.502)	-0.188 (1.086)
Control post-period mean	27.721	21.339	15.853	32.590	10.238	80.756	10.567
Observations	701441	701441	701441	701441	701441	701441	701441
Vehicles	6428	6428	6428	6428	6428	6428	6428
Vehicle FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Calendar-date FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Joint p -value	0.293	0.243	0.213	0.549	0.291	0.499	0.888

Notes: The unit of observation is the vehicle-day. The table repeats Appendix Table D.2, replacing the duration outcomes in Columns 2–3 with excess-over-threshold measures that exclude the first 10 or 30 seconds of each qualifying spell. Social information compares *Private & Social Info* with *Private Info*; payments compare *Private & Social Info + Pay* with *Private & Social Info*; and IdleGuard compares *Private & Social Info + Pay + Reminder* with the matched *Private & Social Info + Pay + No-Reminder* arm. Positive coefficients indicate an increase in the outcome relative to the preceding arm or matched no-reminder arm. The estimation sample, periods, outcome construction, fixed effects, and clustered standard errors follow Appendix Table D.3. The joint p -value tests whether all three displayed incremental effects equal zero. Significance levels are denoted by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

D.3.2 Duration-Bin and Idling-Context Decompositions

Figure D.4 shows the baseline distribution of daily idling minutes contributed by idling spells longer than 120 seconds. The distribution is highly right-skewed, with a large mass of low-idling vehicle-days and a long right tail.

Figure D.4: Baseline distribution of daily idling minutes from spells longer than 120 seconds



Notes: The figure reports the distribution of daily idling minutes contributed by idling spells longer than 120 seconds, computed over observed GPS vehicle-days. Bars show the histogram, the solid line shows the kernel density, and the dashed vertical line marks the sample mean. The sample uses the two clean immediate pre-main-treatment weeks, starting 23 and 30 March 2026. For readability, the figure displays observations through the 99th percentile of the baseline distribution; the underlying outcome is not winsorised. The sample excludes the 25 vehicles whose incentivised choices were implemented before the second-stage *Reminder* versus *No-Reminder* randomisation.

Table D.5: Treatment effects across idling-spell duration bins

	Idling duration bins (minutes)				
	0–10s	10–30s	30–60s	60–120s	> 120s
Private Info $\times$ Post	-0.041 (0.033)	-0.143 (0.131)	-0.159* (0.085)	-0.218** (0.104)	-1.431** (0.563)
Private & Social Info $\times$ Post	-0.055* (0.032)	-0.152 (0.135)	-0.137 (0.088)	-0.241** (0.115)	-0.937* (0.523)
Private & Social Info + Pay $\times$ Post	-0.058* (0.034)	-0.156 (0.133)	-0.105 (0.083)	-0.127 (0.099)	-1.641*** (0.596)
Private & Social Info + Pay + Reminder $\times$ Post	-0.053 (0.040)	-0.043 (0.117)	-0.034 (0.077)	-0.072 (0.092)	-0.624 (0.502)
Private & Social Info + Pay + No-Reminder $\times$ Post	-0.022 (0.030)	-0.128 (0.120)	-0.114 (0.080)	-0.176* (0.100)	-1.435*** (0.537)
Control post-period mean	0.950	5.799	3.699	4.220	13.053
Observations	701441	701441	701441	701441	701441
Vehicles	6428	6428	6428	6428	6428
Vehicle FE	Yes	Yes	Yes	Yes	Yes
Calendar-date FE	Yes	Yes	Yes	Yes	Yes
Joint p -value	0.413	0.745	0.344	0.182	0.026

Notes: The unit of observation is the vehicle-day. The columns decompose total daily idling minutes into mutually exclusive bins defined by the total duration of the underlying spell. Each column reports treatment-arm-by-post coefficients from a regression absorbing vehicle and calendar-date fixed effects; the control arm is omitted. Post covers 6 April through 30 May 2026; the pooled pre-treatment period comprises 12 study weeks observed between 17 November 2025 and 4 April 2026. Outcomes are computed over observed GPS vehicle-days. The sample excludes the 25 vehicles whose incentivised choices were implemented before the second-stage *Reminder* versus *No-Reminder* randomisation. Standard errors are clustered by vehicle. The joint p -value tests whether all displayed coefficients equal zero. Significance levels are denoted by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

Table D.6: Incremental treatment effects across idling-spell duration bins

	Idling duration bins (minutes)				
	0–10s	10–30s	30–60s	60–120s	>120s
Social information	-0.014	-0.008	0.022	-0.023	0.495
	(0.029)	(0.141)	(0.091)	(0.120)	(0.495)
Payments	-0.002	-0.004	0.033	0.114	-0.704
	(0.030)	(0.143)	(0.089)	(0.116)	(0.532)
IdleGuard	-0.031	0.085	0.081	0.104	0.811*
	(0.035)	(0.112)	(0.075)	(0.095)	(0.442)
Control post-period mean	0.950	5.799	3.699	4.220	13.053
Observations	701441	701441	701441	701441	701441
Vehicles	6428	6428	6428	6428	6428
Vehicle FE	Yes	Yes	Yes	Yes	Yes
Calendar-date FE	Yes	Yes	Yes	Yes	Yes
Joint p -value	0.769	0.899	0.671	0.488	0.144

Notes: The unit of observation is the vehicle-day. Each column reports incremental effects constructed from the corresponding treatment-arm-by-post coefficients in Appendix Table D.5. Social information compares *Private & Social Info* with *Private Info*; payments compare *Private & Social Info + Pay* with *Private & Social Info*; and *IdleGuard* compares *Private & Social Info + Pay + Reminder* with the matched *Private & Social Info + Pay + No-Reminder* arm. Positive coefficients indicate increased idling relative to the preceding arm or matched no-reminder arm. The columns decompose total daily idling minutes into mutually exclusive bins defined by the total duration of the underlying spell. The estimation sample, periods, outcome construction, fixed effects, and clustered standard errors follow Appendix Table D.5. The joint p -value tests whether all three displayed incremental effects equal zero. Significance levels are denoted by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

Table D.7: Treatment effects by idling context

	All idling minutes				Idling minutes from stops > 30s			
	Total	Warming	Traffic	Waiting	Total	Warming	Traffic	Waiting
Private Info $\times$ Post	-1.993*** (0.737)	-0.671** (0.296)	-0.702** (0.347)	-0.621* (0.341)	-1.809*** (0.654)	-0.670** (0.283)	-0.510** (0.246)	-0.629* (0.341)
Private & Social Info $\times$ Post	-1.523** (0.722)	-0.320 (0.243)	-0.492 (0.355)	-0.711** (0.342)	-1.316** (0.629)	-0.299 (0.221)	-0.309 (0.254)	-0.708** (0.341)
Private & Social Info + Pay $\times$ Post	-2.087*** (0.756)	-0.678** (0.310)	-0.572* (0.338)	-0.837** (0.367)	-1.873*** (0.673)	-0.644** (0.294)	-0.399* (0.232)	-0.830** (0.366)
Private & Social Info + Pay + Reminder $\times$ Post	-0.826 (0.665)	-0.015 (0.270)	-0.204 (0.305)	-0.607* (0.317)	-0.730 (0.582)	-0.064 (0.243)	-0.059 (0.214)	-0.607* (0.316)
Private & Social Info + Pay + No-Reminder $\times$ Post	-1.876*** (0.718)	-0.502* (0.265)	-0.555* (0.322)	-0.819** (0.338)	-1.725*** (0.631)	-0.507** (0.245)	-0.399* (0.227)	-0.819** (0.337)
Control post-period mean	27.721	5.866	13.874	7.981	20.972	4.504	8.939	7.528
Observations	701441	701441	701441	701441	701441	701441	701441	701441
Vehicles	6428	6428	6428	6428	6428	6428	6428	6428
Vehicle FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Calendar-date FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Joint p -value	0.023	0.053	0.269	0.197	0.017	0.049	0.157	0.198

Notes: The unit of observation is the vehicle-day. Columns 1–4 report total daily idling minutes and their decomposition into warming, traffic, and waiting idling; Columns 5–8 report the corresponding outcomes for stops longer than 30 seconds. Each column reports treatment-arm-by-post coefficients from a regression absorbing vehicle and calendar-date fixed effects; the control arm is omitted. Post covers 6 April through 30 May 2026. Outcomes are computed over observed GPS vehicle-days. The sample excludes the 25 vehicles whose incentivised choices were implemented before the second-stage *Reminder* versus *No-Reminder* randomisation. Standard errors are clustered by vehicle. The joint p -value tests whether all displayed coefficients equal zero. Significance levels are denoted by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

Table D.8: Incremental treatment effects by idling context

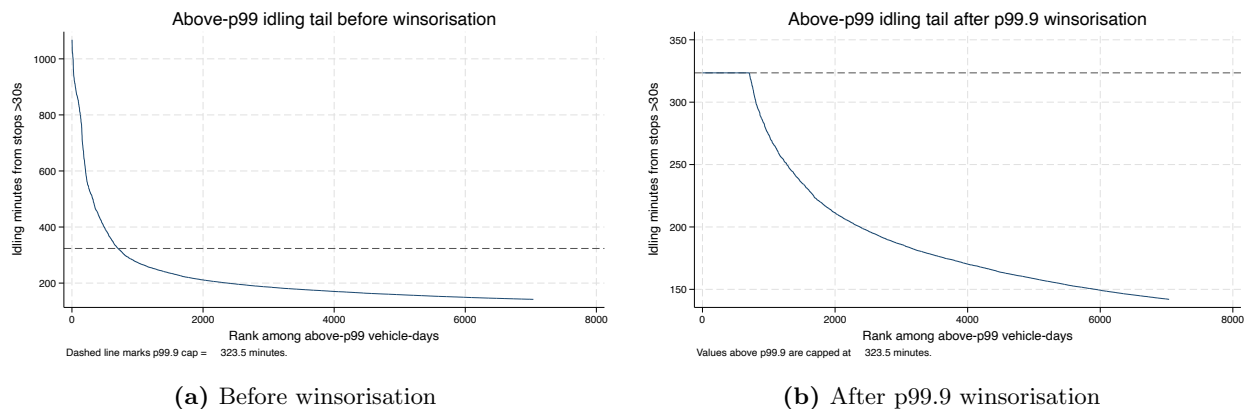
	Idling minutes from stops > 30s			
	Total	Warming	Traffic	Waiting
Social information	0.493 (0.609)	0.371 (0.264)	0.201 (0.269)	-0.079 (0.307)
Payments	-0.557 (0.629)	-0.345 (0.276)	-0.089 (0.256)	-0.123 (0.335)
IdleGuard	0.996* (0.532)	0.443* (0.245)	0.340 (0.211)	0.212 (0.275)
Control post-period mean	20.972	4.504	8.939	7.528
Observations	701441	701441	701441	701441
Vehicles	6428	6428	6428	6428
Vehicle FE	Yes	Yes	Yes	Yes
Calendar-date FE	Yes	Yes	Yes	Yes
Joint p -value	0.212	0.107	0.367	0.812

Notes: The unit of observation is the vehicle-day. The outcomes are daily idling minutes from stops longer than 30 seconds, reported in total and decomposed into warming, traffic, and waiting idling. Social information compares *Private & Social Info* with *Private Info*; payments compare *Private & Social Info + Pay* with *Private & Social Info*; and IdleGuard compares *Private & Social Info + Pay + Reminder* with the matched *Private & Social Info + Pay + No-Reminder* arm. Positive coefficients indicate increased idling relative to the preceding arm or matched no-reminder arm. The estimation sample, periods, outcome construction, fixed effects, and clustered standard errors follow Appendix Table D.7. The joint p -value tests whether all three displayed incremental effects equal zero. Significance levels are denoted by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

D.4 Winsorisation and Extreme Values

Figure D.5 shows the far-right tail of the primary outcome before and after winsorising daily idling minutes from stops longer than 30 seconds at the 99.9th percentile. The cap affects only a small number of extreme vehicle-days while leaving the remainder of the distribution unchanged.

Figure D.5: Far-right tail of daily idling minutes from stops longer than 30 seconds



Notes: The figure plots the top 1% of observed vehicle-days ranked by daily idling minutes from stops longer than 30 seconds. The 99th-percentile cut-off, computed over observed vehicle-days in study weeks 1–20, is approximately 142 minutes. Panel A reports the primary outcome before winsorisation. Panel B reports the same vehicle-days after values above the 99.9th percentile are capped at 323.5 minutes. The dashed horizontal line marks this cap.

Tables D.9 and D.10 repeat the main treatment-arm and incremental-effect specifications using outcomes winsorised at their respective 99.9th percentiles. The estimates are similar to the corresponding primary results, showing that the findings are not driven by extreme vehicle-day observations.

Table D.9: Treatment effects across behavioural margins with p99.9-winsorised outcomes

	Idling duration (minutes)			Idle spells (count)		Vehicle use	
	Total	>10s	>30s	>10s	>30s	Moving minutes	Trips
Private Info $\times$ Post	-1.729** (0.674)	-1.687** (0.659)	-1.523*** (0.587)	-1.197* (0.681)	-0.482** (0.215)	-1.262 (1.734)	-0.440 (0.441)
Private & Social Info $\times$ Post	-1.244* (0.657)	-1.199* (0.641)	-1.037* (0.559)	-1.098 (0.728)	-0.449* (0.231)	-1.669 (1.956)	-0.495 (0.383)
Private & Social Info + Pay $\times$ Post	-1.947*** (0.704)	-1.898*** (0.689)	-1.734*** (0.619)	-1.126 (0.711)	-0.427* (0.222)	-2.435 (1.786)	-0.583 (0.458)
Private & Social Info + Pay + Reminder $\times$ Post	-0.794 (0.608)	-0.744 (0.593)	-0.689 (0.526)	-0.102 (0.624)	-0.0798 (0.203)	-0.434 (1.625)	-0.431 (0.463)
Private & Social Info + Pay + No-Reminder $\times$ Post	-1.817*** (0.660)	-1.805*** (0.645)	-1.669*** (0.573)	-0.949 (0.654)	-0.477** (0.217)	-2.484 (1.702)	-0.167 (0.390)
Control post-period mean	27.560	26.620	20.818	32.576	10.231	80.680	10.414
Observations	701441	701441	701441	701441	701441	701441	701441
Vehicles	6428	6428	6428	6428	6428	6428	6428
Vehicle FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Calendar-date FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Joint p-value	0.027	0.024	0.018	0.227	0.055	0.579	0.717

Notes: The unit of observation is the vehicle-day. Each column reports treatment-arm-by-post coefficients from a regression absorbing vehicle and calendar-date fixed effects; the control arm is omitted. Outcomes correspond to the behavioural margins in the main treatment-arm specification, with each winsorised at its 99.9th percentile in the full observed vehicle-day distribution. Post equals one from 6 April through 30 May 2026; the pooled pre-treatment period comprises 12 study weeks observed between 17 November 2025 and 4 April 2026. Outcomes are computed over observed GPS vehicle-days. The sample excludes the 25 vehicles removed before the second-stage *Reminder* versus *No-Reminder* randomisation. Standard errors are clustered by vehicle. The joint p -value tests whether all displayed coefficients equal zero. Significance levels are denoted by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

Table D.10: Incremental treatment effects across behavioural margins with p99.9-winsorised outcomes

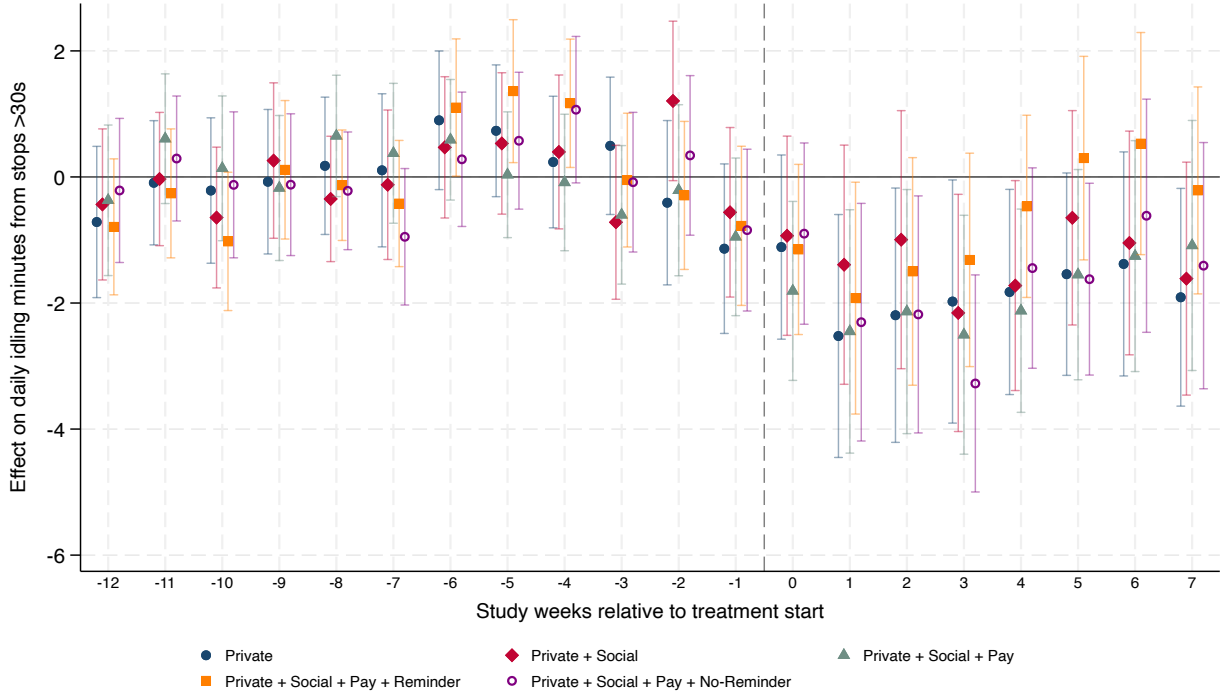
	Idling duration (minutes)			Idle spells (count)		Vehicle use	
	Total	>10s	>30s	>10s	>30s	Moving minutes	Trips
Social information	0.484 (0.657)	0.488 (0.642)	0.486 (0.562)	0.0995 (0.731)	0.0333 (0.227)	-0.407 (1.880)	-0.0552 (0.388)
Payments	-0.703 (0.687)	-0.699 (0.672)	-0.697 (0.596)	-0.0281 (0.760)	0.0219 (0.234)	-0.766 (1.929)	-0.0885 (0.407)
IdleGuard	1.023* (0.592)	1.060* (0.578)	0.980* (0.514)	0.848 (0.599)	0.397** (0.200)	2.049 (1.497)	-0.264 (0.419)
Control post-period mean	27.560	26.620	20.818	32.576	10.231	80.680	10.414
Observations	701441	701441	701441	701441	701441	701441	701441
Vehicles	6428	6428	6428	6428	6428	6428	6428
Vehicle FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Calendar-date FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Joint p-value	0.248	0.208	0.160	0.567	0.262	0.502	0.920

Notes: The unit of observation is the vehicle-day. Social information compares *Private & Social Info* with *Private Info*; payments compare *Private & Social Info + Pay* with *Private & Social Info*; and IdleGuard compares *Private & Social Info + Pay + Reminder* with the matched *Private & Social Info + Pay + No-Reminder* arm. Outcomes correspond to the behavioural margins in Appendix Table D.2, with each winsorised at its 99.9th percentile in the full observed vehicle-day distribution. The estimation sample, periods, outcome construction, fixed effects, and clustered standard errors follow Table D.9. The joint p -value tests whether all three displayed incremental effects equal zero. Significance levels are denoted by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

D.5 Additional Dynamics and Persistence

Figure D.6 extends the treatment-dynamics analysis in the main paper to include the twelve pre-treatment study weeks. Table D.11 then reports incremental treatment effects during the main intervention and the two post-treatment periods, while Figure D.7 complements these pooled estimates with week-specific treatment effects after the main intervention ended.

Figure D.6: Treatment dynamics before and during the main intervention



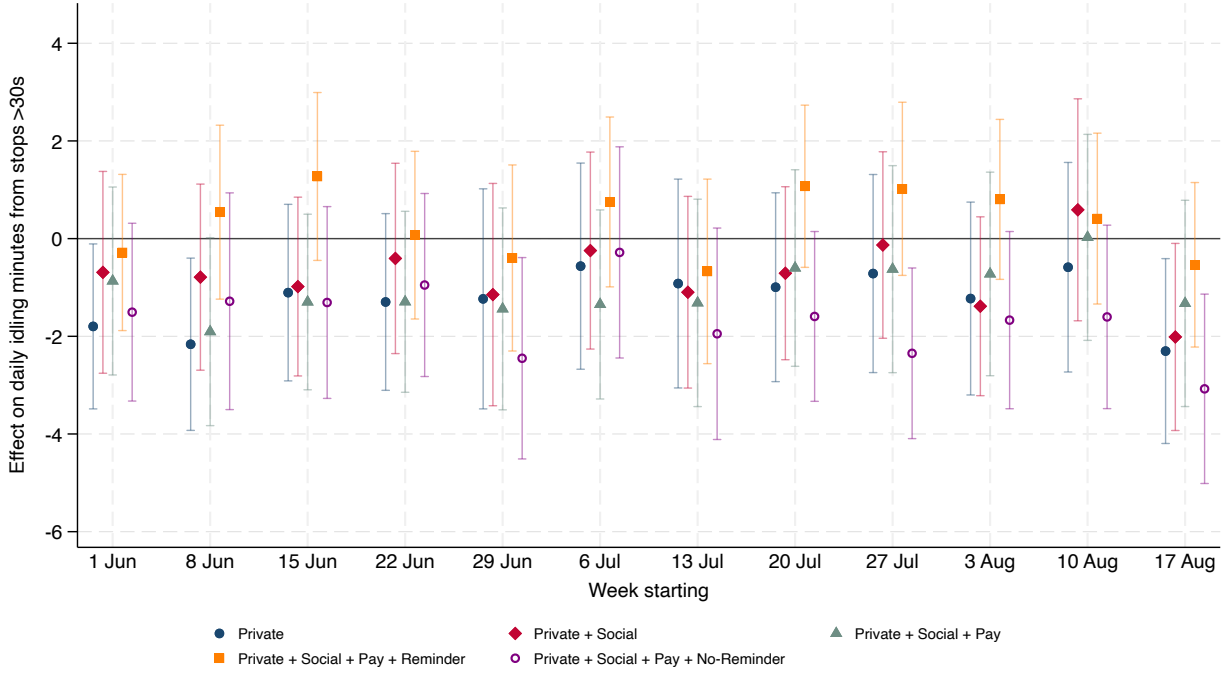
Notes: The figure reports study-week-specific treatment effects on daily idling minutes from stops longer than 30 seconds relative to the control arm. Points report treatment-arm-by-study-week coefficients, and capped lines report 95% confidence intervals. Estimates are normalised relative to the average treatment-control difference across the twelve pre-treatment study weeks, so the displayed pre-treatment coefficients average to zero within each treatment arm. Event time zero denotes the first treatment week, beginning 6 April 2026; the dashed vertical line marks treatment launch. Study weeks index the twelve pre-treatment analysis weeks observed between 17 November 2025 and 4 April 2026 and the eight main-treatment weeks; excluded calendar periods are omitted from the study-week sequence. Estimates are horizontally offset within each study week for readability. Regressions absorb vehicle and calendar-date fixed effects. Outcomes are computed over observed GPS vehicle-days, and standard errors are clustered by vehicle. The sample excludes the 25 vehicles removed before the second-stage *Reminder* versus *No-Reminder* randomisation. A joint test that treatment-control differences are constant across the twelve pre-treatment study weeks yields $p = 0.618$.

Table D.11: Incremental treatment effects during and after the main intervention

	Idling minutes from stops >30s		
	Treatment weeks	First 2 post weeks	Remaining post weeks
Social information	0.541 (0.616)	1.269 (0.908)	0.357 (0.803)
Payments	-0.580 (0.629)	-0.579 (0.988)	-0.235 (0.795)
IdleGuard	0.948* (0.531)	1.541* (0.869)	2.106*** (0.682)
Control period mean	20.972	22.032	22.070
Observations	1,076,021	1,076,021	1,076,021
Vehicles	6,429	6,429	6,429
Vehicle FE	Yes	Yes	Yes
Calendar-date FE	Yes	Yes	Yes
Joint p -value	0.229	0.160	0.021

Notes: The unit of observation is the vehicle-day, and the outcome is daily idling minutes from stops longer than 30 seconds. Social information compares *Private & Social Info* with *Private Info*; payments compare *Private & Social Info + Pay* with *Private & Social Info*; and *IdleGuard* compares *Private & Social Info + Pay + Reminder* with the matched *Private & Social Info + Pay + No-Reminder* arm. The pooled pre-treatment period comprises study weeks 1–12. The main intervention comprises weeks 13–20, followed by weeks 21–22 and 23–32. The social-information comparison pools Early-Off and Late-On *Private Info* vehicles in both post-treatment periods. Regressions absorb vehicle and calendar-date fixed effects. Outcomes are computed over observed GPS vehicle-days, and standard errors are clustered by vehicle. The joint p -value tests whether all three displayed effects in the corresponding period equal zero. Significance levels are denoted by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

Figure D.7: Post-treatment dynamics in idling minutes from stops longer than 30 seconds



Notes: The figure reports week-specific treatment effects on daily idling minutes from stops longer than 30 seconds relative to the control arm. Points report treatment-arm-by-week coefficients, and capped lines report 95% confidence intervals. Study weeks 1–12 form the pooled reference period. The displayed weeks begin on 1, 8, 15, 22, and 29 June; 6, 13, 20, and 27 July; and 3, 10, and 17 August 2026. Estimates are horizontally offset within each week for readability. Regressions absorb vehicle and calendar-date fixed effects. Outcomes are computed over observed GPS vehicle-days, and standard errors are clustered by vehicle. The sample excludes the 25 vehicles removed before the second-stage *Reminder* versus *No-Reminder* randomisation. The first two *Private Info* estimates pool Early-Off and Late-On vehicles.

At the end of the eight-week main intervention, I randomised half of the *Private Info* group to continue receiving the same messages for two additional weeks. Table D.12 compares these Late-On vehicles with the Early-Off vehicles whose messages stopped after 30 May 2026. A negative coefficient would indicate that continuing the messages reduced idling relative to stopping them. The estimate is close to zero but imprecise, providing no detectable evidence that two additional weeks of messages reduced idling while leaving economically meaningful effects possible.

Table D.12: Effect of continuing Private Info messages after the main intervention

	Idling duration (minutes)		
	>30s	>10s	Beyond first 10s
Late-On $\times$ continuation window	0.048	-0.222	-0.019
	(0.881)	(1.036)	(0.880)
Early-Off continuation-period mean	22.765	28.614	23.078
Late-On continuation-period mean	21.408	26.870	21.666
Observations	20,222	20,222	20,222
Vehicles	867	867	867
Early-Off vehicles	428	428	428
Late-On vehicles	439	439	439
Vehicle FE	Yes	Yes	Yes
Calendar-date FE	Yes	Yes	Yes

Notes: The unit of observation is the vehicle-day, and the sample is restricted to *Private Info* vehicles. Columns 1–2 report daily idling minutes from spells longer than 30 and 10 seconds, respectively; Column 3 excludes the first 10 seconds of each spell longer than 10 seconds. Early-Off vehicles stopped receiving messages after 30 May 2026, while Late-On vehicles continued through 13 June. The sample covers study weeks 19–22: weeks 19–20 precede the continuation split, and weeks 21–22 form the continuation period. Each coefficient reports the change for Late-On relative to Early-Off vehicles during the continuation period. Regressions absorb vehicle and calendar-date fixed effects. Outcomes are computed over observed GPS vehicle-days, and standard errors are clustered by vehicle. Significance levels are denoted by $*p < 0.10$, $**p < 0.05$, and $***p < 0.01$.

Table D.13 provides a complementary stacked switch-off test. It compares each *Private Info* switch-off cohort with the control arm around the date when its messages stopped. A positive coefficient would indicate that idling rose after switch-off. The estimate is close to zero but imprecise, providing no detectable evidence of rebound while leaving economically meaningful decay possible.

Table D.13: Private Info decay after message switch-off

	Idling duration (minutes)		
	>30s	>10s	Beyond first 10s
Private Info $\times$ Post-switch-off	0.164	0.318	0.191
	(0.525)	(0.604)	(0.523)
Control post-switch-off mean	21.219	26.665	21.505
Observations	59,916	59,916	59,916
Vehicles	1,727	1,727	1,727
Early-Off vehicles	429	429	429
Late-On vehicles	437	437	437
Vehicle-by-stack FE	Yes	Yes	Yes
Calendar-date-by-stack FE	Yes	Yes	Yes

Notes: The unit of observation is the vehicle-day. Columns 1–2 report daily idling minutes from spells longer than 30 and 10 seconds, respectively; Column 3 excludes the first 10 seconds of each spell longer than 10 seconds. Stack 1 compares Early-Off *Private Info* vehicles with the control arm over study weeks 19–22, with messages stopping after 30 May 2026. Stack 2 compares Late-On vehicles with the control arm over weeks 21–24, with messages stopping after 13 June. Each coefficient reports the post-switch-off change for *Private Info* relative to the control arm within the corresponding stack. Regressions absorb vehicle-by-stack and calendar-date-by-stack fixed effects. Outcomes are computed over observed GPS vehicle-days, and standard errors are clustered by vehicle. In Stack 1, control-arm safety messages also ended after 30 May; interpretation assumes that stopping those messages did not affect idling. Significance levels are denoted by $*p < 0.10$, $**p < 0.05$, and $***p < 0.01$.

Table D.14 reports the Early-Off *Private Info* effect during the active campaign and after switch-off. The pooled twelve-week post-switch-off estimate provides the persistence input used in the cost–benefit calculation.

Table D.14: Private Info effects during and after an eight-week campaign

	Idling duration (minutes)		
	>30s	>10s	Beyond first 10s
Active campaign, weeks 13–20	-2.394*** (0.852)	-2.480*** (0.926)	-2.332*** (0.843)
First two weeks after switch-off, weeks 21–22	-2.638*** (0.915)	-2.488** (1.041)	-2.467*** (0.903)
Remaining ten weeks after switch-off, weeks 23–32	-1.926* (1.084)	-2.131* (1.171)	-1.873* (1.061)
All twelve weeks after switch-off, weeks 21–32	-2.050** (0.989)	-2.193** (1.069)	-1.976** (0.967)
Control mean, active campaign	20.972	26.770	21.339
Control mean, first two weeks after	22.032	27.654	22.298
Control mean, remaining ten weeks	22.070	27.571	22.460
Control mean, all twelve weeks after	22.063	27.586	22.432
Observations	251,288	251,288	251,288
Vehicles	1,512	1,512	1,512
Early-Off vehicles	503	503	503
Vehicle FE	Yes	Yes	Yes
Calendar-date FE	Yes	Yes	Yes

Notes: The unit of observation is the vehicle-day. The sample comprises Early-Off *Private Info* and control-arm vehicles. Columns 1–2 report daily idling minutes from spells longer than 30 and 10 seconds, respectively; Column 3 excludes the first 10 seconds of each spell longer than 10 seconds. The active-campaign period comprises study weeks 13–20, followed by weeks 21–22 and 23–32 after switch-off. The pooled post-switch-off estimate combines weeks 21–32 in a separate specification. Each coefficient reports the Early-Off *Private Info* effect relative to the control arm, using weeks 1–12 as the pooled reference period. Regressions absorb vehicle and calendar-date fixed effects. Outcomes are computed over observed GPS vehicle-days, and standard errors are clustered by vehicle. The pooled twelve-week estimate provides the persistence input for the cost–benefit calculation. Significance levels are denoted by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

D.6 Fuel-Expenditure Effects and Robustness

Appendix Tables D.15 and D.16 report fuel-expenditure effects for the estimands that do not exhibit differential valid-outcome availability in Appendix Tables C.7 and C.8. Table D.15 reports the primary estimates, while Table D.16 assesses robustness to winsorising the outcome at the 99.9th percentile.

Table D.15: Effects on daily fuel expenditure

	Daily fuel expenditure (Rs.)
<i>Panel A. Treatment effects</i>	
Private Info	-872.946** (383.668)
Private & Social Info	-781.747* (407.416)
Control mean	3730.432
Vehicle-weeks	2,474
Vehicles	880
Joint <i>p</i> -value	0.067
<i>Panel B. Incremental effects</i>	
Social information	91.199 (312.378)
IdleGuard	3.946 (247.572)
No-Reminder mean	3073.208
IdleGuard-comparison vehicle-weeks	2,750
IdleGuard-comparison vehicles	825
Survey-week FE	Yes
Assigned-day FE	Yes
Randomisation-stratum FE	Yes

Notes: The unit of observation is the vehicle-week. The outcome is cleaned fuel expenditure, in Pakistani rupees, for the randomly assigned survey day. Panel A reports effects for *Private Info* and *Private & Social Info* relative to the control arm. Panel B reports the incremental effects of social information and *IdleGuard*. Social information compares *Private & Social Info* with *Private Info*; *IdleGuard* compares *Private & Social Info + Pay + Reminder* with the matched *Private & Social Info + Pay + No-Reminder* arm. Payment-containing treatment effects relative to the control arm and the incremental effect of payments are omitted because assignment changes valid-outcome availability for those comparisons. Regressions include survey-week, assigned-day, and original randomisation-stratum fixed effects. Standard errors are clustered by vehicle. Significance levels are denoted by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

Table D.16: Effects on daily fuel expenditure with a p99.9-winsorised outcome

	Daily fuel expenditure (Rs.)
<i>Panel A. Treatment effects</i>	
Private Info	-707.158** (320.061)
Private & Social Info	-616.264* (348.422)
Control mean	3559.409
Vehicle-weeks	2,474
Vehicles	880
Joint p -value	0.072
<i>Panel B. Incremental effects</i>	
Social information	90.894 (311.868)
IdleGuard	24.699 (240.483)
No-Reminder mean	3045.635
IdleGuard-comparison vehicle-weeks	2,750
IdleGuard-comparison vehicles	825
P99.9 cap (Rs.)	50,000
Survey-week FE	Yes
Assigned-day FE	Yes
Randomisation-stratum FE	Yes

Notes: The outcome is cleaned fuel expenditure for the randomly assigned survey day, winsorised at the 99.9th percentile of all non-missing observations in the full six-arm fuel-survey panel. Values above the reported cap are set equal to the cap; no observations are dropped. The retained estimands, samples, regression specifications, fixed effects, and clustered standard errors follow Appendix Table D.15. Significance levels are denoted by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

E Additional Mechanism and Heterogeneity Results

E.1 Belief Correction and Decision Rules

Table E.1 summarises baseline beliefs about the minimum stop duration for which shutting off the engine saves fuel or reduces environmental harm. Tables E.3 and E.4 then examine whether treatment responsiveness varies with drivers' stop-duration decision rules and baseline fuel-saving threshold beliefs.

Table E.1: Baseline beliefs about when turning off the engine saves fuel and reduces environmental harm

Belief domain	N	Threshold (minutes)		Share of respondents (%)	
		Mean	Median	≤ 10 seconds	> 30 seconds
Fuel savings	6,466	6.60	5.00	3.1	93.6
Environmental harm	6,452	5.01	2.00	8.4	84.6

Notes: The unit of observation is the vehicle. The sample comprises the 6,596 vehicles in the final six-arm randomised sample, excluding the 25 vehicles removed before the second-stage *Reminder* versus *No-Reminder* randomisation. The table reports baseline beliefs about the minimum stop duration for which turning off the engine saves fuel or reduces environmental harm. Thresholds are reported in minutes. The final two columns report the shares stating a threshold of 10 seconds or less and greater than 30 seconds.

Table E.2: *Private Info* effects by baseline beliefs

	Idling minutes from stops > 30 s		
	Warm-up engine life	Fuel-saving threshold	Threshold + confidence
Less mistaken	-1.660 (1.084)	-1.204 (0.803)	-1.741** (0.694)
More mistaken	-1.934*** (0.714)	-2.310*** (0.791)	-2.333** (1.117)
Difference: more minus less	-0.275 (1.099)	-1.106 (0.898)	-0.592 (1.124)
Control post-period mean, less mistaken	19.600	20.858	21.067
Control post-period mean, more mistaken	21.479	21.097	20.472
Observations	213670	216447	216447
Vehicles	1969	1989	1989
Vehicle FE	Yes	Yes	Yes
Calendar-date FE	Yes	Yes	Yes

Notes: The unit of observation is the vehicle-day, and the outcome is daily idling minutes from stops longer than 30 seconds. Each column reports the effect of *Private Info* relative to the control arm separately for drivers with less- and more-mistaken baseline beliefs. In Column 1, more mistaken denotes believing that warming up the engine increases engine life; the comparison group comprises drivers who reported that warming up decreases engine life or has no effect, with unsure responses excluded. In Column 2, more mistaken denotes reporting a fuel-saving shutoff threshold of at least 300 seconds; Column 3 additionally requires being confident or very confident in that belief. Difference rows test whether the *Private Info* effect differs between more- and less-mistaken drivers. Regressions absorb vehicle and calendar-date fixed effects. Outcomes are computed over observed GPS vehicle-days, and standard errors are clustered by vehicle. Significance levels are denoted by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

Table E.3: Treatment effects by stop-duration decision rule

	Idling minutes from stops >30s
<i>Panel A. Stop duration not very important</i>	
Private Info	-1.979*** (0.702)
Social information	0.450 (0.694)
Payments	-0.958 (0.716)
IdleGuard	0.931 (0.601)
<i>Panel B. Stop duration very important</i>	
Private Info	-1.181 (1.007)
Social information	0.656 (1.259)
Payments	0.991 (1.292)
IdleGuard	1.253 (1.140)
<i>Panel C. Very important minus not very important</i>	
Private Info	0.798 (1.026)
Social information	0.206 (1.438)
Payments	1.949 (1.477)
IdleGuard	0.322 (1.289)
Control post-period mean, not very important	20.868
Control post-period mean, very important	21.420
Observations	701441
Vehicles	6428
Vehicle FE	Yes
Calendar-date FE	Yes

Notes: The unit of observation is the vehicle-day, and the outcome is daily idling minutes from stops longer than 30 seconds. The table reports the *Private Info* effect relative to the control arm and the incremental effects of social information, payments, and *IdleGuard*. The heterogeneity indicator equals one if the respondent reports that expected stop duration is very important for deciding whether to keep the engine on or turn it off. The comparison group comprises respondents with valid answers who report that duration is important, not important, or not at all important. Difference rows test whether each effect varies with the decision rule. Regressions absorb vehicle and calendar-date fixed effects. Outcomes are computed over observed GPS vehicle-days, and standard errors are clustered by vehicle. Respondents with missing or invalid baseline responses are excluded. Significance levels are denoted by $*p < 0.10$, $**p < 0.05$, and $***p < 0.01$.

Table E.4: Treatment effects by long fuel-saving threshold belief

	Idling minutes from stops >30s
<i>Panel A. Fuel-saving threshold below 300 seconds</i>	
Private Info	-1.207 (0.803)
Social information	-0.856 (0.949)
Payments	-0.625 (1.087)
IdleGuard	2.285*** (0.746)
<i>Panel B. Fuel-saving threshold at least 300 seconds</i>	
Private Info	-2.313*** (0.791)
Social information	1.661** (0.798)
Payments	-0.623 (0.758)
IdleGuard	-0.0170 (0.761)
<i>Panel C. Long threshold minus short threshold</i>	
Private Info	-1.106 (0.898)
Social information	2.517** (1.240)
Payments	0.00139 (1.325)
IdleGuard	-2.302** (1.066)
Control post-period mean, threshold below 300s	20.858
Control post-period mean, threshold at least 300s	21.097
Observations	687912
Vehicles	6303
Vehicle FE	Yes
Calendar-date FE	Yes

Notes: The unit of observation is the vehicle-day, and the outcome is daily idling minutes from stops longer than 30 seconds. The table reports the *Private Info* effect relative to the control arm and the incremental effects of social information, payments, and *IdleGuard*. A long fuel-saving threshold is defined as reporting that the engine must be turned off for at least 300 seconds to save fuel. The comparison group comprises respondents with valid thresholds below 300 seconds. Difference rows test whether each effect varies with the baseline threshold belief. Regressions absorb vehicle and calendar-date fixed effects. Outcomes are computed over observed GPS vehicle-days, and standard errors are clustered by vehicle. Significance levels are denoted by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

E.2 Social-Information Mechanisms

Table E.5 examines whether social-information effects vary with baseline environmental-harm misperceptions. Table E.6 then decomposes the incremental effect of social information across warming, traffic, and waiting idling.

Table E.5: Social-information effects by environmental-harm misperception

	Idling minutes from stops >30s	
	Threshold at least 300s	Threshold at least 300s + confidence
<i>Panel A. Less mistaken</i>		
<i>Private Info</i>	-1.841** (0.805)	-1.737** (0.689)
<i>Private & Social Info</i>	-2.453*** (0.774)	-1.555** (0.667)
Social information	-0.613 (0.884)	0.182 (0.674)
<i>Panel B. More mistaken</i>		
<i>Private Info</i>	-1.826** (0.780)	-2.562** (1.229)
<i>Private & Social Info</i>	0.103 (0.731)	0.268 (0.950)
Social information	1.930** (0.823)	2.830** (1.395)
<i>Panel C. Difference: more minus less</i>		
<i>Private Info</i>	0.014 (0.888)	-0.826 (1.233)
<i>Private & Social Info</i>	2.557*** (0.817)	1.823* (0.938)
Social information	2.542** (1.207)	2.648* (1.549)
Control post-period mean, less mistaken	20.244	20.747
Control post-period mean, more mistaken	22.048	23.363
Observations	323763	323763
Vehicles	2964	2964
Vehicle FE	Yes	Yes
Calendar-date FE	Yes	Yes

Notes: The unit of observation is the vehicle-day, and the outcome is daily idling minutes from stops longer than 30 seconds. Column 1 defines more mistaken as reporting an environmental-harm shutoff threshold of at least 300 seconds; Column 2 additionally requires being confident or very confident in that belief. Panels A and B report effects for less- and more-mistaken drivers, respectively, and Panel C reports differences between them. Within each panel, rows report the effects of *Private Info* and *Private & Social Info* relative to the control arm and the incremental effect of social information. Regressions absorb vehicle and calendar-date fixed effects. Post covers 6 April through 30 May 2026. Outcomes are computed over observed GPS vehicle-days, and standard errors are clustered by vehicle. The sample excludes the 25 vehicles removed before the second-stage *Reminder* versus *No-Reminder* randomisation. Significance levels are denoted by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

Table E.6: Social-information effects by environmental-harm threshold belief and idling context

	Total	Warming	Traffic	Waiting
Social information: threshold below 300s	-0.613 (0.884)	0.141 (0.415)	-0.280 (0.407)	-0.474 (0.425)
Social information: threshold at least 300s	1.931** (0.822)	0.664** (0.288)	0.796** (0.331)	0.471 (0.453)
At least 300s minus below 300s	2.544** (1.207)	0.523 (0.505)	1.076** (0.525)	0.945 (0.621)
Control post-period mean, threshold below 300s	20.244	4.674	8.686	6.884
Control post-period mean, threshold at least 300s	22.048	4.256	9.351	8.442
Observations	685895	685895	685895	685895
Vehicles	6286	6286	6286	6286
Vehicle FE	Yes	Yes	Yes	Yes
Calendar-date FE	Yes	Yes	Yes	Yes

Notes: The unit of observation is the vehicle-day. Outcomes are daily idling minutes from stops longer than 30 seconds, reported in total and decomposed into warming, traffic, and waiting idling. The table reports the incremental effect of social information, comparing *Private & Social Info* with *Private Info*, separately by whether the respondent reports an environmental-harm shutoff threshold of at least 300 seconds. The comparison group comprises respondents with valid thresholds below 300 seconds. Difference rows test whether the social-information effect varies with the threshold belief. Regressions absorb vehicle and calendar-date fixed effects. Outcomes are computed over observed GPS vehicle-days, and standard errors are clustered by vehicle. Significance levels are denoted by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

Table E.7: Effects of social information on endline environmental beliefs

	Endline environmental beliefs	
	Believes warm-up harms environment	Environmental-harm shutoff threshold ≤ 30 seconds
Social information	-0.031 (0.024)	-0.010 (0.019)
<i>Private Info</i> mean	0.451	0.169
Observations	1570	1552
Baseline outcome control	Yes	Yes

Notes: The unit of observation is the vehicle. The sample comprises vehicles assigned to *Private Info* or *Private & Social Info* with a usable endline outcome and corresponding baseline belief. Column 1 equals one if the respondent believes that warming up the engine harms the environment; Column 2 equals one if the respondent reports an environmental-harm shutoff threshold of 30 seconds or less. Each column reports the incremental effect of social information, comparing *Private & Social Info* with *Private Info*, and adjusts for the corresponding baseline belief. Endline availability does not differ significantly between the two groups for either outcome. Robust standard errors are shown in parentheses. Significance levels are denoted by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

Table E.8: Endline diagnostics for social-information mechanisms

	Correct recall	Low action relevance	Too much information	More likely to follow	Chooses messages
Social information	-0.010 (0.007)	-0.018 (0.032)	0.009 (0.016)	0.047 (0.038)	0.048 (0.031)
<i>Private Info</i> mean	0.028	0.255	0.043	0.531	0.626
Observations	1584	740	740	696	951

Notes: The unit of observation is the vehicle. Each column reports the incremental effect of social information, comparing *Private & Social Info* with *Private Info*, on the indicated endline outcome. Correct 10-second recall equals one if the respondent correctly recalls the private-cost shutoff threshold; respondents who report not remembering or not knowing the threshold are coded as zero. Low action relevance equals one if the respondent describes the messages as not requiring personal action, containing too much information, being too general to act on, or as telling them what to do. More likely to follow and chooses messages over Rs. 250 are indicators for the corresponding survey responses. Columns 2–4 are observed among respondents routed to the message-reader follow-up questions; Column 5 is observed among respondents eligible for the incentivised continued-message choice. Robust standard errors are shown in parentheses. Significance levels are denoted by $*p < 0.10$, $**p < 0.05$, and $***p < 0.01$.

Table E.9: Social-information effects on message credibility by baseline environmental prior

	Low credibility
<i>Panel A. Baseline environmental-harm threshold below 300 seconds</i>	
Social information	-0.046*** (0.016)
<i>Panel B. Baseline environmental-harm threshold at least 300 seconds</i>	
Social information	-0.001 (0.017)
<i>Panel C. At least 300 seconds minus below 300 seconds</i>	
Social information	0.045* (0.024)
<i>Private Info</i> mean, threshold below 300s	0.056
<i>Private Info</i> mean, threshold at least 300s	0.022
Observations	736

Notes: The unit of observation is the vehicle. Low credibility equals one if the respondent reports that the treatment messages were not very believable or not believable at all. The outcome is observed among respondents who reported reading all, most, or some of the treatment messages. Panels A and B report the incremental effect of social information, comparing *Private & Social Info* with *Private Info*, for drivers whose baseline environmental-harm shutoff threshold was below and at least 300 seconds, respectively. Panel C reports the corresponding at-least-300-seconds-minus-below-300-seconds difference in the social-information effect. Robust standard errors are shown in parentheses. Significance levels are denoted by $*p < 0.10$, $**p < 0.05$, and $***p < 0.01$.

E.3 Payment Mechanisms

Table E.10 assesses whether pooling the two paid arms without an active reminder improves precision for estimating the incremental effect of payments.

Table E.10: Pooling diagnostic for payment effects across behavioural margins

	Idling duration (minutes)			Idle spells (count)		Vehicle use	
	Total	> 10s	> 30s	> 10s	> 30s	Moving minutes	Trips
Payments	-0.564 (0.712)	-0.562 (0.698)	-0.557 (0.629)	0.030 (0.768)	0.064 (0.244)	-0.762 (1.940)	-0.228 (0.476)
Pooled paid no-reminder arms	-1.981*** (0.647)	-1.941*** (0.628)	-1.799*** (0.571)	-0.986* (0.595)	-0.434** (0.192)	-2.506 (1.537)	-0.465 (0.452)
Incremental effect of pooled payments	-0.458 (0.595)	-0.474 (0.583)	-0.483 (0.519)	0.101 (0.656)	0.045 (0.216)	-0.798 (1.707)	0.087 (0.338)
<i>p</i> -value, paid-arm effects equal	0.765	0.800	0.814	0.837	0.865	0.965	0.196
Control post-period mean	27.721	26.770	20.972	32.590	10.238	80.756	10.567
<i>Private & Social Info</i> post-period mean	28.742	27.808	21.845	33.730	10.918	84.082	9.442
Pooled paid no-reminder arms post-period mean	27.054	26.128	20.434	32.039	10.169	80.059	9.913
Observations	434,844	434,844	434,844	434,844	434,844	434,844	434,844
Vehicles	3,966	3,966	3,966	3,966	3,966	3,966	3,966
Vehicle FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Calendar-date FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes: The unit of observation is the vehicle-day. The table assesses whether pooling the two paid arms without an active reminder improves precision for the incremental effect of payments. Payments compare *Private & Social Info + Pay* with *Private & Social Info*; pooled payments compare the pooled *Private & Social Info + Pay* and *Private & Social Info + Pay + No-Reminder* arms with *Private & Social Info*. The reminder arm is excluded because activating *IdleGuard* changes the attention and self-monitoring environment. The reported *p*-value tests whether the effects of the two paid arms are equal. Regressions absorb vehicle and calendar-date fixed effects. Post equals one from 6 April through 30 May 2026; the pooled pre-treatment period comprises 12 study weeks observed between 17 November 2025 and 4 April 2026. Outcomes are computed over observed GPS vehicle-days, and standard errors are clustered by vehicle. Significance levels are denoted by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

E.4 *IdleGuard* Beliefs and Baseline Behaviour

Tables E.11–E.13 examine whether the incremental effect of *IdleGuard* varies with baseline beliefs and warming behaviour and how any differences are distributed across warming, traffic, and waiting idling.

Table E.11: *IdleGuard* effects by strong fuel-saving threshold misperception and idling context

	Total	Warming	Traffic	Waiting
Incremental effect of <i>IdleGuard</i> : no strong misperception	2.154*** (0.737)	1.005*** (0.369)	0.430 (0.290)	0.719* (0.412)
Incremental effect of <i>IdleGuard</i> : strong misperception	0.0547 (0.771)	-0.0611 (0.335)	0.278 (0.309)	-0.162 (0.374)
Difference: strong minus no strong misperception	-2.100** (1.066)	-1.066** (0.499)	-0.152 (0.424)	-0.882 (0.556)
Control post-period mean: no strong misperception	20.872	4.755	8.521	7.597
Control post-period mean: strong misperception	21.091	4.296	9.292	7.502
Observations	687912	687912	687912	687912
Vehicles	6303	6303	6303	6303
Vehicle FE	Yes	Yes	Yes	Yes
Calendar-date FE	Yes	Yes	Yes	Yes

Notes: The unit of observation is the vehicle-day. Outcomes are daily idling minutes from stops longer than 30 seconds, reported in total and decomposed into warming, traffic, and waiting idling. The table reports the incremental effect of *IdleGuard*, comparing *Private & Social Info + Pay + Reminder* with the matched *Private & Social Info + Pay + No-Reminder* arm, separately by strong fuel-saving threshold misperception. Strong fuel-saving threshold misperception is defined as reporting that the engine must be turned off for at least 300 seconds to save fuel and being confident or very confident in that answer. The comparison group comprises respondents with valid fuel-saving threshold and confidence responses who do not satisfy this definition. Difference rows test whether the *IdleGuard* effect varies with strong fuel-saving threshold misperception. Regressions absorb vehicle and calendar-date fixed effects. Outcomes are computed over observed GPS vehicle-days, and standard errors are clustered by vehicle. Significance levels are denoted by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

Table E.12: *IdleGuard* effects by warm-up engine-life belief and idling context

	Idling minutes from stops >30s			
	Total	Warming	Traffic	Waiting
Incremental effect of <i>IdleGuard</i> : no warm-up belief	0.549 (1.460)	1.068 (0.826)	-0.134 (0.493)	-0.384 (0.747)
Incremental effect of <i>IdleGuard</i> : warm-up belief	0.893 (0.548)	0.205 (0.219)	0.392* (0.235)	0.296 (0.281)
Difference: belief minus no belief	0.344 (1.559)	-0.863 (0.855)	0.526 (0.546)	0.681 (0.797)
No-reminder post-period mean: no belief	20.934	4.797	8.829	7.308
No-reminder post-period mean: belief	19.640	4.276	8.625	6.740
Observations	247734	247734	247734	247734
Vehicles	2139	2139	2139	2139
Vehicle FE	Yes	Yes	Yes	Yes
Calendar-date FE	Yes	Yes	Yes	Yes

Notes: The unit of observation is the vehicle-day. Outcomes are daily idling minutes from stops longer than 30 seconds, reported in total and decomposed into warming, traffic, and waiting idling. The table reports the incremental effect of *IdleGuard*, comparing *Private & Social Info + Pay + Reminder* with the matched *Private & Social Info + Pay + No-Reminder* arm, separately by whether the respondent reported at baseline that warming up the engine before driving increases engine life. The comparison group comprises respondents who reported that warming up decreases engine life or has no effect; unsure responses are excluded. Difference rows test whether the *IdleGuard* effect varies with the warm-up engine-life belief. Regressions absorb vehicle and calendar-date fixed effects. Outcomes are computed over observed GPS vehicle-days, and standard errors are clustered by vehicle. Significance levels are denoted by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

Table E.13: *IdleGuard* effects by baseline warming intensity and idling context

	Idling minutes from stops >30s			
	Total	Warming	Traffic	Waiting
Incremental effect of <i>IdleGuard</i> : low baseline warming	-0.193 (0.513)	0.142 (0.242)	-0.109 (0.221)	-0.226 (0.254)
Incremental effect of <i>IdleGuard</i> : high baseline warming	1.999** (0.915)	0.747* (0.402)	0.698** (0.351)	0.554 (0.472)
Difference: high minus low baseline warming	2.191** (1.049)	0.605 (0.470)	0.807* (0.414)	0.780 (0.536)
No-reminder post-period mean: low baseline warming	12.736	2.068	6.496	4.172
No-reminder post-period mean: high baseline warming	27.551	6.838	10.843	9.870
Observations	254782	254782	254782	254782
Vehicles	2200	2200	2200	2200
Vehicle FE	Yes	Yes	Yes	Yes
Calendar-date FE	Yes	Yes	Yes	Yes

Notes: The unit of observation is the vehicle-day. Outcomes are daily idling minutes from stops longer than 30 seconds, reported in total and decomposed into warming, traffic, and waiting idling. The table reports the incremental effect of *IdleGuard*, comparing *Private & Social Info + Pay + Reminder* with the matched *Private & Social Info + Pay + No-Reminder* arm, separately by whether the vehicle had above-median baseline warming idling. Baseline warming idling is the vehicle-level average of warming idling minutes from stops longer than 30 seconds over the two clean immediate pre-main-treatment GPS weeks. Low baseline warming denotes vehicles at or below the median, and high baseline warming denotes vehicles above the median. Difference rows test whether the *IdleGuard* effect varies with baseline warming intensity. Regressions absorb vehicle and calendar-date fixed effects. Outcomes are computed over observed GPS vehicle-days, and standard errors are clustered by vehicle. Significance levels are denoted by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

E.5 *IdleGuard* Threshold-Crossing and Long-Spell Decompositions

Tables E.14–E.18 examine where the incremental *IdleGuard* effect arises and whether the threshold-crossing pattern persists after treatment ends. They assess whether more spells cross the 30-second reminder threshold, how the resulting duration effect is distributed across portions and terminal durations of spells, and whether the increase in very long spells reflects entry into the long-spell category or continued idling beyond 120 seconds.

Table E.14: *IdleGuard* effects on spells crossing the 30-second threshold

	Daily number of idling spells > 30s			
	All	Traffic-like	Waiting-like	Warming-like
Reminder arm $\times$ Post	-0.066 (0.206)	0.087 (0.155)	-0.061 (0.039)	-0.092* (0.049)
No-Reminder arm $\times$ Post	-0.453** (0.222)	-0.252 (0.163)	-0.098** (0.046)	-0.104** (0.050)
<i>IdleGuard</i>	0.387* (0.207)	0.338** (0.152)	0.037 (0.043)	0.012 (0.048)
No-Reminder post-period mean	9.893	6.347	1.719	1.826
Observations	372,907	372,907	372,907	372,907
Vehicles	3,412	3,412	3,412	3,412
Vehicle FE	Yes	Yes	Yes	Yes
Calendar-date FE	Yes	Yes	Yes	Yes

Notes: The unit of observation is the vehicle-day, and the outcomes are daily numbers of idling spells longer than 30 seconds. Columns report all idling and its traffic, waiting, and warming components. The first two rows report effects for *Private & Social Info + Pay + Reminder* and *Private & Social Info + Pay + No-Reminder* relative to the control arm; the third reports the incremental effect of *IdleGuard*. Positive coefficients indicate that more spells cross the 30-second threshold. Regressions absorb vehicle and calendar-date fixed effects. Post covers 6 April through 30 May 2026; the pooled pre-treatment period comprises 12 study weeks observed between 17 November 2025 and 4 April 2026. Outcomes are computed over observed GPS vehicle-days. The sample excludes the 25 vehicles removed before the second-stage *Reminder* versus *No-Reminder* randomisation. Standard errors are clustered by vehicle. Significance levels are denoted by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

Table E.15: Persistence of *IdleGuard* effects on spells crossing the 30-second threshold

	Daily number of idling spells >30s			
	All	Traffic-like	Waiting-like	Warming-like
Prior Reminder arm $\times$ Post	0.303 (0.269)	0.278 (0.172)	0.026 (0.050)	-0.002 (0.103)
Prior No-Reminder arm $\times$ Post	-0.582** (0.256)	-0.360** (0.179)	-0.113* (0.058)	-0.109* (0.061)
Prior <i>IdleGuard</i> assignment	0.885*** (0.288)	0.639*** (0.181)	0.139** (0.058)	0.107 (0.107)
No-Reminder post-period mean	9.813	6.214	1.741	1.857
Observations	430,812	430,812	430,812	430,812
Vehicles	3,412	3,412	3,412	3,412
Vehicle FE	Yes	Yes	Yes	Yes
Calendar-date FE	Yes	Yes	Yes	Yes

Notes: The unit of observation is the vehicle-day, and the outcomes are daily numbers of idling spells longer than 30 seconds. Columns report all idling and its traffic, waiting, and warming components. The first two rows report post-treatment effects for vehicles previously assigned to *Private & Social Info + Pay + Reminder* and *Private & Social Info + Pay + No-Reminder* relative to the control arm; the third reports the corresponding prior-*IdleGuard*-assignment difference, defined as *Reminder* minus *No-Reminder*. Positive coefficients in the third row indicate more threshold-crossing spells among vehicles previously assigned to *IdleGuard*. Regressions absorb vehicle and calendar-date fixed effects. The post-treatment period comprises study weeks 21–32, from 1 June through 22 August 2026, after all treatment delivery and the *IdleGuard* reminder had ended; the pooled pre-treatment period comprises 12 study weeks observed between 17 November 2025 and 4 April 2026. Outcomes are computed over observed GPS vehicle-days. The sample excludes the 25 vehicles removed before the second-stage *Reminder* versus *No-Reminder* randomisation. Standard errors are clustered by vehicle. Significance levels are denoted by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

The threshold-crossing difference persists after the reminder is removed. Over the twelve post-treatment weeks, vehicles previously assigned to the no-reminder arm continue to have fewer spells longer than 30 seconds than the control arm, while the corresponding estimate for vehicles previously assigned to *IdleGuard* is close to zero. The resulting prior-*IdleGuard*-assignment difference is 0.9 additional threshold-crossing spells per vehicle-day and is statistically significant at the 1% level. Most of this difference arises in traffic-like idling. Because the reminder was no longer active during this period, the persistence of the gap is consistent with more durable preventive self-monitoring among drivers who had not been able to rely on the external cue.

Table E.16: Decomposing the incremental *IdleGuard* effect on idling duration

	Idling minutes from stops > 30s by duration layer							Daily spell count
	Full duration	First 30s	Beyond 30s	30–35s	35–60s	60–120s	> 120s	> 30s
Reminder arm $\times$ Post	-0.730 (0.581)	-0.033 (0.103)	-0.697 (0.512)	-0.013 (0.017)	-0.073 (0.063)	-0.099 (0.082)	-0.512 (0.408)	-0.066 (0.206)
No-Reminder arm $\times$ Post	-1.726*** (0.631)	-0.227** (0.111)	-1.500*** (0.551)	-0.038** (0.018)	-0.154** (0.069)	-0.236*** (0.090)	-1.071** (0.432)	-0.453** (0.222)
<i>IdleGuard</i>	0.996* (0.532)	0.194* (0.104)	0.803* (0.457)	0.025 (0.017)	0.082 (0.064)	0.137 (0.084)	0.559 (0.346)	0.387* (0.207)
Share of <i>IdleGuard</i> duration effect (%)	100.0	19.4	80.6	2.5	8.2	13.7	56.1	
No-Reminder post-period mean	19.874	4.946	14.928	0.802	2.952	3.497	7.677	9.893
Observations	372,907	372,907	372,907	372,907	372,907	372,907	372,907	372,907
Vehicles	3,412	3,412	3,412	3,412	3,412	3,412	3,412	3,412
Vehicle FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Calendar-date FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes: The unit of observation is the vehicle-day. Columns 1–7 decompose daily idling minutes from spells longer than 30 seconds into mutually exhaustive duration components; Column 8 reports the daily number of such spells. Full duration equals the first 30 seconds of each qualifying spell plus all idling beyond 30 seconds, which is further divided into the 30–35-, 35–60-, 60–120-, and beyond-120-second layers. The first two rows report effects for *Private & Social Info + Pay + Reminder* and *Private & Social Info + Pay + No-Reminder* relative to the control arm; the third reports the incremental effect of *IdleGuard*. The share row reports each duration component’s contribution to the full incremental *IdleGuard* effect. Regressions absorb vehicle and calendar-date fixed effects. Post covers 6 April through 30 May 2026; the pooled pre-treatment period comprises 12 study weeks observed between 17 November 2025 and 4 April 2026. Outcomes are computed over observed GPS vehicle-days. The sample excludes the 25 vehicles removed before the second-stage *Reminder* versus *No-Reminder* randomisation. Standard errors are clustered by vehicle. Significance levels are denoted by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

Table E.17: *IdleGuard* effects by terminal spell duration

	Idling minutes from stops > 30s by total spell duration				Daily spell count		
	Full duration	30–60s	60–120s	>120s	> 30s	> 60s	> 120s
Reminder arm $\times$ Post	-0.730 (0.581)	-0.033 (0.077)	-0.072 (0.092)	-0.624 (0.502)	-0.066 (0.206)	-0.085 (0.114)	-0.056 (0.060)
No-Reminder arm $\times$ Post	-1.726*** (0.631)	-0.114 (0.080)	-0.176* (0.100)	-1.436*** (0.537)	-0.453** (0.222)	-0.304** (0.125)	-0.182*** (0.065)
<i>IdleGuard</i>	0.996* (0.532)	0.081 (0.075)	0.104 (0.095)	0.811* (0.442)	0.387* (0.207)	0.219* (0.117)	0.126** (0.060)
Share of <i>IdleGuard</i> duration effect (%)	100.0	8.1	10.5	81.4			
No-Reminder post-period mean	19.874	3.553	4.116	12.205	9.893	5.148	2.264
Observations	372,907	372,907	372,907	372,907	372,907	372,907	372,907
Vehicles	3,412	3,412	3,412	3,412	3,412	3,412	3,412
Vehicle FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Calendar-date FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes: The unit of observation is the vehicle-day. Columns 1–4 report daily idling minutes from spells longer than 30 seconds, decomposed into mutually exclusive bins defined by total spell duration; Columns 5–7 report cumulative daily spell counts crossing the 30-, 60-, and 120-second thresholds. The first two rows report effects for *Private & Social Info + Pay + Reminder* and *Private & Social Info + Pay + No-Reminder* relative to the control arm; the third reports the incremental effect of *IdleGuard*. The share row reports each terminal-duration bin’s contribution to the full incremental *IdleGuard* effect on idling duration. Regressions absorb vehicle and calendar-date fixed effects. Post covers 6 April through 30 May 2026; the pooled pre-treatment period comprises 12 study weeks observed between 17 November 2025 and 4 April 2026. Outcomes are computed over observed GPS vehicle-days. The sample excludes the 25 vehicles removed before the second-stage *Reminder* versus *No-Reminder* randomisation. Standard errors are clustered by vehicle. Significance levels are denoted by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

Table E.18: Entry versus continuation within long *IdleGuard* spells

	Full > 120s min	First 120s of > 120s spells	Excess beyond 120s	Count > 120s
IdleGuard vs Placebo	-0.522	-0.049	-0.474	-0.024
	(0.448)	(0.121)	(0.359)	(0.060)
No IdleGuard vs Placebo	-1.360***	-0.325**	-1.035***	-0.163**
	(0.455)	(0.137)	(0.341)	(0.068)
IdleGuard added	0.837*	0.276**	0.561	0.138**
	(0.497)	(0.136)	(0.390)	(0.068)
Share of > 120s duration gap (%)	100.0	33.0	67.0	
NoIG post mean	12.205	4.528	7.677	2.264
Observations	179,604	179,604	179,604	179,604
Vehicles	3,133	3,133	3,133	3,133
Vehicle FE	Yes	Yes	Yes	Yes
Calendar-date FE	Yes	Yes	Yes	Yes

Notes: The unit of observation is the vehicle-day. The sample comprises the control arm, *Private & Social Info + Pay + Reminder*, and the matched *Private & Social Info + Pay + No-Reminder* arm. The table decomposes daily idling minutes from spells longer than 120 seconds into an entry component and a continuation component. The entry component equals the first 120 seconds of each qualifying spell, while the continuation component equals idling beyond 120 seconds in those spells. The count column reports the daily number of spells longer than 120 seconds. The incremental *IdleGuard* row compares the reminder arm with the matched no-reminder arm. The share row reports each component's contribution to the incremental *IdleGuard* effect on minutes from spells longer than 120 seconds. Regressions absorb vehicle and calendar-date fixed effects. Post covers 6 April through 30 May 2026; the pooled pre-treatment period comprises 12 study weeks observed between 17 November 2025 and 4 April 2026. Outcomes are computed over observed GPS vehicle-days, and standard errors are clustered by vehicle. Significance levels are denoted by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

E.6 *IdleGuard* Endline Mechanism Evidence

Tables E.19 and E.20 provide complementary endline evidence on the mechanisms underlying the *IdleGuard* effect. The first reports drivers’ experiences with the reminder, while the second tests whether *IdleGuard* changed beliefs about required engine warm-up.

Table E.19: Endline reports of experience with *IdleGuard*

	Relied on beep	Delayed correction	More comfortable
Share	0.314	0.596	0.421
Observations	309	329	321

Notes: The unit of observation is the vehicle. The table reports descriptive shares among drivers assigned to the *IdleGuard* arm who provide a substantive response to the corresponding endline question; responses indicating uncertainty, not noticing or remembering the beep, or an incomplete interview are excluded. Column 1 equals one if the respondent reports relying more on the beep, rather than continuing to monitor oneself or relying on both equally. Column 2 equals one if the respondent reports not usually turning the engine off or starting to drive promptly after the beep. Column 3 equals one if the respondent reports becoming more comfortable leaving the engine running because the beep would remind them; among substantive responses to this question, 30% report no change and 28% report becoming less comfortable.

Table E.20: Effects of *IdleGuard* on endline warm-up beliefs

	Any warm-up	At least 30 seconds	At least 1 minute
IdleGuard	-0.017 (0.019)	0.004 (0.025)	0.016 (0.026)
No-Reminder mean	0.864	0.709	0.362
Observations	1413	1413	1413

Notes: The unit of observation is the vehicle. The sample comprises vehicles assigned to the *Reminder* or matched *No-Reminder* arm with a usable endline outcome. Columns 1–3 equal one if the respondent believes, respectively, that any engine warm-up is needed, that at least 30 seconds is needed, and that at least one minute is needed. Each column reports the randomised effect of *IdleGuard*, comparing the *Reminder* arm with the matched *No-Reminder* arm. Endline completion and outcome availability do not differ significantly between the two groups. Robust standard errors are shown in parentheses. Significance levels are denoted by $*p < 0.10$, $**p < 0.05$, and $***p < 0.01$.

E.7 Pro-Social Preference Heterogeneity

Table E.21 examines whether treatment responsiveness varies with revealed pro-social preference, measured using the respondent’s incentivised baseline donation to WWF-Pakistan.

Table E.21: Treatment effects by revealed pro-social preference

	Idling minutes from stops >30s
<i>Panel A. WWF donation = Rs. 50</i>	
Private Info	-0.941 (0.972)
Social information	0.262 (1.109)
Payments	-1.030 (0.963)
IdleGuard	1.112 (0.793)
<i>Panel B. WWF donation = Rs. 100</i>	
Private Info	-2.004*** (0.682)
Social information	0.501 (0.652)
Payments	-0.550 (0.684)
IdleGuard	0.991* (0.587)
<i>Panel C. Rs. 100 minus Rs. 50</i>	
Private Info	-1.063 (0.874)
Social information	0.239 (1.144)
Payments	0.480 (0.992)
IdleGuard	-0.121 (0.831)
Control post-period mean	20.985
Observations	688864
Vehicles	6313

Notes: The unit of observation is the vehicle-day, and the outcome is daily idling minutes from stops longer than 30 seconds. The table reports the *Private Info* effect relative to the control arm and the incremental effects of social information, payments, and *IdleGuard*. The heterogeneity variable is the respondent’s baseline donation to WWF-Pakistan, scaled from zero to one. Panels A and B report implied effects evaluated at donations of Rs. 50 and Rs. 100 out of the Rs. 100 endowment, respectively; Panel C reports the implied change in each effect when moving from Rs. 50 to Rs. 100. Social information compares *Private Info* with *Private Info*; payments compare *Private Info* & *Social Info* + *Pay* with *Private Info* & *Social Info*; and *IdleGuard* compares *Private Info* & *Social Info* + *Pay* + *Reminder* with the matched *Private Info* & *Social Info* + *Pay* + *No-Reminder* arm. Respondents with missing or invalid donation amounts are excluded. Regressions absorb vehicle and calendar-date fixed effects. Outcomes are computed over observed GPS vehicle-days, and standard errors are clustered by vehicle. Significance levels are denoted by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

F Cost–Benefit Analysis

This appendix documents the construction of the cost–benefit estimates reported in Section 7 of the main paper. The exercise evaluates the *Private Info* campaign because it isolates the low-cost informational intervention that generated the clearest behavioural response without requiring social-information, payment, or reminder components. Benefits include avoided private fuel expenditure, avoided global climate damages, and avoided local PM_{2.5}-related health damages. Reduced engine wear and other possible maintenance benefits are not monetised.

F.1 Treatment-Effect Inputs

The calculation uses the treatment effect on the full duration of idling spells longer than 10 seconds. During the eight-week treatment period, the estimated reduction in the *Private Info* arm, expressed as a positive magnitude, is

$$\hat{\beta}_{\text{active}}^{PI} = 1.952 \text{ minutes per observed vehicle-day.} \quad (\text{F.1})$$

The active-period calculation covers eight treatment weeks and six study days per week. To express benefits per treated vehicle, I apply the estimated effect per observed vehicle-day to the 48 possible study days during this period, treating temporary tracker non-reporting as missing measurement and assuming that the estimated effect applies similarly on unobserved study days. Appendix Table C.6 finds no evidence of differential post-launch observability across treatment arms.

To incorporate the twelve directly observed post-switch-off weeks, I use the *Early-Off* subgroup of the *Private Info* arm. These vehicles stopped receiving messages after the eight-week campaign. Appendix Table D.14 estimates an active-period effect of 2.480 minutes and a pooled twelve-week post-switch-off effect of 2.193 minutes for this subgroup. The estimated post-switch-off effect is therefore

$$\frac{\hat{\beta}_{\text{post}}^{EO}}{\hat{\beta}_{\text{active}}^{EO}} = \frac{2.193}{2.480} = 0.884. \quad (\text{F.2})$$

of the corresponding active-period effect. I apply this persistence ratio to the more precisely estimated full-arm effect in Equation F.1. The resulting full-arm-equivalent post-switch-off reduction, expressed as a positive magnitude, is

$$\hat{\beta}_{\text{post}}^{PI} = 1.952 \times \frac{2.193}{2.480} = 1.726 \text{ minutes per observed vehicle-day.} \quad (\text{F.3})$$

This scaling preserves the post-to-active relationship estimated within the randomly assigned *Early-Off* subgroup while anchoring the treatment-period calculation to the full *Private Info* arm.

F.2 Physical-Benefit and Monetisation Inputs

For each evaluation horizon h , let M_h denote the cumulative reduction in idling minutes per treated vehicle. Fuel savings and avoided emissions are calculated as

$$F_h = M_h \times 0.0246, \quad (\text{F.4})$$

$$CO_{2,h} = M_h \times 0.06133, \quad (\text{F.5})$$

$$PM_{2.5,h} = M_h \times 0.02282, \quad (\text{F.6})$$

where F_h is measured in litres, $CO_{2,h}$ in kilograms, and $PM_{2.5,h}$ in grams.

Table F.1: Inputs to the cost–benefit calculation

Input	Value	Unit	Use in calculation
Idling fuel consumption	1.476	litres/hour	Conversion from avoided idling minutes to fuel saved
Petrol price	310.71	PKR/litre	Valuation of avoided private fuel expenditure
Idling CO ₂ emissions	3.68	kg/hour	Conversion from avoided idling minutes to CO ₂ avoided
Social cost of carbon	229.31	2025Q4 USD/tonne	Valuation of avoided global climate damages
Exchange rate	278.12	PKR/USD	Conversion of climate and health values into PKR
Idling PM _{2.5} emissions	0.02282	g/minute	Conversion from avoided idling minutes to PM _{2.5} avoided
Local PM _{2.5} health damage	137.0	PKR/g	Valuation of avoided local health damages
Programme cost	375.68	PKR/vehicle	Eight-week app, email, and WhatsApp delivery cost

Notes: The idling fuel-consumption rate follows the engineering input used to construct the treatment messages (Gaines and Rask, 2012). The petrol price is taken from Pakistan State Oil’s price archive (Pakistan State Oil, 2026). The CO₂ emissions rate combines the fuel-use rate with the average carbon content of petrol and diesel reported by the U.S. Environmental Protection Agency (U.S. Environmental Protection Agency, 2005). The social cost of carbon begins from the USD 185-per-tonne estimate in Rennert et al. (2022) and is updated from 2020 to 2025Q4 dollars using the U.S. GDP implicit price deflator (U.S. Bureau of Economic Analysis, 2026). The exchange rate is the USD-to-PKR rate observed on 14 July 2026 (XE, 2026). The derivation of the PM_{2.5} emissions and health-damage inputs is described in Appendix F.3. Programme cost includes the marginal app, email, and WhatsApp delivery costs of the eight-week campaign.

Table F.2: Implied physical benefits per treated vehicle

Evaluation horizon	Fuel saved (litres)	CO ₂ avoided (kg)	PM _{2.5} avoided (g)
8-week treatment period	2.30	5.75	2.14
20-week observed period	5.36	13.37	4.97
1 year: faster decay	7.15	17.81	6.63
1 year: slower decay	9.44	23.53	8.76

Notes: The table reports physical benefits per average treated *Private Info* vehicle. Fuel savings use an idling consumption rate of 0.0246 litres per minute. Avoided emissions use rates of 0.06133 kg of CO₂ and 0.02282 g of PM_{2.5} per idling minute. The eight-week and 20-week calculations use observed experimental effects. In the faster-decay scenario, the treatment effect declines linearly to zero 26 weeks after switch-off. In the slower-decay scenario, it declines linearly to zero by campaign week 52.

Private fuel benefits are

$$B_h^F = F_h \times 310.71. \quad (\text{F.7})$$

Global climate benefits are

$$B_h^{CO_2} = \frac{CO_{2,h}}{1000} \times 229.31 \times 278.12. \quad (\text{F.8})$$

Local air-pollution health benefits are

$$B_h^{PM} = PM_{2.5,h} \times 137.0. \quad (\text{F.9})$$

Total gross benefits, net benefits, and the benefit–cost ratio are therefore

$$B_h = B_h^F + B_h^{CO_2} + B_h^{PM}, \quad (\text{F.10})$$

$$NB_h = B_h - C, \quad (\text{F.11})$$

$$BCR_h = \frac{B_h}{C}, \quad (\text{F.12})$$

where $C = 375.68$ PKR is the eight-week programme cost per treated vehicle.

F.3 Local PM_{2.5} Health Benefits

The local-health calculation first converts avoided idling fuel into avoided primary PM_{2.5} emissions. The calculation uses a simple average of petrol and diesel densities. The petrol-density input is taken from laboratory analysis of commercial motor gasoline in Pakistan (Yasin et al., 2008), while the diesel-density input is taken from laboratory analysis of Pakistan State Oil high-speed diesel (Ashraf et al., 2016). Their midpoint is 0.7795 kilograms per litre. I combine this density with the

fleet-average emissions factor of 1.19 grams of PM_{2.5} per kilogram of fuel estimated from near-road vehicle measurements in Mumbai (Raparathi et al., 2023). Together with the idling fuel-use rate, these inputs imply

$$0.0246 \times 0.7795 \times 1.19 = 0.02282 \text{ grams of PM}_{2.5} \text{ per idling minute.} \quad (\text{F.13})$$

I map avoided emissions into ambient exposure using the transport contribution to annual PM_{2.5} emissions in Punjab’s emissions inventory (FAO, 2018). The exposure calculation uses the combined population of Punjab, Sindh, and the Islamabad Capital Territory from the 2023 Population and Housing Census (PBS, 2023). I then translate the implied change in annual average exposure into life-expectancy gains using the Air Quality Life Index concentration–life-expectancy relationship (Greenstone et al., 2025).

Mortality-risk reductions are valued using the OECD value of a statistical life for lower- and middle-income countries (OECD, 2025). Because the Air Quality Life Index relationship represents the long-run effect of sustained exposure, I conservatively prorate the estimated life-expectancy gain linearly over an assumed 68-year life expectancy. This procedure implies a central local health value of approximately

$$137.0 \text{ PKR per gram of avoided PM}_{2.5}. \quad (\text{F.14})$$

This valuation is illustrative. It relies on an emissions-to-exposure mapping rather than a local atmospheric-dispersion model and assumes that the health consequences of a temporary exposure reduction can be approximated by linear prorating of the long-run concentration–life-expectancy relationship.

F.4 Evaluation Horizons

I report four evaluation horizons.

Eight-week treatment period. This benchmark counts benefits over the same eight weeks in which messages were delivered. It uses the full-arm effect in Equation F.1 over 48 possible study days and requires no assumption about persistence after treatment.

Twenty-week observed period. This horizon adds the twelve directly observed post-switch-off weeks. The first eight weeks use the full-arm active-period estimate, while the subsequent twelve weeks use the scaled post-switch-off estimate in Equation F.3. The programme cost remains fixed because no additional messages are delivered during these twelve weeks.

One year: faster decay. The faster-decay scenario begins from the observed post-switch-off effect and assumes that it declines linearly to zero 26 weeks after switch-off. Because treatment

ends after campaign week 8, the effect reaches zero at campaign week 34. Benefits after that point are set to zero.

One year: slower decay. The slower-decay scenario begins from the same observed post-switch-off effect and assumes that it declines linearly to zero by campaign week 52. No benefits are counted beyond the one-year evaluation horizon.

The eight-week and 20-week estimates are based on experimentally observed effects. The two one-year estimates are extrapolation scenarios rather than estimated one-year treatment effects. Both scenarios impose eventual complete decay and assume neither continued message delivery nor permanent treatment persistence.

F.5 Illustrative National Scaling

Let N_{PV} denote Pakistan’s reported stock of registered motor cars, jeeps, and station wagons. The Pakistan Economic Survey reports

$$N_{PV} = 4,889,104 \quad (\text{F.15})$$

registered vehicles in this category in 2025 ([Government of Pakistan, Ministry of Finance, 2026](#)).

Let s denote the share of this stock reached by the campaign. The number of reached vehicles is

$$N_{\text{reached}} = N_{PV}s. \quad (\text{F.16})$$

For evaluation horizon h , national gross benefits and programme costs are

$$B_h^N = B_h N_{PV}s, \quad (\text{F.17})$$

$$C^N = C N_{PV}s. \quad (\text{F.18})$$

National net benefits are

$$NB_h^N = B_h^N - C^N. \quad (\text{F.19})$$

Because benefits and costs scale proportionally with programme reach, the national benefit–cost ratio is equal to the corresponding per-vehicle ratio:

$$BCR_h^N = \frac{B_h^N}{C^N} = \frac{B_h}{C}. \quad (\text{F.20})$$

The preferred illustration uses $s = 0.25$, corresponding to

$$4,889,104 \times 0.25 = 1,222,276 \quad (\text{F.21})$$

reached vehicles. This is intended as an illustrative programme scale rather than a forecast of realised national coverage.

Table F.3: Illustrative national physical benefits at 25% programme reach

Evaluation horizon	Fuel saved (million litres)	CO ₂ avoided (tonnes)	PM _{2.5} avoided (tonnes)
8-week treatment period	2.82	7,024	2.61
20-week observed period	6.55	16,341	6.08
1 year: faster decay	8.73	21,776	8.10
1 year: slower decay	11.54	28,763	10.70

Notes: The illustration applies the per-vehicle *Private Info* intent-to-treat effects to 1,222,276 vehicles, equal to 25% of Pakistan’s registered motor cars, jeeps, and station wagons. The eight-week and 20-week calculations use observed experimental effects. The one-year estimates assume linear decay to zero either 26 weeks after switch-off or by campaign week 52.

Table F.4: Monetised benefits in the preferred national-scale application

Evaluation horizon	Gross benefit	Programme cost	Net benefit	BCR
8-week treatment period	1.68	0.46	1.22	3.66
20-week observed period	3.91	0.46	3.45	8.52
1 year: faster decay	5.21	0.46	4.75	11.35
1 year: slower decay	6.89	0.46	6.43	14.99

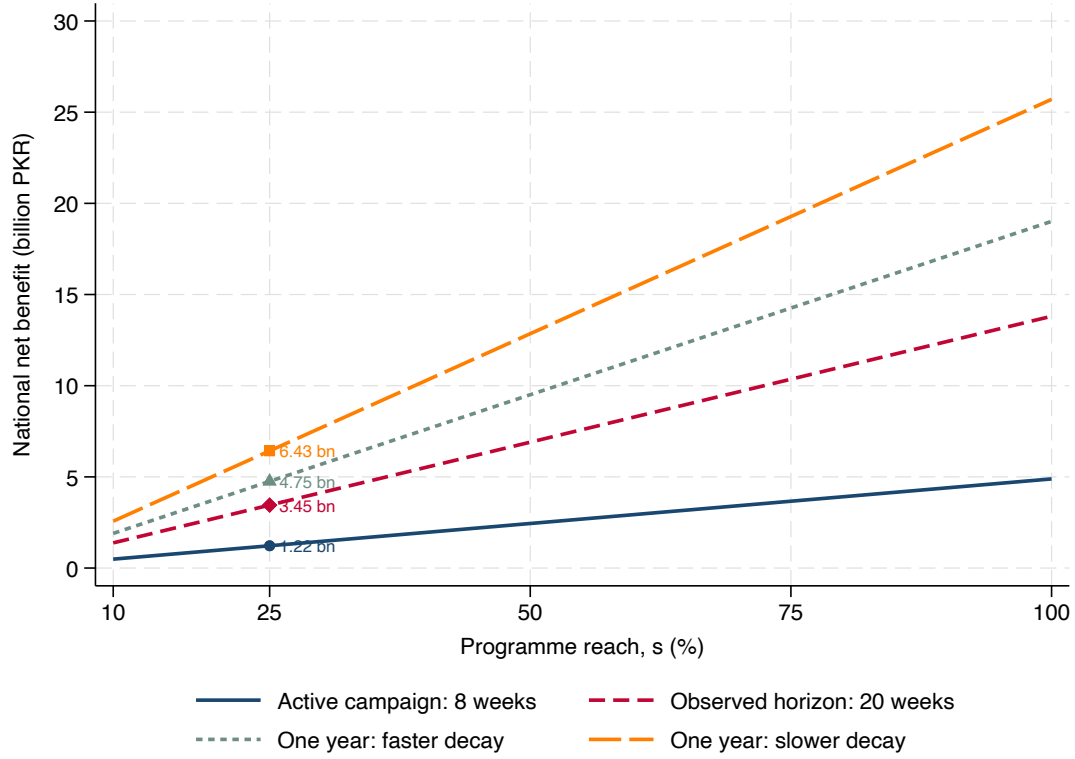
Notes: Monetary values are billions of PKR. The illustration assumes that the programme reaches 25% of Pakistan’s registered motor cars, jeeps, and station wagons, corresponding to 1,222,276 vehicles. The experimental per-vehicle intent-to-treat benefit is applied directly to each reached vehicle. Benefits and programme costs scale proportionally with programme reach, so the national benefit–cost ratios equal the corresponding per-vehicle ratios. The eight-week and 20-week calculations use observed experimental effects. The one-year calculations apply the faster- and slower-decay scenarios described in Appendix F.4.

F.6 Interpretation and Limitations

The eight-week estimate is the most conservative benchmark because it counts benefits only while the campaign is actively delivered. The 20-week estimate additionally incorporates twelve weeks of directly observed persistence without adding programme costs. The one-year estimates show how returns vary under two transparent assumptions about decay beyond the observed period.

The exercise monetises private fuel savings, global climate benefits, and local PM_{2.5}-related health benefits. It excludes reduced engine wear and other possible maintenance gains, making the calculation conservative in the set of benefits included. Conversely, the national calculations assume that the experimental per-vehicle intent-to-treat effect applies directly to each vehicle reached. They

Figure F.1: Illustrative national net benefits under alternative programme reach



Notes: The figure reports illustrative national net benefits across four evaluation horizons and alternative programme-reach shares. Programme reach is defined as the share of Pakistan’s registered motor cars, jeeps, and station wagons reached by the campaign. The per-vehicle *Private Info* intent-to-treat estimates are applied directly to each reached vehicle. Markers identify the preferred illustrative case of 25% reach. The eight-week and 20-week calculations use observed experimental effects. The one-year estimates assume linear decay to zero either 26 weeks after switch-off or by campaign week 52. The calculations are illustrative scaling exercises rather than forecasts of realised national programme coverage.

should therefore be interpreted as illustrative scaling exercises rather than forecasts of take-up, implementation fidelity, or realised national effects.

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